

Psychiatric history taking

Dr. Nitin Sethi



Contents

1. Introduction	1
2. Personal information	3
3. History of presenting complains	5
4. Past History	9
5. Family History	10
6. Personal History	11
7. Physical Examination	15
8. Mental Status Examination	17
9. MSE of an uncooperative patient	53
10. Diagnostic formulation	55
11. Diagnosis and Diagnostic nomenclature systems	58
12. Special populations	59
12.1 Children and adolescent	59
12.2 Substance use disorders	63
12.3 Epilepsy	66
13. Appendices	73
13.1 Symptom analysis	73
13.2 Mood graph & Illness graph	75
13.3 Cognitive function assessment	77
13.4 Mini Mental Status Examination	89
13.5 Bush Francis Catatonia Rating Scale	91
13.6 Abnormal Involuntary Movement Scale (AIMS)	93
13.7 Intoxication states for substance of abuse	95
13.8 Withdrawal states for substance of abuse	99
13.9 Epilepsy classifications	101

Introduction

If a person has physical illness, he goes to a doctor with certain complaint. The complaint points to the direction in which its cause is to be sought and narrows the field of enquiry. A careful physician makes a full examination of all the systems, but with a mind that is sensitized to a limited number of possibilities, his history taking will be governed by same principle. The mental health professional proceeds in the same way, but in the field which has been left almost untouched by the physician. He will pay attention to matters which the latter has dismissed as personal or accidental. He sees beyond abnormalities of structure, into the complexities of behaviour or deviations of mood.

In behavioural sciences, nearly everything of clinical importance is derived from the study of the patient as an individual, and precise and detailed knowledge is therefore required of the way in which the patient's personality differed from that of other people, how it had grown, and how it had been influenced by all the events of his life. Thus the interviewer will necessarily be interested in circumstances of patient's life, his hopes, fears, conflicts & disappointments.

In history taking and the examination of the patient, two methods may be followed. They are not alternatives, and it is best if they are combined, for each has its own deficiencies. The *free interview* may deteriorate into a conversation into social level, and may provide only hints and indications, rather than solid facts; certain important themes may go untouched. The method of *questionnaire* has other defects. It is uncomfortable for the informant or the patient and he may feel as a pupil in presence of school-master; and the information obtained may consist in a mass of detail, without highlight or relief, which is very difficult to organize into a coherent picture.

Thus the best plan is to have a framework of questionnaire in mind, but to allow the patient to tell his story. As the story unfolds, it is fitted into the framework, so that any gaps that are still left are apparent. Further questioning will then fill up the gaps and will clarify point of salient importance. The plan should be subject to modification as the information pours in. The interviewer should avoid giving the patient any feeling that he is being treated as a "case" only. He should be both *neutral and sympathetic*. As far as possible, he should keep his mind open and guard himself against pre-conceived ideas. So, as with a physician, the examination by a mental health professional needs to be methodological and should be described under proper headings and should follow a *general schema*. But the most important caution in use of the schema is not to be too rigidly bound by it.

Only in the course of time, one can develop the art of eliciting, by tactful questioning, of all he has to know. Long training is required to learn how to overcome the patient's resistance, to be aware of where his tale is biased, where information has been withheld and where it has been coloured by an emotional attitude. It is better for the beginner to be too circumstantial in his descriptions rather than being too selective.

A general schema and description of terms used in history taking are given in following pages.

The aim of this write-up is to facilitate the learning of an art. For that the learner will encounter certain terms, whose meaning and significance he would want to know. This write-up is nowhere intended to replace any of the standard text material available in context.

The books which will be handy for a trainee, and which are directly or indirectly source of information in this write-up includes:

- Fish Psychopathology (2nd and 3rd editions)
- SIMS (Symptoms in the mind) 4th edition
- SCAN glossary and interview schedule
- Hutchison's Clinical Methods 23rd edition
- Bickerstaff's Neurological Examination in Clinical Practice 7th adapted edition
- The Mental Status Examination in Neurology (Strub & Black) 4th edition
- The International Classification of diseases (ICD)-10 Classification of Mental and Behavioural Disorders- Diagnostic criteria for research (DCR)- WHO, 1993.
- The ICD-10 Classification of Mental and Behavioural Disorders- Clinical descriptions and diagnostic guidelines (CDDG)- WHO, 1992
- Diagnostic and Statistical Manual of Mental Disorders- Fourth Edition- Text revision (DSM-IV-TR)- APA, 2000
- Diagnostic and Statistical Manual of Mental Disorders- Fifth Edition (DSM-5)- APA, 2013
- New Oxford Textbook of Psychiatry 2nd edition
- Kaplan and Sadock's Comprehensive Textbook of Psychiatry 9th edition
- Kaplan and Sadock's Synopsis of Psychiatry 10th edition
- Introduction to Psychology (Morgan & King) 7th edition

Personal information

Name of the individual: It is very important to know the name of the person before an interview is started. It is the primary identification of an individual. It may be asked as the opening sentence of interaction or it may be read from the documents. It is advisable to call the patient by his name (with due respect and regard) in subsequent interaction, because it gives a feeling of being familiar to the patient and helps in making him at ease throughout the interaction.

Father's/Husband's Name: It is advisable to document, as identification becomes difficult with own name if there are two individuals with same name.

Sociodemographic profile: All socio demographic variables (age, sex, education, occupation, socio-economic status, marital status, religion and area of residence) should be adequately and correctly noted as all of these factors have a role to play in onset, course, presentation, treatment and prognosis of various illnesses.

Age: Age should be noted and further corroborated while asking duration of illness and age at onset of illness. Various mental as well as physical illnesses have a particular age of onset. Thus knowing current age as well as age at onset of illness becomes important.

Sex: Certain disorders are common in one sex than the other. Certain socio cultural factors might have more importance for one sex than the other.

Education: It would help in assessing the overall knowledge of the patient and also to base our testing based on educational status of the patient. Signs and symptoms can also vary according to educational background of the patient. In intervention, especially non pharmacological methods, the modality should be adjusted according to the educational level of the patient.

Occupation: Knowing the past as well as current occupation of the patient is important as it will have direct implication in socio-economic status of the patient. Impact of illness on occupation can be assessed only if we know what the patient used to do before illness.

Socio-economic status: One needs to know the SES of the patient to be aware how much one can afford to spend on treatment and required investigations.

Marital status: It is an important prognostic factor. It also helps us to get an information on social support of the individual. Proper marital history needs to be taken if individual is/was married.

Religion: Customs vary from one religion to other significantly. Also the examiner needs to have basic idea about customs of prevalent religions in his/her area.

Residence: Customs and beliefs are significantly different for rural/urban population. One should also know how far from the patient is coming, so that to formulate the frequency of follow up visits accordingly.

Language spoken as mother tongue: Psychiatric interview is all about talking and observing. An initial enquiry needs to be made as to whether patient knows the language in which examiner intends to conduct the interview. It is preferable to conduct the interview in mother tongue of the patient. If possible, mental health professional should attempt to have a basic idea of languages spoken in his/her area.

Address: Proper address (both permanent as well as present) needs to be noted along with Phone no and email id, so that any postal/electronic/telephonic contact can be made with the patient or the guardians as per need in the future.

Source of referral: It will hint us about the awareness of the patient and his/her caregivers about the current condition of the patient. If referred from some authority, proper mention should be made.

Type of admission, if admitted: It needs to be mentioned as it has legal implications as per Mental Health Act, 1987

Identification marks: They have role in identifying the patient and have medico-legal importance. Traceable and permanent marks should be mentioned. Common moles should be avoided. Proper anatomical location should be mentioned.

History of Present illness

Patient's report of his illness: It should take precedence over the other informants' report. Patient should be asked about his version of the illness. Using the patient's own words gives insight into his state of mind and how he himself views his symptoms. In case information from patient is unsatisfactory or he denies symptoms or is non communicative, it should be mentioned with reasons and then should be proceeded for history taking from other available informants.

Informant's report: Details of all the available informants should be documented first, including their names, relationship with the patient, acquaintance, length of contact, consistency of the information. Also mention the reason for seeking help at current point of time and what do they expect from the treatment they are seeking currently.

Reliability- It refers to the likelihood that similar results will be obtained by different observers. The verification of the information especially factual data given by an informant can be cross- checked by talking to another informant. Essentials for reliability are (remembered by acronym of 5Cs)

- Contact- between the patient and informant
- Closeness- of relationship between the patient and informant
- Continuity- of the account given by informant
- Consistency- of the verbatims of the informant
- Corroborativeness- between various sources of information

Adequacy- it refers to the amount of information given by the informant and assessment that whether this information is sufficient for forming a diagnosis or not.

Chief complaints- The chief complaints, often recorded verbatim states, why he has come or has been brought in for help. It usually describes present symptoms, including the duration of each and an account of the development of the illness. Complaints should be in chronological order with the earliest complaint first and recent most being last in list.

History of present illness- Popularized by the acronym of HOPI, this history forms the backbone of psychiatric case work up. This provides a comprehensive and chronological picture of the events leading up to the current moment in the patient's life. The evolution of the patient's symptoms should be determined and summarized in an organised and systematic way.

Factors in illness

Predisposing factors- Factors operating from early life that determines a person's vulnerability to develop a disorder or likelihood that person will develop certain symptoms under given stress conditions.

- Biological (delayed milestones, head injury, family history of psychiatric illness)
- Psychological (impaired premorbid personality)
- Social (home atmosphere in childhood, neglect, abuse, low education level)

Precipitating factors- Events that occur shortly before the onset of a disorder and act as physical or psychosocial stressors and lead to the onset of disorder in a person who may be predisposed to develop the disorder

- Biological (fever, accident, onset of severe medical illness)
- Psychological (stress intolerance, poor impulse control)
- Social (trauma, loss of job/partner)

Perpetuating factors- Factors due to which the disorder is maintained or aggravated.

- Biological (chronic medical illness, substance use)
- Psychological factors (poor insight, poor impulse control, low intelligence)
- Social (social isolation, unemployment, ongoing expressed emotions in family)

Limiting factors- Factors which limit the illness from an extensive progress and may include factors such as good social support or treatment during the course of illness.

Modifying factors- Factors which modify natural or expected course of the illness. This includes factors such as use of substance by a patient with Schizophrenia which may lead to affective colouring of illness, use of antidepressants causing a manic switch in patient with depressive illness.

Mode of onset: It is assessed as time from being asymptomatic to symptomatic

Abrupt- Sudden appearance of signs and symptoms within 48 hours e.g. delirium

Acute- Rapid onset of signs and symptoms within 2 weeks e.g. ATPD

Insidious- Onset of signs and symptoms takes more than 2 weeks e.g. Schizophrenia

Course of illness

Continuous- Characterised by uninterrupted change without breaks or with steps infinitely small and thus not detectable e.g. Schizophrenia.

Episodic- An illness can be said episodic when it has an onset and an offset of signs and symptoms of the disease with periods of recovery inbetween at least for a period of 2 months e.g. affective illness, non affective remitting psychosis

Fluctuating- When the course is waxing and waning especially under the effect of treatment. e.g. Obsessive compulsive disorder, Schizophrenia

Progress of illness- To what extent has the patient's symptomatology represented an evolution over time

Improving- Improving from the date of onset e.g. Depression (with treatment)

Deteriorating-Condition is getting worse by time e.g. Schizophrenia

Static- Condition remains same no change happens e.g. Dysthymia

Symptom analysis needs to be done at level of each symptom, so as to rule out (as far as possible) all differential reasons for that particular symptom at that point itself. When explaining a particular complaint or symptom, it should be dealt in with as ABC model, i.e. assessing the antecedent to that behaviour, then the behaviour itself and then the consequence of

that particular behaviour. Adequate and explanatory examples for each behaviour should be mentioned, in chronological order. How and when a particular behaviour is noticed and how it ends, all needs to be explained in the words of the informant, avoiding use of technical words.

Other symptoms which the examiner expects to be present should be asked for, even if the informant does not give spontaneous account. Certain symptoms (such as suicidal attempt) should be asked for and ruled out in all cases. Any such relevant symptom, thus present, should be adequately mentioned. One also needs to rule out all other major psychiatric illness, which can be thought of as possible differentials or co-morbid illnesses.

Presence or absence of fever or other physical illness, endocrinal disorder, drug (which can cause psychiatric illness for example antimalarials, steroids etc.) intake, significant head injury, substance use disorder, epilepsy should be enquired about. When mental disorder is thought secondary to these enumerated causes, the decision to classify a clinical syndrome as organic/induced is supported by the following

- a) Evidence that the cause in question is known to be associated with one of the mental disorder
- b) A temporal relationship (weeks or a few months) between the development of the underlying cause and the onset of the mental disorder
- c) Recovery from the mental disorder following removal or improvement of the underlying presumed cause
- d) Absence of evidence to suggest an alternative cause of the mental disorder (such as a strong family history or precipitating stress)

Conditions (a) and (b) justify a provisional diagnosis; if all four are present, the certainty of diagnostic classification is significantly increased.

Persistent and pervasive mood during the course of illness should be asked for and mentioned.

Impact of illness on patient's attitudes should be noted. This should include mixing with people, interest in work, self confidence, enthusiasm & optimism, experiencing pleasure in light of current problems.

Role functioning and biological functions during the period of illness should be asked for.

Role functioning refers to any pattern of behaviour involving certain rights, obligation, duties which an individual is expected, trained and indeed encouraged to perform in a social situation. This would include duties such as studies (in case of a student), occupation (in case of a working adult), taking care of children or household duties.

Biological functions are considered in this section because they are subjective phenomena appropriately considered with other subjective symptomatology. These include bowel and bladder control, sleep, appetite, weight and libido.

Insomnia is a subjective complaint of difficulty falling or staying asleep or of poor sleep quality. It can be initial insomnia i.e. difficulty in falling sleep; middle insomnia i.e. awakening in the middle of the night and eventually falling asleep but with difficulty or terminal insomnia i.e. wakening before one's waking time and being unable to return to sleep.

Activities of daily living and personal care & hygiene should be asked for. This would include activities such as brushing teeth, taking bath, taking care of bowel and bladder habits, having food and water.

To summarize, HOPI should cover major headings under acronym of ABCD

- A- Affect
- B- Biological functions
- C- Complaints in behaviour
- D- Daily living activities

In the end of HOPI, all relevant histories which were ruled out should be mentioned under the rubric of negative history.

Treatment history: It includes details of the treatment obtained in the present episode. It is noted whether help was sought and if help was sought, following are noted. When was the first contact; whether treatment was voluntary/ involuntary; who saw the patient and for how long (Psychiatrist/Psychologist/Physician/Faith healer/Traditional practitioner); the nature of the treatment (Pharmacological/Psychotherapy/faith healing/traditional); modality that was helpful (psychopharmacological interventions, individual/ group therapy); medication, if any that were prescribed, details should be mentioned including doses, duration, compliance, response, adverse effects (tabulate details as much as possible); length of treatment; reason for discontinuing treatments or poor compliance; day treatment/ hospitalization if done, all of these domains should be elaborated.

Compliance- Compliance is defined as the extent to which the patient's behaviour (in terms of taking medications, following diets or executing other lifestyle changes) coincides with medical recommendations. Physicians prescribe medication, diets and expect patient to follow them faithfully. Many, perhaps most, patients comply with a physician's instruction, but other patients do not; in fact some patients may pay little attention to a doctor's guidance, and this is especially the case when they begin to feel better or when symptoms are not obvious. Compliance requires comprehension by the patient and communication is the key for avoiding noncompliance. The motivation to be healthy, a prescribed vulnerability to an illness, the potential for negative consequences, effectiveness of the treatment, sense of personal control, and effective communication are the strongest influences for compliance.

Another closely related entity is adherence. Adherence is defined as the extent to which the patient's behaviour (in terms of taking medications, following diets or executing other lifestyle changes) matches medical recommendations "*jointly agreed*" between patient and prescriber. The difference is minimal and essentially concerns the physician's degree of authority. It is more of a western concept and has been adopted by many as an alternative to compliance in an attempt to emphasise that the patient is free to decide whether to adhere to the doctor's recommendations and failure to do so should not be a reason to blame the patient. This definition thus emphasises on the need for agreement between patient and the prescriber.

Past history

Past medical history: This includes an account of major medical illness and conditions, including common as well as uncommon chronic childhood illness, conditions leading to frequent medical consultation and treatment and those requiring emergency department visits, and those requiring hospitalization.

Past Psychiatric History: Take a detailed history of previous episodes, symptoms, duration, probable diagnoses, all available treatment details including hospitalization, inter-episodic functioning, deficits.

Clinical course indicators- Different task forces have come up with definitions for course indicators in different psychiatric illnesses. The ones worthy of mention are as follows

MacArthur Foundation Research Network task force proposed following definitions for *unipolar depressive disorder*, based on the assumption that major depression was episodic and that the episodes in the illness have an ending.

- Remission- Treatment of a depressive episode, if successful, would lead to a significant reduction of symptoms (“response”) and ideally to “remission,” a state of minimal or no symptoms.
- Relapse- If symptoms reemerged following remission, this would be considered a “relapse” within the index episode.
- Recovery- If remission were stable over a number of months (i.e. there was no relapse or sub-syndromal symptomatic exacerbations), then recovery would result. “Recovery” essentially meant that the index depressive episode had ended at both the clinical and neurobiological levels.
- Recurrence- After recovery, a subsequent emergence of symptoms would be regarded as a “recurrence,” or a new depressive episode.

Work-group of experts in *bipolar disorder* developed these consensus operational definitions.

- Response- A 50% reduction in a score from a standard rating scale of symptomatology from an appropriate baseline, regardless of index episode type (manic, depressed, or mixed) is defined as response. In addition, the other pole cannot be significantly worsened during response.
- Remission was defined as absence or minimal symptoms of both Mania and Depression for at least 1 week. Sustained remission requires at least eight consecutive weeks of remission, and perhaps as many as 12 weeks.
- A relapse/recurrence was defined as a return to the full syndrome criteria of an episode of mania, mixed episode, or depression following a remission of any duration.
- Roughening was defined as a return of symptoms at a subsyndromal level, perhaps representing a prodrome of an impending episode.

For *Schizophrenia*, The Schizophrenia Work Group recommended that remission include attaining minimal levels of psychoticism, disorganization, and negative symptoms as assessed by the Scale for the Assessment of Positive Symptoms (SAPS), Scale for the Assessment of Negative Symptoms (SANS), Positive and Negative Syndrome Scale (PANSS), and Brief Psychiatric Rating Scale (BPRS) for a minimum of 6 months.

Family history

Parents and siblings: Age now or at death (if dead, the cause), occupation, personality, quality of relationship with parents, psychiatric and medical history.

A brief statement about any psychiatric illness, hospitalization, and treatment of the patient's immediate family members should be placed in the family history part of the report.

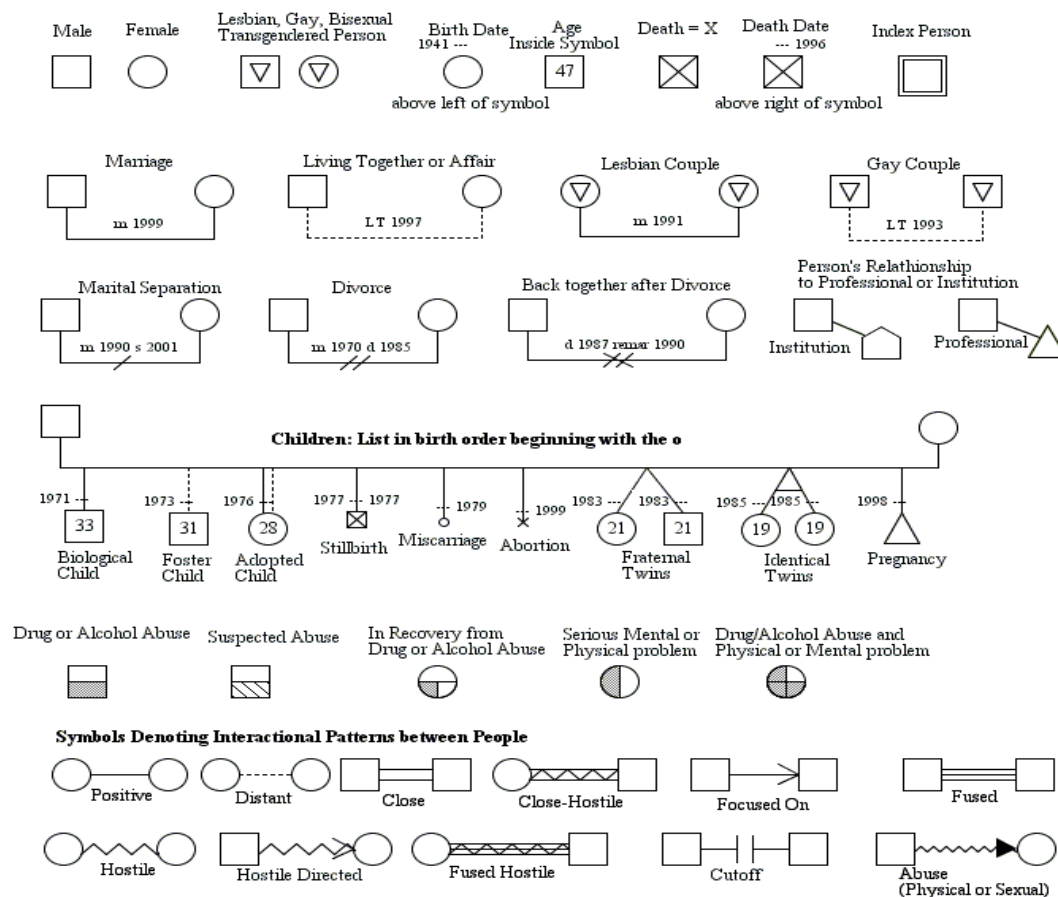
Any family history of alcohol or substance abuse or of personality problems should be documented. In addition, the family history should provide a description of the personalities and intelligence of the various households in which the patient lived.

Consanguinity: Relation by blood/descent from a common ancestor within the same family stock. If present than degree of the consanguinity should be noted.

Relationships amongst family members: Patient's relationship with family members, interpersonal relationship among family members; family squabbles, attitude of family towards patient's illness; family support system should all be noted in family history part.

Genogram: The genogram is a valuable assessment tool for learning about a family's history over a period of time. Based upon the concept of a family tree, it usually includes data about three or more generations of the family, which provides a longitudinal perspective. The genogram provides a graphic picture of family genealogy, including significant life events (birth, marriage, separation, divorce, illness, death); occupations; places of family residence

Standard Symbols for Genograms



Personal history

It comprises of a chronological account of the person's personal experiences starting with his birth and birth details. The personal history is usually divided into perinatal, early childhood, late childhood and adulthood. The predominant emotions associated with the different life periods (e.g. painful, stressful and conflictual) should be noted.

Birth and Early Development: Antenatal history should start from presence of any illness, medication, drugs, alcohol use, trauma or bleeding, exposure to X-rays, any physical/psychiatric illness during pregnancy. Illness can include infectious disease which can present as fever with or without rash, sexually transmitted diseases, diabetes, hypertension, jaundice etc. For medications used in pregnancy, one should be aware of teratogenic effects of common drugs.

How was the home situation into which the patient was born. Whether he/she was wanted? Whether it was a planned or unplanned conception? Whether a failed abortion attempt was made?

What was the mother's emotional and physical state at the time of pregnancy as well as delivery. Whether and from when fetal movements were perceived by the mother.

Whether the delivery was full term, preterm or postterm? Place (home/ hospital/ other) and type of delivery (normal/ instrumental/ episiotomy/ caesarian section), any injury at the time of birth, birth weight, normal or delayed cry should be documented. Any other complication during delivery such as abnormal presentation, cord around neck, prolapsed cord, multiple pregnancy or congenital anomaly noticed immediately after birth and presence of neonatal jaundice and its extent should be enquired about.

What was the mode of feeding after birth, any problems associated with feeding, age at weaning, recurrent infections, significant injury, convulsions in period immediate after birth and early childhood should be reported

Developmental milestones should be probed into. Any delay in developmental milestones should be documented. For this the interviewer himself should be well verse with normal developmental milestones. Important milestones are mention below:

<i>Gross Motor</i>	
Neck holding	3 months
Sits alone	6 months
Crawls	8 months
Stands alone	10 months
Walks alone	12 months
Runs stiffly	18 months
Walks up and down stairs	24 months
Rides tricycle	36 months

<i>Fine motor</i>	
Follows moving object	2 months
Grasps rattle	4 months
Reaches for object	6 months
Thumb finger grasp	9 months
Pincer grasp	12 months
Scribbles	18 months
Explores drawers & baskets	18 months
Block games	24 months

<i>Language</i>	
Cooing	2 months
Turns to sound	4 months
Monosyllabic	6 months
Bisyllabic	8 months
One word	12 months
Jargon	15 months
Phrases	18 months
Sentences	24 months
Tells a story	36 months

<i>Social skills</i>	
Social smile	3 months
Vocalizes in response	5 months
Stranger anxiety	8 months
Waves bye-bye	9 months
Indicates desire, hugs parents	15 months
Plays with other child	24 months
Tells his name	30 months
Gender identity	36 months
Rules of games	60 months

Adaptive skills

Feeds self in any way	8 months	Drink from bottle	34 months
Helps in house	15 months	Bladder control night	36 months
Feeds properly	20 months	Brushes teeth, wash hands	36 months
Bowel control day	20 months	Helps to dress	36 months
Bowel control night	22 months	Visit key places around	42 months
Bladder control day	24 months	Plays with several children	48 months
Helps to undress	24 months	Dresses and undresses self	60 months

Presence of childhood disorders: Comment on presence of hyperactivity, attention deficits and impulsivity which are usually noticed from age of 2-3 years.

Conduct problems during childhood should be probed into and will include disobedience, lying, stealing, truancy (running away from school), cruelty towards animals, bossy attitude towards younger children, not obeying rules while playing etc. If these symptoms are found in childhood, do make an attempt to look for dissocial personality traits in adolescent period.

Temper tantrums are very common in children; when present, extent and intensity should be carefully noted. Neurotic traits (nail-biting, thumb sucking, food-faddiness, stammering, mannerisms, bedwetting, phobias, night-terrors, sleep walking, etc.) during childhood should be probed into and if present, the details should be mentioned.

A comment on social relation with peers, elderly and authority figures and younger children should be made.

Home atmosphere in childhood and adolescence: A comment should be made on emotional environment at home in formative years (disturbed/congenial/any abnormality of family situation viz. desertion by a parent, broken home, step-parents, adopted sibs etc. and also patient's attitude towards parents)

Scholastic and extracurricular activities: Comment on age and class of entry in school, type of school, scholastic performance and progress in studies, regularity in school, failures if any, disciplinary problems/actions if any, relational problems with peers/authorities, any

discontinuity or change in school/college with reasons, involvement in games and extra-curricular activities. Also mention special interests in games if any during childhood.

Vocational/Occupational history: Mention the age at which the individual started working professionally for the first time. Duration at each work place, positions held, reasons for leaving, relation with work mates and superiors, promotion (in comparison to colleagues) should be commented upon. Impact of illness on occupation will form a part of history of presenting illness itself.

Menstrual history: Age of menarche should be asked. What was the reaction of patient towards it and also information and attitude towards mensuration subsequently? Regularity and duration of usual cycle, whether associated with psychological and physical change (pain or any other). Date of last menstruation, duration and reasons of amenorrhea, if any.

Sexual and marital history: How and when sexual information and knowledge was first obtained and of what kind, masturbatory history (fantasy and activity), sex play if any, adolescent sexual activity, premarital and extramarital sexual relationship if any, sexual disorders (normal and abnormal), presence of any gender identity disorder are areas to inquire about. Also probe for any history of childhood sexual abuse.

Marital history includes all enduring intimate relationships. Ask for age at marriage and parental consent for marriage. The spouse's age, occupation, personality and state of health are relevant to the patient's circumstances should be documented. Also ask for role allocation, sharing of responsibilities and decision making, perceived adequacy of sexual relation. Knowledge and use of contraception should be documented.

Forensic history: Trouble with police, law; charges and convictions (sections), status of cases should be adequately mentioned here as per the available information.

General pattern of living: Physical environment of the individual should be mentioned here (accommodation, number of rooms, ownership). Also make a comment on ways of handling adversity in home environment.

Premorbid personality: It can be viewed as individualised styles of dealing with the environment that is characteristic to each person prior to the onset of psychiatric disorder. It is important to elicit details regarding the personality of the individual. Assess from patient/relatives/others who know the patient well. Mention source of information and its reliability.

Try to give illustrative anecdotes and detailed statements. Aim at a picture of individual, not a type. The following is merely a collection of hints, not a scheme. Describe as under:

1. Social relations: How were his relation to family (attachment, dependence); to friends, groups, societies, clubs; to work and work-mates (leader or follower, aggressive or submissive, organizer, ambitious, adjustable, independent)?

2. Intellectual activities, hobbies and use of leisure time: Comment on books, plays, pictures preferred; memory, observation, judgement, critical faculty.

3. Predominant Mood: What used to be persistent mood like, was it cheerful or despondent; worrying or placid; strung up or relaxed; optimistic or pessimistic; self-depreciative or satisfied? Was mood changeable- could he express feelings of love, anger, frustration or sadness, did he loses control over feelings, had he been violent? Was mood stable or unstable (with or without any reason)

4. Character:

a. Attitude to Self: How does patient describe self? What were his strengths and abilities, shortcomings, ability to plan ahead, resilience in face of adversity, hopes and ambitions? Was the level of aspiration high or low? Was he self critical and perfectionist or self approving and complacent in relation to own behaviour and achievements? Was he steadfast in face of difficulties or intolerant to frustration? Were his interests sustained or evanescent?

b. Attitude to Work & Responsibility: Did he welcome responsibility or was worried by it; made decisions easily or with difficulty? Was he methodological or haphazard in his approach? Was he flexible or rigid? Was he cautious, foresightful and given to checking or impulsive & slipshod? Was he determined towards goal or used to get bored or discouraged easily?

c. Interpersonal relationships: Was he self confident or shy and timid? Was he insensitive or sensitive to criticism? Was he trusting or suspicious and jealous? Was he selfish and egotistical or unselfish and altruistic? Was he emotionally controlled or irritable and quick-tempered? Was he tactful or outspoken? Did he use to enjoys or avoids self-display? Was he quiet and restrained or expressive and demonstrative in speech and gesture? Was he tolerant or intolerant to others? Was he adaptable or unadaptable? Did he use to prefer company or solitude? Was he shy or used to make friends easily, were relationships close and lasting? How he used to handle others' mistakes, did he always want to be center of attention? How was the relation with work-mates or superiors, any affiliations to any society?

d. Standards in moral, religious and health matters: What were his religious and moral attitudes? Was he given to much or little concern about own health?

e. Energy, initiative: Was he energetic or sluggish? Was output sustained or fitful? Did he used to get easily fatigued? Were there regular or irregular fluctuations in energy or work output?

5. Fantasy life: What was the frequency and content of day dreaming?

6. Habits: Use of alcohol, drugs, tobacco; comment on food and sleep pattern

Physical Examination

Whenever a patient comes to a physician, it is mandatory for the physician to conduct a thorough physical examination. For a psychiatrist, he is a doctor first and then a psychiatrist. For the respect of the medical knowledge, one should be and remain thorough with the skills of physical examination. For doing physical examination of a female patient, a male doctor need to have a female attendant at the time of examination. Here is a brief summary of what all things should be looked into.

A detailed method of physical examination should be followed from *Hutchison's Clinical methods*. For neurological examination, one should follow *Bickerstaff's Neurological examination*. These two books remain standard books for physical and neurological examination.

General Survey: One need to start by taking height, weight and body temperature of the patient. Then comment on build, posture, skin colour, eruptions, petechiae, vitiligo, spider naevi, nutrition, oedema, hair, nails, clubbing, lymph nodes, swelling, deformities, thyroid gland, injuries, scars on whole of the body after doing a thorough inspection.

Cardiovascular System: Start with taking the pulse of the patient. Look for at least radial and femoral pulse. Comment on rate, rhythm, volume, character, arterial wall, radio-radial and radio-femoral delay.

Take the blood pressure in right upper arm in both supine and erect position.

Look for neck veins, any engorgement and jugular venous pressure.

Conduct a thorough inspection, palpation and auscultation of precordium (Comment on position and force of cardiac impulse, apex beat, heart sounds and murmurs, if any).

Respiratory system: Bare the chest of the patient and go for inspection (rate and character of respiration, chest wall movements); palpation (position of trachea, swelling, tenderness, fremitus); percussion (character of note, symmetry); auscultation (breath sounds, added sounds, if any).

Abdomen: Again on a bare abdomen, comment on inspection (shape, distention, movements, veins, umbilicus, visible peristalsis); palpation (tenderness, rigidity, organomegaly, hernial orifices, genitalia); percussion (character of note, shifting dullness, fluid thrill) and auscultation (peristaltic sounds, arterial bruits, succession splash).

Central Nervous System: Start with higher mental functions; look for consciousness and comprehension, attention and concentration and orientation.

Examine cranial nerves one by one. Here one should be ready with all the apparatus required for cranial nerve examination such as for smell (soap, toothpaste, coffee, asafetida, ginger etc.), for vision (Ishihara Charts, Snellen's chart, Jaguar chart/ newspaper), for fundus examination (ophthalmoscope), for eye reflexes (cotton wisp, a torch), for deep tendon reflexes (knee hammer), for temperature testing (test tubes with hot and cold water), for taste

(salt, sugar, lemon and quinine solution, drinking water, cotton swabs, placards with names of different tastes), for hearing (tuning forks of different frequencies), for touch (a blunt pin)

Proceed for examination of motor system. Look for strength, bulk, tone, co-ordination of all major muscles and any involuntary movements.

In sensory system, look for pain, touch, temperature, position, vibration in all possible dermatomes. In case of deficiency, try to find out level of deficiency.

Look for both deep tendon and superficial reflexes. Check for bilateral biceps, triceps, supinator, knee, ankle reflexes and plantar, abdominal and cremasteric reflex

Look for cortical sensations including astereognosis, graphesthesia, two-point discrimination and sensory inattention.

Look for signs of autonomic system instability. This includes a thorough examination of skin (and its appendages) and mucous membrane, pulsations in extremities, gland functions (sweat, salivary, lacrimatory), endocrine system, postural fall in blood pressure, other cardiovascular reflexes, valsalva maneuver, deep breath test, genito-urinary functions and skin reaction to various stimuli.

Cerebellar Signs should be looked into and include rebound phenomenon, finger-nose test, heel-shin test, rapid alternating movements (dysdiadochokinesis).

Signs of meningeal irritation including neck rigidity, Kernig's sign and Brudzinski's Neck sign should be checked.

Make a comment on handedness, skull, spine, gait & stance.

The central nervous system will become all the more important in a neurological case work-up. This will be of great help in clinically localizing the lesion in brain or spinal cord.

Results of Investigation, if any available: If the patient has got any investigation done prior to current consultation, which might include blood investigations such as blood counts, endocrinological profile (Blood sugar level, thyroid function test, etc.), function tests (liver, kidney etc.), blood electrolyte levels (sodium, potassium, chloride, calcium, etc.), lipid profile, antibody levels; X ray of any body part; CT scan; MRI; EEG; EMG or any other relevant investigation, that should be noted.

Physical findings or diagnosis: Based on detailed physical examination and available investigations, physical findings or diagnosis or differentials should be reached at.

Mental Status Examination

Mental Status Examination: It is the part of clinical assessment that describes the sum total of the examiner's observation and impressions of a psychiatric patient at the time of the interview (SOP).

It is a process of clinical observation of the patient for evaluation of psychological signs and symptoms. It is analogous to physical examination in medicine and follows a definite procedure. In psychiatry we are largely dependent upon the patient's subjective account of symptoms in order to reach a diagnosis, with few opportunities to do objective diagnostic tests. This can be difficult task for the patient, struggling to put complex feelings and experience into words, and for the interviewer, looking for diagnostic signs among all the information given. The mental status examination helps to overcome these difficulties by providing a structure for a detailed, systematic description of the patient's symptoms and behaviour. Herein, one starts with basic functions (level of consciousness) to more complex ones.

Four assessment techniques are used to take a mental status examination of the patient

1. **Observation:** The interviewer observes appearance, level of consciousness, psychomotor activity, body language and affect.
2. **Conversation:** The interviewer can draw conclusions from conversation about speech, concentration, memory, intelligence, affect, thought process, judgement and insight.
3. **Exploration:** Exploration requires patient's willingness to disclose information about mood, thought content, obsessions, compulsions, suicidal ideations, perceptual experiences etc.
4. **Testing:** Testing of patient's mental functions, whether intact or impaired requires higher degree of co-operation.

Getting an expertise at mental status examination is the most difficult and challenging part of psychiatric case work-up. As already mentioned, it is a structured set up which should not be rigidly followed. For beginners, a performa of examination given in SCAN i.e. present state examination (PSE-9) is a helpful guide to proceed for mental status examination and thus it should be loosely followed.

MSE can be viewed on following parameters:

General Appearance: It is the evaluation of the patient's manner of presentation at the time of interview. The description should be as vivid as possible. By listening to the description, one should be able to pin point that individual from a crowd. A rich deal of information can be elicited from examination of the general appearance and behaviour. While examining, it is important to remember patient's sociocultural background and personality. It is significant in assessing the duration of illness and in some cases, the severity of the disorder.

General physical appearance: External attributes need to be examined. Body build, hygiene and grooming should also be commented upon. Following hints should be considered.

Kempt: Trimmed (beard) & combed (hair) and Tidy: Orderly and neat in appearance. Usually found in normal individuals, Obsessive Compulsive personality disorder.

Overtly made up: Exaggerated concern or preoccupation with appearance and dress. May be found in Mania/Hypomania; Histrionic and Narcissistic personality disorder; some cases of Schizophrenia.

Unkempt and untidy: This refers to neglect of personal appearance with regard to dress and hygiene. This may be found in Organic psychosis, Dementia, Substance abuse disorders, Severe Depression, Schizophrenia.

Sickly: Refers to a patient who looks ill or has complete neglect of his health. This may be found in Substance abuse disorders, severe and long-standing Dementia.

Perplexed: This is used to describe a confusional state in which a patient has inability to decide on a task or a solution.

Estimate of age: It should be commented whether the patient appears appropriate to his stated or real age. This also gives insight into the overall manner of lifestyle.

Appropriate to age: Seen in normal individuals

Younger than stated age: May be seen in Histrionic personality disorder

Older than stated age: May be seen in Depression or Chronic Schizophrenia

Body built: Make a comment on body built of a person. Ernst Kretschmer proposed a system of body typing:

- Pyknics (stocky, rounded shapes)
- Leptosomes (long, linear physiques)
- Athletics (broad-shouldered, muscular types)

William Herbert Sheldon devised another system of body type classification most widely used today. Sheldon's system, known as somatotyping, is based on three components (endomorph, mesomorph, ectomorph) of body shape. Any given individual is said to be a mixture of these three in various proportions.

The extreme endomorph is as spherical as humanly possible. He has a round head, a large, fat abdomen predominating over his chest and weak, floppy, penguin like arms and legs with heavy upper arms and thighs but slender wrists and ankles.

The extreme mesomorph is represented by the classical "Hercules". He has a massive, cubical head, broad shoulders and chest, and heavily muscular arms and legs.

The extreme ectomorph is the linear man. He has a thin face with a receding chin and high forehead; a thin, narrow chest and abdomen and spindly arms and legs.

In Sheldon's system, the amount of each component that a person has, is rated on a scale ranging from 1 to 7. Most people have mid-range somatotypes, such as 3-4-4 or 4-3-3.

Besides rating body types, Sheldon's system assumes a close relationship between body build and behavior and temperament. For example, a person high in ectomorphy, is said to be usually socially inhibited, secretive, and prone to seek solitude when troubled. A person with a high endomorph rating supposedly loves physical comfort, needs approval and affection, and seeks people when troubled.

Touch with surroundings: In this we evaluate the patient's perception of self in respect to his surroundings.

Present: Patient is oriented; has normal perception of self in respect to surroundings.

Partial: Some aspect of his surroundings or their significance to the patient is lost.

Absent: Patient is unable to orient himself and behaves in a manner inappropriate to the situation. For example, he may ask the doctor to carry out his orders. Seen in Dementia, Substance abuse disorders (in intoxicated state)

Eye contact with the examiner: This is useful in assessing the establishment of rapport, truthfulness, insight and concentration of the patient.

Partial: Fleeting eye contact with the examiner, which is not adequate for the continuation of a successful interview. Seen in Depression- lowered eyes; Anxiety- Shifting gaze.

Absent: Here, there is complete loss of eye contact with the examiner. This is found in Paranoid Schizophrenia, Acute delusional disorder.

Dress: Dress is the key to person's appearance and gives the interviewer an impression of the patient's cultural background and his economic position.

Appropriate: Dress is properly worn, clean and in conformity with the situation. This is found in normal people.

Shabby: Neglect or decreased care for dress occurs concurrently with neglect of other aspects of appearance. Found in Dementia, Substance abuse disorders.

Inappropriate: Dressing is said to be inappropriate when it is not done in conformity with the situation, for example rain hood worn on a dry day. It is seen in Mania, Histrionic personality disorder.

Facial expression: It provides information and a rough estimate about certain diagnosis.

In depression, the corners of mouth are turned down and there are vertical furrows on the brow

Anxious patients have horizontal creases on the forehead, widened palpebral fissures and dilated pupils.

Facial expression may reflect elation, irritability and anger.

There may be fixed wooden expression due to drugs with parkinsonian side effects.

The facial expressions may also suggest physical disorders e.g. thyrotoxicosis.

Posture: Posture and movements also reflect mood.

Depressed patients characteristically sit with hunched shoulders, with head and gaze inclined downwards.

An anxious patient may sit on the edge of the chair with hands gripping its sides.

Anxious patients and those with agitated depression may be tremulous and restless, touching their jewelry or picking at their fingernails.

Manic patients are restless.

Catatronics can maintain odd postures.

Attitude towards examiner: Attitude is a mental and neural state of readiness organized through experience; exerting a directive and dynamic influence upon the individual's response to all objects and situations with which it is related.

Co-operative: Helps the examiner conduct the interview smoothly.

Attentive: Patient pays attention to the interviewer. It is a normal response.

Defensive: It is the kind of behaviour that turns the examiner's attention away from one's deficiencies, or behavior that might cause him guilt or embarrassment. Seen in Paranoid Schizophrenia, Delusional disorder.

Frank: This behaviour helps to conduct an open conversation that includes all the deficiencies without guilt or embarrassment.

Hostile: It is characterized by behaviour of covert aggression, so that it tends to create strong negative feelings or anger in the examiner or the patient himself shows anger towards the examiner. Seen in Paranoid Schizophrenia, Antisocial personality disorder

Seductive: The patient (mostly a female) tries attention-seeking behaviour that uses verbal or non-verbal seductive clues towards the examiner. Found in Histrionic personality disorder.

Guarded: Patient will restrict his information and weigh the information as per his/her ideas of importance. Seen in Paranoid Schizophrenia, Delusional disorder.

Evasive: Patient attempts to escape from an argument and shifts topics. Seen in Organic Psychoses, Substance abuse disorders.

Rapport: It is a bidirectional empathetic relationship, which the examiner shares with the patient. Ekkehard Othmer and Sieglinde Othmer defined the development of rapport as encompassing six strategies:

1. Putting patients and interviewers at ease.
2. Finding patients' pain and expressing compassion.
3. Evaluating patients' insight and becoming an ally.
4. Showing expertise.
5. Establishing authority as physicians and therapists.
6. Balancing the roles of empathic listener, expert, and authority.

Easily established: Seen in Normal persons, Mania

Established with difficulty: Seen in Schizophrenia, Dementia and Depression

Not possible: Seen in Paranoid Schizophrenia, Dementia and Depression

Motor Behaviour: It denotes both the quantitative and qualitative aspects of a person's motor behaviour and the level of his activity. For example: psychomotor agitation or retardation.

Retardation: Motor retardation implies slowness of the initiation, execution and completion of physical activity. It is frequently associated with the retardation of thought (SIMS). Psychomotor retardation is experienced subjectively as a feeling that all actions have become much more difficult to initiate and carry out (Fish). Seen in Depression.

Hyperactive: Hyperactivity is a state in which there is increased motor activity, possibly with aggressiveness, over-talkativeness or uncoordinated physical activity. The term is descriptive of behaviour, rather than of subjective psychological state.

Preoccupied: Preoccupied is the state of being excessively focused on one task with neglect or avoidance of any other thought. Found in Depression, Paranoid Schizophrenia.

Mannerisms: Unusual repeated performances of a goal directed motor action or the maintenance of an unusual modification of an adaptive posture are known as mannerisms. Example: unusual hand movements while shaking hands.

Restless: Restlessness is purposeless movement of extremities, limbs; fiddling, stretching, shifting, cannot sit still, standing up and sitting again.

Stereotypy: A stereotyped movement is repetitive, non-goal directed action, which is carried out in a uniform way (Fish); Repetitive, seemingly driven and non-functional motor behaviour. (DSM-IV-TR)

Grimace: It is a specific facial expression, which is non-goal directed & spontaneous. For example: *Schnauzerampf* (snout spasm) found in Schizophrenia.

Awkward: Clumsy moments having a little skill in dealing with the surrounding items/ events. For examples: apraxia, mental retardation.

Destructive: Breaking/ throwing or disrupting items nearby. For example: catatonic excitement, manic excitement.

Self-injurious: Behaviour characterized by doing self-harm or inflicting injuries on oneself. It is usually viewed as having a psychological meaning i.e. attention seeking. For example: Borderline personality disorder.

Silly Smiling: Apparently spontaneous and childish laughter on little provocation. For example: Mania, Hebephrenia.

Tics: Tics are sudden, involuntary twitchings of small group of muscles and are usually reminiscent of extensive movements or defensive reflexes. Commonly the face is affected so that the tic takes the form of blinking; distortions of the forehead, nose or mouth; but clearing of the throat and twitching of the shoulders may also be tics (Fish). Tics are usually rapid, repetitive, coordinated and stereotyped movements, most of which can be mimicked and are usually reproduced faithfully by the individual. For example: Gilles de la Tourette syndrome (SIMS).

Aggressive- Hurting others either as end in itself (hostile aggression) or to achieve some other goal (instrument aggression). It may also be symbolic of some underlying conflicts or pathology. For example: Delirium, Frontal lobe syndrome, Mania, Paranoid Schizophrenia.

Odd posturing- Voluntary assumption of inappropriate or bizarre posture. The position may be maintained voluntarily or imposed by the examiner. The maintenance should be at least for one minute. For example: Catatonia, Akinesia.

Rigidity- Assumption of a rigid posture against all attempts to move. For example: Catatonia, Mania, Depression.

Touching the examiner- The patient may touch the examiner in a way different from formal greetings. E.g.: touching feet- depression; kicking/caressing- Histrionic patient.

Gestures- A mode of non-verbal communication in which information is conveyed by movements of hands, arms or parts of the body. For example: Mania, Hypomania

Hallucinatory behaviour- Behaviour suggestive of active hallucinations. It may be in the form of suddenly making postures of listening, looking intently at some point or talking in response to imaginary voices.

Perseveration- Perseveration is a induced movement because it is senseless repetition of a goal directed action which has already served its purpose (Fish). Thus, when a patient is asked to put his tongue out, he puts it out then puts it in when told to, but continues to put it out and in thereafter.

Waxy Flexibility- Condition in which person maintains the body position into which they are placed (CTP). In waxy flexibility when the limbs of the patient are put into any posture by the interviewer, they will be retained in that posture for a sustained period (a minute or more) (SIMS).

Ambivalence- Co-existence of two opposing impulses towards the same thing in the same person at the same time. Seen in Schizophrenia, borderline state, OCD (CTP).

Automatic obedience- The phenomenon of undue compliance with instructions (without awareness of what one is doing, irrespective of consequences). Seen in catatonia.

Echolalia- Psychopathological repeating of words or phrases of one person by another; tends to be repetitive and persistent. Seen in certain kinds of Schizophrenia, particularly catatonic (CTP).

Echopraxia- Repetition by imitation of the movements of another. The action is not a willed or voluntary one and has a semiautomatic and uncontrollable quality (DSM-IV)

Negativism- Verbal or non-verbal opposition or resistance outside suggestions and advice; commonly seen in Catatonic Schizophrenia in which the patient resists any effort to be moved or does the opposite of what is asked (CTP).

Dystonia- Uncontrolled muscle spasms leading to involuntary movements of the eyelids, face, jaw, hands and the other parts. Slow and sustained contractions of the trunk and limbs.

Dyskinesia- Difficulty in performing voluntary movements.

Chorea- Movement disorder characterised by random and involuntarily quick, jerky, purposeless movements. (SOP).

Athetosis- Spontaneous movements that are slow, twisting and writhing; which bring about strange postures of the body, especially of the hands.

Speech: This part describes the physical characteristics of speech. Speech can be described in terms of its intensity, pitch, quality, prosody, reaction time, speed, ease, coherence, relevance, goal directedness, rate of production, manner of relation and deviations .

Intensity: How loudly the words are spoken

Audible- The examiner can listen to the voice of the patient. This occurs in normal conversation.

Excessively loud- Intensity of speech is louder than required. For example: Excited patient, Mania, Hypomania.

Abnormally soft- Intensity of speech is softer than required. For example: Vocal cord palsy, Depression, Paranoid Schizophrenia.

Pitch: The relative highness or lowness of a tone as perceived by the ear.

Monotonous- Speech without change in pitch or lack of modulation. For example- Parietal lobe damage, Chronic Schizophrenia, Depression.

Quality: It denotes how the sound seems to listener's ears.

Soft- Spoken politely, usually in low volume and with slow speed. Example: Obsessive compulsive personality disorder, Anxious avoidant personality disorder.

Hoarse- Spoken forcefully, usually in a husky tone. Example: Mania after they have shouted for long periods of time, certain normal individuals.

Prosody: Use of pitch, loudness, tempo and rhythm in speech to convey information about the structure and meaning of an utterance.

Reaction Time: The time taken by the patient from listening the question to answering.

Increased reaction time- Time taken is increased or response to stimulus is delayed. Example: Hypothyroidism, Depression, Schizophrenia.

Decreased reaction time- Time taken is decreased. Sometime patient doesn't listen to the examiner properly and is in a hurry to answer. For example: Mania.

Speed: The rate of production and output of speech.

Very slow- The output of speech is slow e.g. Depression, Dementia, Hypothyroidism

Rapid- The rate of speech output and production is more than normal. It is associated with prolixity and flight of ideas. Example: Mania, Hypomania, Hyperthyroidism.

Pressure of speech- Speech that is increased in amount, accelerated and difficult or impossible to interrupt. Usually it is also loud and emphatic. Frequently the person talks without any social stimulation and may continue to talk even when no one is listening (DSM-IV-TR). Increase in the amount of spontaneous speech; rapid, loud, accelerated speech as occurs in Mania, Schizophrenia and cognitive disorders (CTP).

Ease of speech: Ease with which words are uttered

Hesitant- Speech characterized by frequent pauses & breaks. There is a defect in verbal fluency. May be tested by asking the patient to enumerate the name of animals during a 60 second period. Examples: Head injury, anxiety states.

Mutism- Complete loss of speech; comprehension may be fully preserved; the patient may be able to communicate by writing his/her thoughts. Examples: Stupor, dementia.

Slurring- A form of speech in which the words are pronounced with prolongation of syllables. Example: Cerebellar damage.

Stuttering/Stammering- The disorder of rhythm of speech in which the normal flow is interrupted by pauses, prolongations or repetition of sounds, or fragments of words i.e. syllables. Example: Severe anxiety, Schizophrenia.

Whispering: Production of sound by using breath but not vocal cords. Example: Pseudobulbar palsy.

Muttering- Speaking in a low voice, not meant to be heard (using lip movement).
Example: Schizophrenia, Dementia.

Speaking only when questioned- Speaks only when very much required. Patient uses words very economically.

Relevance: Whether the answer is relevant to the question asked?

Relevant- There is relevance between the question and the answer given by the patient but the answer may not be correct. E.g. Examiner's question- where is your home? Patient's answer- America (His answer is matching with the question but if we ask the informant we find that it is not true).

Irrelevant- Grammatically correct but meaning of speech sample is unrelated to the question asked/topic of the discussion. Example, seen in Schizophrenia, Dementia, Delirium, drug intoxication.

Coherence: Understandability of the speech

Coherent- Speech is understandable by the examiner. For example: Normal individual

Incoherent speech- Not understandable by the examiner due to lack of logical connection between words, phrases, sentences; idiosyncratic words or distorted grammar. For example: Schizophrenia, Catatonic excitement, organic psychoses.

Goal Direction: A speech can be said to be goal directed when it is reaching the goal and answering what examiner has asked.

Productivity: The overall content of speech, whether adequate/less/more.

Increased productivity/overabundant- Copious, pressured coherent speech which appears like excessive uncontrollable talking. Example: Hypomania, Mania.

Decreased Productivity/scant speech- Speech production is less than adequate in amount. Either the production of speech is less or there is difficulty in final verbalization. Example: Negative Schizophrenia, Depression, Catatonia.

Manner of relating: The way in which the patient speaks/interacts with the examiner.

Excessively formal- Patient using more than required number of formal gestures (formalities) during the interview. Example: Obsessive compulsive personality disorder, Antisocial personality disorder, Hypomania.

Tensed up- Showing features of anxiety i.e. wringing of hands, sweating, fidgeting etc. during conversation. Example- anxious trait/state, Avoidant personality disorder.

Inappropriately familiar- Patient acts as if he is very much used to the present set of examination/situation. Example: Histrionic personality disorder, Antisocial personality disorder, Hypomania.

Disinterested- Not showing any interest in the proceedings of the interview. The patient may look here and there or tries to get up on slight indication. Example: Schizophrenia, Schizoid personality disorder, Severe Depressive disorder.

Deviation: Following deviations should be noted here:

Rhyming and punning- Rhyme is sameness of the sounds of the endings of two or more words. i.e. *I am going... rowing...* especially at the end of lines or verses; Punning is humorous use of words with similar meanings of a word with double meaning i.e. both me and my bike need fluid. Example: Mania, Hypomania.

Talking past the point- Deliberate answering of an associated topic related to the answer. The patient answers to the questions quite readily but mostly these are incorrect answers. Such as: *what is the colour of grass?* Replies- *white*. Examples: Ganser's Syndrome, Schizophrenia.

Clang association- Association or speech directed by the sound of a word rather than its meaning; words have no logical connections, punning and rhyming may dominate the verbal behaviour. Seen most frequently in Schizophrenia or Mania. (CTP). A pattern of speech in which 'sounds' rather than 'meaningful relationship' appear to govern word choice so that intelligibility of the speech is impaired and redundant words are introduced (TLC).

Stereotypy- Continuous mechanical repetition of speech or physical activities. Observed in Catatonic Schizophrenia (CTP).

Perseveration- Pathological repetition of the same response to different stimuli as in a repetition of a same verbal response to different questions. Persistent repetition of a specific word or concept in the process of speaking (CTP). Persistent repetition of words, ideas or subject so that once a patient begins a particular subject or uses a particular word, he continually returns to it in the process of speaking (TLC). Seen in cognitive disorders, Schizophrenia and other mental disorders.

Cognitive Functions

Orientation- It is defined as a state of awareness of oneself and one's surrounding in terms of time, place and person. Orientation to time, place and person are actually measure of recent memory, as they test the patient's ability to learn three continually changing facts. (Strub & Black). Loss of orientation occurs in Delirium, Schizophrenia, Severe Depression.

Attention- The ability to focus in a sustained manner on a particular stimulus or activity. A disturbance in attention may be manifested by easy distractibility or difficulty in finishing tasks or concentrating on work (DSM-IV-TR). The aspect of consciousness that relates to the amount of effort exerted in focussing on certain aspects of an experience, activity or task. Usually impaired in anxiety or depressive disorders. Attention is the ability to attend to a specific stimulus without being distracted by extraneous, internal or environmental stimulus (Strub & Black).

Concentration- Concentration is the ability to maintain attention to specific stimuli over an extended period. Concentration is the ability to maintain focus on the task at hand (SOP).

Memory: Mental process that allows the individual to store information for last recall (Strub & Black). Process whereby what is experienced or learned is established as a record in the CNS (Registration) where it persists with available degree of permanence (Retention) and can be recollected or retrieved from storage at will (Recall) (CTP). Memory is the encoding, storage and retrieval of what was learned earlier (Morgan & King).

- Immediate memory- Reproduction, recognition and recall of perceived material within seconds after presentation. (CTP). Immediate memory/ recall is used to recall a memory trace after an interval of few seconds, as in the repetition of series of digits (Strub & Black).
- Recent Memory: Recall of events over the past few days (CTP). Recent memory is the patient's capacity to remember current day-to-day events. More strictly defined, recent memory is the ability to learn new material and to retrieve that material after an interval of minutes, hours or days (Strub & Black).
- Remote memory: Traditionally refers to the recall of facts or events that occurred years back, such as the names of teacher and old school friends, birthdates and historical dates (Strub & Black). Recall of events in the distant past (CTP).

Abstract Ability: It is the ability to deal with concepts. Patients can have disturbances in the manner of conceptualizing or handling ideas. The appropriateness of answers and the manner in which they are given also be noted (SOP). Abstract ability refers to ability to shift voluntarily from one aspect of a situation to another (at the same time) keeping in mind simultaneously the various aspects of a situation. In Piaget's theory of cognitive development, the capacity for abstract thinking is acquired around 12 years i.e. stage of formal operations. Abstract thinking is synonymous for conceptual thinking.

Disturbance in abstract thinking are seen in Schizophrenia, Dementia, past head injury.

- Functional Thinking- This type of thinking is characterised by the some degrees of abstraction relating only towards the functional aspects of an object or event. For e.g. Q: What is a knife; A: An instrument that cuts fruits. Found in Schizophrenia, MR.
- Concrete thinking- This refers to the thinking characterized by actual things or events and immediate experience rather than by abstraction. The person cannot get the abstract meaning of a situation or problem and deals with the event in a superficial way. For e.g. Q: What is the similarity between a table and a chair? A: Both are of wood. Found in Schizophrenia, young children (Piaget's concrete operational stage 7-11 years)
- Overabstraction- This refers to a pattern of thinking characterised by excessive manifestation of a concept beyond the point of relevance. The problem at hand loses its value and remain unsolved. For e.g. The similarity between a table and chair can be answered as: Number of electrons of an atom of a molecule in wood of chair and table are equal. Found in: Schizophrenia, people with philosophical preoccupations.

Intelligence: It is the ability to think logically, act rationally and deal effectively with environment. It has following components:

- Comprehension- Understanding and reacting intelligently in a problematic situation. Conditions having poor comprehension are inattentive states, Dementia.
- General information- This refers to a person's knowledge of general information or general knowledge giving due concession to one's socio-economic status, education and current mental status. Conditions with poor general information are mental retardation, Dementia.
- Vocabulary- This refers to the amount of words the patient uses in his day-to-day dealings. Poor vocabulary is seen in mental retardation, Autism and Dementia.
- Calculation- Calculations are complex neuropsychological functions that involve somewhat distinct components of number sense and manipulation. Components of calculation include rote table, basic arithmetic of carrying/borrowing, recognition of the signs (+, -), correct spatial alignment for written calculation (Strub & Black).

Mood and Affect

Mood- Mood is perceived as a persistent and sustained emotion that colours the patient's perception of the world (SOP). Mood is the prevailing and conscious emotional feeling expressed by the patient (Strub & Black). Mood is an emotional state which usually lasts for some time and which colours the total experience of the subject. It is also referred as a "mood state" (Fish).

Affect- Affect can be defined as the patient's present emotional responsiveness, inferred from the patient's facial expression, including the amount and range of expressive behaviour (SOP). Affect is a wave of emotion in which there is a sudden exacerbation of emotion usually as a response to some event (Fish). It is immediate experience of emotion attached to idea or some event. It had both subjective and objective manifestations.

In the absence of a psychopathological process, affect fluctuates with time and context and ranges from sadness to anger to elation, depending on the emotional state.

Affect can be expressed through autonomic responses, body movements and alterations in speech to concrete or abstract stimuli. Speech changes that reflect affect include tone of voice, vocalization, and word selection. Visible autonomic changes that may reflect changes in affect include sweating, trembling, blushing and becoming flush. Changes in posture, alterations in facial expression, reactive responses and grooming movements are body changes seen in expression of affect. Reactive movements include movements of the body and face made in response to a novel stimulus, such as in a startle response, when an individual jumps or turns and looks at the stimulus. Changes in facial movements of the mouth, nose, and eyes are found with different affective states. Manipulation of one's appearance is common in states of discomfort; individuals may fix their hair, clean their nails, scratch or straighten their clothes.

Evaluation of affect consists of monitoring gestures, body movements, and facial expressions. Because adults are frequently capable of controlling facial expression in attempts to

intentionally or unintentionally suppress their affect, other behavioural gestures may give clues to the underlying affect.

Affect can be assessed under the following domains-

Quality of Affect- the label or valence of the affect. Assessed on two criteria.

- Subjective evaluation- 'how do u feel'.
- Objective evaluation- based on the observation of interviewer.

The quality of affect can be:

Dysphoric- An affect characterised by sustained emotional states such as sadness, anxiety or irritability.

Anxious- Feeling of apprehension caused by anticipation of danger which may be internal or external. Anxiety is an unpleasant affective state with the expectation, but not the certainty of something untoward happening.

Irritable- A state of poor control over aggressive impulses directed towards others; most frequently to those nearest and dearest. May manifest in outbursts in which a person is easily annoyed and provoked to anger.

Depressed- Emotional mood tending towards sorrow. Relative passivity and diminished muscular tone with weeping.

Elevated- An exaggerated feeling of well-being out of keeping with the life situation.

Euphoric- Increased sense of well-being with cheerful thoughts and lack of response to depressing influence so that everything is seen in the best possible light.

Elated- Feeling of well-being and euphoria leading to faulty judgement, general over activity and disinhibited behaviour.

Exalted- Feeling of intense elation and grandeur; seen in Mania.

Ecstatic- Intense sense of rapture or blissfulness; seen in Delirium; stuporous Mania.

Euthymic- Mood in normal range implying absence of depressed or elated mood.

Intensity of affect- It is the strength of the emotional expression. It normally varies according to the situation. Those with a limited intensity of emotional experience may have-

Shallow affect- When there is lack of depth in emotion.

Blunted affect- Greatly diminished emotional response or expressionless face and a uniform voice, irrespective of the topic of conversation, patient is indifferent to distressing topics.

Flat affect- When no affect is displayed, it is reported to be flat or absent in emotional response. Or it may be understood as an absence of appropriate outwardly thoughts.

Mobility of affect- It is the ease and speed with which one moves from one type to another type of emotion. Change in type and intensity of emotional expression normally occurs gradually.

Constricted affect- Reduced mobility is also referred as constricted affect.

Fixed affect- When affect is extremely constricted to one emotion it is called fixed or immobile.

Labile affect- Pathologically increased mobility of affect is referred to as liable, marked by a rapid shift from one type to another emotion without persistence of any affect.

Range of affect- The range of the affect is characterized by the variety of emotional expression noted in a session. Ordinarily, there are different feelings experienced at the different times. The criterion for assessing range are

Full Range- appropriately expressed many emotions depending on the context have a full or broad range of affect.

Restricted range- person shows only a fixed emotion, or limited range.

Reactivity- The reactivity is the extent to which affect changes in response to environmental stimuli. When patient does not respond to examiner's provocation in the form of joking, for instances, the affect is said to be non-reactive.

Communicability- The ability of the expression of affect to communicate to another one's emotional response to events, interaction, behaviour, and situation. The capacity to connect with the interviewer. Usually present in Mania (infectious jocularity), absent in Schizophrenia

Appropriateness- It refers to the congruence or fit between the expressed quality of emotion and the content of speech, thought, expected degree of intensity and the overall situation.

Paramimia- Lack of unity between various modes of expression of emotion.

Parathymia- Expression of an emotion that is exactly opposite of what is expected under the circumstances, for example, laughing at a tragic news.

Diurnal variation of affect- The change in affect occurring with passage of the day.

Worse in morning- Seen in Endogenous Depression, Headache.

Worse in the evening- Seen in Anxiety, Delirium (sun downing).

Worse at night- Seen in Uremia.

Thought: It is an idea produced by thinking or occurring suddenly in mind or the action or process of thinking (Oxford Dictionary). We gain access to the patient's thoughts via their speech, and it is important to listen carefully to factors, which however can be understood.

The division of disorders can be understood, if we compare thought to a flowing river. The source of water, be it hills, rains or waste from community is equivalent to *possession*; the flow of river is *stream*; the water content in it is *content* and the liquid form of water is its *form*. Thus, the abnormality in thoughts can be assessed on following points.

Stream- Flow of thoughts

Volubility- Copious, coherent, pressured speech; uncontrollable excessive talking; observed in Mania. Also called Tachylogia, Verbomania, Logorrhea.

Acceleration- Flow of thinking becomes rapid and increase in amount. Number of ideas/mental images per minute are more than normal. It may result in pressured speech, flight of ideas or prolixity.

Pressured Speech- (Detailed under heading of Speech- Speed)- Increase in the amount of spontaneous speech, rapid, loud, accelerated speech. Occurs in Mania, Schizophrenia and cognitive disorders.

Flight of ideas: In flight of ideas thoughts follow each other rapidly; there is no general direction of thinking; and the connections between successive thoughts appear to be due to chance factors which, however, can usually be understood. The patient's speech is easily diverted to external stimuli and by internal superficial associations. The absence of a determining tendency to thinking allows the associations of the train of thought to be determined by chance relationships, verbal associations of all kinds (such as assonance, alliteration and so on), clang associations, proverbs, maxims, clichés. The chance linkage of thoughts in flight of ideas is demonstrated by the fact that one could completely reverse the sequence of the record of a flight of ideas, and the progression of thought would be understood just as well (Fish).

Rapid succession of fragmentary thoughts or speech in which content changes abruptly and speech may be incoherent (CTP). A nearly continuous flow of accelerated speech with abrupt changes from topic to topic that are usually based on understandable associations, distracting stimuli or play of words. When severe, speech may be disorganized and incoherent (DSM-IV-TR).

Flight of ideas is seen in Mania (typically), Schizophrenia, organic states, mixed affective states (flight of ideas without pressure of speech).

Prolixity: 'Ordered flight of ideas' or marginal variety of flight of ideas has been called as prolixity. In prolixity, despite many irrelevances, the patient is able to return to the task in hand; clang and verbal associations are not so marked; the speed of emergence of thoughts is not as fast as in flight of ideas; although patients cannot keep accessory thoughts out of the main stream, they only lose the thread for a few moments and finally reach their goal; unlike the tedious elaboration of details in circumstantiality, there is a lively embellishment of the thinking. Seen in Hypomania (Fish).

Retardation (bradyphrenia): The train of thought is slowed down and the number of ideas and mental images which present themselves is decreased. This is experienced by the patient as difficulty in making decisions, loss of concentration and loss of clarity of thinking (Fish). In retardation (as occur in depression), thinking, although goal directed, proceed so slowly with such morbid preoccupation with gloomy thoughts that the person may fail to achieve these goals. The patient is likely to show very little initiative to begin planning or spontaneous activity (SIMS). Seen in Schizophrenia, depression.

Poverty of speech: Restriction in amount of speech used; replies might be monosymbolic. Also called laconic speech.

Circumstantiality- Thinking proceeds slowly with many unnecessary and trivial details, but finally the point is reached. The goal of thinking is never completely lost and thinking proceeds towards it by an intricate and convoluted path (Fish). Disturbance in the associate thought and speech processes in which a patient digresses into unnecessary details and inappropriate thoughts before communicating the central idea (CTP). A pattern of speech which is very indirect and delayed in reaching its goal idea. In the process of explaining something, the speaker brings in many tedious details and sometimes make parenthetical remarks (TLC). Seen in epileptic personality change, dullards who are trying to be impressive, pedantic obsessional personality, Schizophrenia, certain cases of dementia.

Tangentiality- Refers to replying to a question in an oblique, tangential or even irrelevant manner. The reply may be related to the question in some distant way or the reply may be unrelated and seem totally irrelevant (TLC). Oblique, digressive or even irrelevant manner of speech in which the central idea is not communicated (CTP).

Perseveration- (Also refer to speech- deviation)- It is a disturbance in the flow of thinking in which the patient retains a constellation of ideas long after they have to be appropriate. (SIMS). Mental operations tend to persist beyond the point at which they are relevant and thus prevent progress of thinking (Fish). Perseveration is common in generalised and local organic disorders of the brain, and when present, provides strong support for such a diagnosis.

Thought blocking- Thought blocking occurs when there is a sudden arrest of the train of thought, leaving a 'blank'. An entirely new thought may then begin (Fish). Interruption of a train of speech before a thought or idea has been completed. After a period of silence which may be from a few seconds to minutes, the person indicates that he can not recall what he had been saying or meant to say. Blocking should only be judged to be present either if a person voluntarily describes losing his thought or if upon questioning by the interviewer the person indicates that that was his reason of pausing (TLC). Abrupt interruption in train of thinking before a thought or idea is finished; after a brief pause, person indicates no recall of what was being said or was going to be said; also called as thought deprivation (CTP). "Snapping off" is the experience that a patient with Schizophrenia has, of his chain of thoughts quite

unexpectedly and unintentionally breaking off or ceasing. It may occur in the middle of sorting out a problem or even mid-sentence (SIMS). While they are flowing freely, the respondent experiences a sudden unexpected stopping of thought. When this occurs, it is dramatic and usually happens on several occasions. The experience is passive (SCAN). When thought blocking is clearly present it is a terrifying experience and highly suggestive of Schizophrenia. However, similar thing may occur in persons who are exhausted and anxious and may appear to have thought blocking.

Form- Form of thought means “the arrangement of parts”. Disturbance in the form of thought are disorder in the logical connections between ideas.

Formal thought disorder- Disorder of form of thinking is also called formal thought disorder. This is disorder of conceptual or abstract thinking, which occur in Schizophrenia and coarse brain disease. Formal thought disorder, from the subjective phenomenological standpoint is abnormality in the mechanism of thinking described by the patient in his own words as a process of thinking which is clearly abnormal to the outside observer (SIMS). Disturbance in form of thought rather than content of thought, is thinking characterized by loosened associations, neologisms and illogical constructs; thought process is disordered and the person is described as psychotic. This is characteristic of Schizophrenia (CTP)

Here there are two types of formal disorder:

- Negative type – the patient loses his previous ability to think and cannot produce a concept.
- Positive type – in this, the patient produces false concept by blending together incongruous elements

Loosening of association: Characteristic Schizophrenic thinking or speech disturbance involving a disorder in the logical progression of thoughts, manifested as a failure to communicate verbally adequately; unrelated and unconnected ideas shift from one subject to another (CTP). Loosening of association denotes a loss of normal structure of thinking. To the interviewer, the patient’s discourse seems muddled and illogical and it does not become clearer when the patient is questioned further; there is a lack of general clarity, and the interviewer has the experience that the more he/she tries to clarify the patient’s thinking the less it is understood (OTP).

Three kinds of loosening of association have been described:

- Knight’s move thinking or derailment where there are odd tangential associations between ideas.
- Talking past the point (vorbeireden) where the patient seems to get close to the point of discussion, but skirts around it and never actually reaches it
- Verbigeration (word salad/schizophasia/paraphrasia) where speech is reduced to a senseless repetition of sounds and phrases.

Derailement: A pattern of speech in which a person’s idea slip off from one track to another that is completely unrelated or only obliquely related. In moving from one

sentence or clause to another, the person shifts the topic idiosyncratically from one point of reference to another and things may be said in juxtaposition that lack a meaningful relationship. This disturbance occurs between clauses, in contrast to incoherence, where the disturbance is within clauses. An occasional change of topic without warning or obvious connection does not constitute derailment (DSM-IV-TR). Gradual or sudden deviation in the train of thought without blocking; sometime used synonymously with loosening of association. There is a breakdown in both the logical connection between ideas and the overall sense of goal-directedness. The words make sense, but the sentences do not make sense. (CTP). In derailment, the thought slides on to a subsidiary thought (Fish). In derailment, there is a breakdown in association, so that there appears to be an interpolation of thoughts bearing no understandable connection with the chain of thoughts (SIMS).

Neologism- These are new words which are constructed by the patient or ordinary words which he uses in a special way (Fish). This creation of a new word becomes necessary in Schizophrenia to fill a semantic gap (SIMS). The inventions of new words/ phrases or the use of conventional words in idiosyncratic ways (CTP). A neologism is defined as a completely new word or phrase whose derivation can not be understood (TLC).

Over inclusion- refers to a widening of the boundaries of concepts such that things are grouped together that are not often closely connected.

Possession: Normally one experiences one's thinking as being one's own, although this sense of personal possession is never in the foreground of one's consciousness and is taken for granted. One also has the feeling that one is in control of one's thinking. In some psychiatric illnesses there is a loss of control or sense of possession of thinking.

Obsession: Persistent and recurrent idea, thought or impulse that cannot be eliminated from consciousness by logic or reasoning (CTP). Obsessions are involuntary and ego-dystonic. According to Schneider, an obsession occur when one cannot get rid of a content of consciousness, although when it occurs he realises it is senseless or atleast it is dominating and persisting without a cause (Fish). An obsession (also termed rumination) is defined as a thought that persists and dominates an individual's thinking despite the individual's awareness that the thought is either entirely without purpose or else has persisted and dominated their thinking beyond the point of relevance or usefulness (Fish 3rd edition). According to Lewis, three essential features are- a feeling of subjective compulsion, a resistance to it and presence of insight. The sufferer knows that it is his own thought or act, that it arises from within himself and that it is subject to his own will whether he continues to think or perform it, he can decide not to think it on this particular occasion, but it does and will recur (SIMS).

Thus to summarize, the essential feature of the obsession are

- Own and ego-dystonic
- Intrusive
- Deemed as irrational or senseless

- Appears against the will (involuntary)
- Tries to resist
- May have temporary relief by yielding to compulsions, but the thoughts recur
- Associated with dysfunction

Obsessions can take various forms-

- Obsessional thoughts- They are repeated intrusive words or phrases which are upsetting to the patient.
- Obsessional images- These are repetitive and vivid images that occupy the patient's mind. At times they may be so vivid that they can be mistaken for pseudo-hallucinations.
- Obsessional ruminations- They are repeated worrying themes of a more complex kind.
- Obsessional doubts- they are repeated themes expressing uncertainty about previous actions, e.g. whether or not the person turned off an electrical appliances that might cause a fire. Whatever the nature of the doubt, the person realizes that the degree of uncertainty and consequent distress is unreasonable.
- Obsessional impulses- They are repeated ways to carry out actions, usually actions that are aggressive, dangerous or socially embarrassing. Whatever the urge, the person has no wish to carry it out, resists it strongly.
- Obsessional phobias- Denotes a symptom associated with avoidance as well as anxiety.
- Obsessional fear of illnesses called illness phobias.
- Obsessional slowness- Many obsessional patients perform actions slowly because their compulsive rituals or repeated doubts take time and distract them from the main purpose.

Obsessions occur in obsessional states, Depression, Schizophrenia, organic states.

Rumination: It is a train of thoughts, usually unproductive and prolonged, on a particular topic, repeatedly experienced and is felt to be less intrusive. Ruminations are linked to abnormal emotion, the valence typically reflected in the content of the thoughts. Ruminations can occur in OCD, depression, melancholia.

Depressive ruminations (in comparison to obsessive ruminations)- Depressive individuals ruminate about every day, real-life events whereas obsessive-compulsive individuals will tend to have obsessions about unusual and neutral topics which are mostly unrelated to the individual. Obsessive thoughts tend to center around a current or future event, whereas depressive rumination typically involves a past incident. OCD individuals often describe their thoughts as intrusive, senseless and unwanted, and often report an attempt to resist them. In direct contrast, depressive ruminators maintain that their thoughts are non-intrusive and are rarely resisted. Another distinction between ruminations and obsessions, *prima facie*, is the behavioural outcome: obsessions lead to compulsions.

Compulsion- compulsions are in fact merely obsessional motor acts. They may result from an obsessional impulse which leads directly to the action or they may be mediated by an obsessional mental image or thought (Fish). The word obsession is usually reserved for the thought and compulsion for the act. Compulsions may occur in form of acts, rituals or behaviours (SIMS). Compulsions are repetition and seemingly purposeful behaviours, performed in a stereotyped way in response to an obsession. They are accompanied by a subjective sense that the behaviour must be carried out and by an urge to resist. (OTP)

Thought alienation: Patient has the experience that his thoughts are under the control of an outside agency or that others are participating in his thinking (Fish). The Schizophrenic experiences his thoughts as foreign or alien; not emanating from himself and not within his control. There is a breakdown in the way he thinks of the boundary between himself and the outer world so that he can no longer discriminate between the two (SIMS)

Thought insertion: In thought insertion, the person experiences thoughts that do not have feeling of familiarity, of being his own, but he feels that those have been put in his mind, without his volition, from outside himself (SIMS). In thought insertion, the patient knows that the thoughts are being inserted into his mind and he recognizes those as being foreign and coming from without (Fish). Delusion that thoughts are being implanted in one's mind by other people or forces (CTP). It is the delusion that certain thoughts are not the patient's own but implanted by an outside agency. Often there is an explanatory delusion, for e.g. the persecutors have used radio waves to insert the thoughts. (OTP). The essence of the symptom is that respondents lack the normal sense of ownership of the thoughts in their mind. Their thoughts are experienced as alien and not their own. (SCAN).

Thought withdrawal: The patient may describe his thoughts being taken away from himself against his will (SIMS). In thought deprivation (as termed in Fish), the patient finds that as they are thinking, their thoughts suddenly disappear and are withdrawn from their mind by a foreign influence. It has been suggested that this is the subjective experience of thought blocking and 'omission' (Fish). Delusion that one's thoughts are being removed from one's mind by other people or forces (CTP). It is the delusion that thoughts have been taken out of the mind. The delusion usually accompanies thought blocking: the patient experienced a sudden break in the flow of thoughts and believes that in "missing" thoughts have been taken away by some outside agency. (OTP). Respondents say that their thoughts have been taken out of their minds so that they have no thoughts. The experience is passive, i.e., it is not willed but experienced. No thoughts are left behind and there is experience of actual withdrawal which often leads to explanatory delusions (SCAN)

Thought broadcasting: Occurs in 'Schizophrenia' when the patient describes his thoughts as leaving himself and being diffused widely out of his control. It is also a passivity expression of first rank (SIMS). In thought broadcasting, the patient knows that as he is thinking, everyone else is thinking in 'unison' with him. This term has

also been used to describe the belief that one's thoughts are quietly escaping from one's mind and other people might be able to access them. Another one is the experience of hearing one's thoughts spoken aloud and believing that, as a result, other people can hear them (Fish). Feeling that one's thoughts are being broadcast or projected into the environment (CTP). The essence of the symptom is that respondents experience their thoughts as diffusing out of their minds so that they can be experienced by others. The experience is passive, i.e. it is not willed, but experienced (SCAN).

Thought echo- One type of auditory hallucination is hearing one's own thoughts spoken aloud and is also one of the first rank symptom of Schizophrenia. Known in German as *Gedankenlautwerden*, it describes hearing one's thoughts spoken just before or at the same time as they are occurring. *Echo de la pensée* (French) is phenomenon of hearing them spoken after the thoughts have occurred. Best English term for them is 'thought echo' or 'thought sonorisation (Fish). Respondents experience their own thoughts as repeated or echoed (not spoken aloud) with very little interval between the original and the echo. The repetition may not be perfect, however, but subtly or grossly changed in quality (SCAN).

Content: It is elicited by listening to content of speech of the patient and described under following headings

Worry – It is a subjective sense of tension or uneasiness. It has three central components-a round of painful and unpleasant thought, not controlled by attending to usually absorbing subjects, often out of proportion to the context. The content of worry is not relevant to recognition of its form (SCAN).

Phobia – Phobias are fears restricted to a specific object, situation or idea (Fish). Persistent, pathological, unrealistic, entire fear of an object or situation. The phobic person may realise that fear is irrational but nonetheless, cant dispel it (CTP). Criteria for phobia according to Marks (1969) are (SIMS):

- Fear is out of proportion to demands of the situation
- It cannot be explained or reasoned away
- It is not under voluntary control
- The fear tends to an avoidance of the feared situation

Impulse- Impulse is defined as a sudden spontaneous inclination or incitement to some usually unpremeditated action. Although everyone acts on impulse at one point or another, individuals who have a pattern of acting on impulse have a problem with *impulsivity*, which has been defined as the tendency to act with less forethought than do most individuals of equal ability and knowledge, or a predisposition toward rapid unplanned reactions to internal or external stimuli without regard to the negative consequences of these reactions. (CTP). The essential feature of Impulse control disorders is the failure to resist an impulse, drive, or temptation to perform an act that is harmful to the person or to others. The individual feels an increasing sense of tension or arousal before committing the act and then experiences pleasure,

gratification, or relief at the time of committing the act. Following the act there may or may not be regret, self-reproach, or guilt (DSM-IV-TR).

Somatic symptoms- Bodily complaints which are varied in anatomical location and usually not associated with any underlying physical pathology. Symptoms may be referred to any part or system of the body, but gastrointestinal sensations (pain, belching, regurgitation, vomiting, nausea, etc.), skin sensations (itching, burning, tingling, numbness, soreness, etc.), cardiovascular symptoms (breathlessness without exertion, chest pains), bodily pains and sexual and menstrual complaints are common. Seen in somatoform disorder, depression, anxiety spectrum disorders

Somatic syndrome associated with depression (4/8 should be present for diagnosis)- Marked loss of interest or pleasure in activities that are normally pleasurable; lack of emotional reactions to events or activities that normally produce an emotional response; waking in the morning 2 hours or more before the usual time; depression worse in the morning; objective evidence of marked psychomotor retardation or agitation; marked loss of appetite; weight loss (5% or more of body weight in the past month); marked loss of libido.

Religious pre-occupations- It is thinking that predominantly centres around ethical and religious matters. Seen in OCD.

Preoccupation with precipitating factor- It is the thinking that centres around the precipitating factor (which is responsible for current illness/exacerbation) and coloured by an affective tone relating to precipitating event. Seen in reactive depression.

Excessive day dreaming- This refers to excessive continuous indulgence in fantasising or engaging in imaginative, speculations regarding the future, which otherwise are beyond the means of the concerned individual. It occurs in most parts of the working hours hampering normal activities. Example: Schizotypal personality disorder.

Antisocial urges- These are sudden and episodic behaviours characterised by aggressiveness, impulsiveness, rage notwithstanding social rules and norms and are not associated with any guilt feeling or remorse for the acts. Seen in antisocial personality disorder

Homicidal ideas- This means the idea of killing someone or causing grievous injury. These ideas might have been expressed for the first time during the interview either as a response to a hallucination or active delusion of persecution. Seen in Schizophrenia, antisocial personality disorder.

Philosophical ideas- This refers to pre-occupation with the thoughts regarding philosophical issues example existence of God, creation of universe, the difference between mind and matter. Seen in normal individuals, mild depression, OCD, Schizotypal personality disorder.

Magical thinking- The person believes that apparently irrelevant actions can make a difference to reality, and some patients (for e.g. OCD) engage in compulsive

behaviours that are often unrealistically connected to the dreaded event that they are trying to prevent. At times magical thinking may reach near-delusional proportions

Depressive cognition- There are negative views about self (worthlessness), world (helplessness) and future (hopelessness), seen in patients with depression. It includes the following:

- **Idea of worthlessness-** This is the feeling that one is good for nothing and cannot tackle the problems of life with due skill. It is frequent underestimation of one's knowledge, talent and capabilities.
- **Ideas of helplessness-** An attitude characterised by an untowards and unfavourable expectation from the environment. The person thinks that there is no one who will help him when danger strikes, which is very likely.
- **Ideas of hopelessness-** There is a bleak and pessimistic view of future. Either the patient is not able to think about the future or he sees complete dark future, have no hopes whatsoever from his life and himself.

Suicidal ideas- This implies the plan of act of ending one's life permanently. The methods are usually self inflicted or self intentioned. Seen in depression.

Death wishes- A wish that something happens and the person's life is ended so that all his agonies are finished along with that. It is just praying for death rather than thinking to end own life.

Deliberate self-harm (DSH) includes self-injury (SI) and self- poisoning and is defined as the intentional, direct injuring of body tissue without suicidal intent. Although suicide is not the intention of self-harm, the relationship between self-harm and suicide is complex, as self-harming behaviour may be potentially life threatening. There is also an increased risk of suicide in individuals who self-harm (SCAN).

Inflated self esteem- In social sciences, self esteem is a hypothetical construct that is quantified as sum of evaluations across salient attributes of one's self or personality. It is the overall affective evaluation of one's own worth, value or importance. Inflated self esteem is inflation of the self esteem and seeing oneself capable of doing things beyond one's abilities. The person may boast of himself as being the best in whatever things he do. It is seen in Mania and Narcissistic personality disorder. In literature, it is often used synonymously with grandiosity.

Delusion- A false firm belief based on incorrect inference about external reality that is firmly sustained despite what almost everyone else believes and despite what constitutes incontrovertible and obvious proof or evidence to the contrary. The belief is not one ordinarily accepted by other members of the person's culture or subculture (DSM-IV-TR). A delusion is a belief that is firmly held on inadequate grounds, is not affected by rational arguments or evidence to the contrary and is not a conventional belief that the person might be expected to hold given his educational and cultural background. False is omitted from this definition because in some cases a delusional belief can be true or subsequently become true e.g. pathological jealousy (OTP).

Delusion is defined as “a false, unshakeable belief that is out of keeping with the patient’s social and cultural background. A delusion is the product of internal morbid process and this is what makes it unamenable to external influences. The fact that a delusion is false makes it easy to recognise but this is not its essential quality. A very common delusion among married persons is that their spouses are unfaithful to them. In the nature of things, some of these spouses will indeed have been unfaithful; the delusion will therefore be true, but only by coincidence (Fish).

Rather than suggesting a unitary definition for delusion, Kendler et al. (1983) have proposed several poorly correlated dimensions or vectors of delusions. (ABCDE P.S.)

- **Affective response:** the degree to which the patient’s emotions are involved with the beliefs.
- **Bizarreness:** the degree to which the delusional beliefs depart from culturally determined consensual reality
- **Conviction:** the degree to which the patient is convinced of the reality of the delusional beliefs.
- **Disorganization:** the degree to which the delusional beliefs are internally consistent, logical and systematized.
- **Deviant behaviour:** acting out on beliefs
- **Extension:** the degree to which the delusional belief involves areas of the patient's life.
- **Pressure (Preoccupation):** the degree to which the patient is preoccupied and concerned with the expressed delusional beliefs.
- **Systematization:** the framework created around the primary delusion.

Overvalued idea- An overvalued idea is a thought that, because of the associated feeling tone, takes precedence over all other ideas and maintains this precedence permanently or for a long period of time. Even though overvalued ideas tend to be less fixed than delusions and tend to have some degree of basis in reality, it may at times be difficult to distinguish between overvalued ideas and delusions. It can occur in individuals both with and without mental illness.

There is also a distinction between true delusions and delusion-like ideas. True delusions are the result of a primary delusional experience that cannot be deduced from any other morbid phenomenon, while the delusion like idea is secondary and can be understandably derived from some other morbid psychological phenomenon – these are also described as secondary delusions. Thus to summarize, delusions are divided into true (primary) delusions and delusion like ideas (secondary delusions).

Primary delusion- a primary delusion is one that appears suddenly and with full conviction but without any mental events leading up to it. The essence of the primary delusional experience (also termed apophany) is that a new meaning arises in connection with some other psychological event. Primary delusional experiences tend to be reported in acute Schizophrenia but are less common in chronic Schizophrenia, where they may be buried under a mass of secondary delusions arising from primary

delusional experiences, hallucinations, formal thought disorder and mood disorders. Schneider (1959) suggested that these experiences could be reduced to these forms of primary delusional experience:

In the **delusional mood** (or atmosphere) the patient has the knowledge that there is something going on around him that concerns him, but he does not know what it is. Usually the meaning of the delusional mood becomes obvious when a sudden delusional idea or a delusional perception occurs.

In the **sudden delusional idea** (delusional intuition) a delusion appears fully formed in the patient's mind. This is also known as an autochthonous delusion. It is not in itself diagnostic of Schizophrenia because sudden ideas '*out of blue*' or 'brain-waves' occur in various mental illnesses such as depression, personality disorders, organic and epileptic psychosis and even in normal individuals.

The **delusional perception** is the attribution of a new meaning, usually in the sense of self-reference, to a normally perceived object. The new meaning cannot be understood as arising from the patient's affective state or previous attitudes. This last provision is important because the delusional perception must not be confused with delusional misinterpretation. Schneider emphasised the importance of this symptom's '*two memberedness*', as there is a link from the perceived object to the subject's perception of this object, and a second link to the new significance of this perception (sometimes also called delusional significance). Using this criterion, Schneider (1959) divided delusional memories into delusional perceptions and sudden delusional ideas. For example, if the patient says that they are of royal descent because they remember that the spoon they used as a child had a crown on it, this is really a delusional perception because there is the memory and also the delusional significance, i.e. the 'two memberedness'. On the other hand, if the patient says that they are of royal descent because when they were taken to a military parade as a small child the king saluted them, then this is a sudden delusional idea because the delusion is contained within the memory and there is no 'two memberedness'.

Delusional memories- These can be distorted or false memories coming spontaneously into the mind like delusional intuitions. In other cases they occur, like delusional percepts, in two stages, which mean that normal memories are interpreted with delusional meaning. Delusional awareness is an experience, which is not sensory in nature, in which ideas, or events take on an extreme vividness as if they had additional reality (SIMS).

Delusional Misinterpretation- This item is a further extension of the delusion of reference and persecution; wherein whole situations are interpreted in a self-referential way. The arrangement of objects may seem to have special significance. Things seem to be arranged to test respondents, street signs or advertisements on buses, or patterns of color seem to have been put there in order to give messages. This may go so far that whole armies of people may seem to be preoccupied with respondent

Secondary delusion- Secondary delusions are derived apparently from a preceding morbid experience. They may accumulate until there is a complicated and stable delusional system. When this happens, the delusions are said to be systematized. Thus delusions are commonly divided into systematized and non-systematized. In the completely systematized delusions there is a basic delusion & remainder of system is logically built on this error. Completely systematized delusions are extremely rare. Systematization is not a question of all or nothing; but of more or less.

In Schizophrenia, once the primary delusional experiences have occurred they are commonly integrated into some sort of delusional system. This elaboration of delusions has been called 'delusional work'.

The morbid experiences leading to secondary delusions may be of many kinds-including:

- Hallucinations- e.g. a person who hears voices may believe that he is being followed.
- Low mood- e.g. a profoundly depressed woman may believe people think she is worthless.
- Or an existing delusion e.g. a person convinced he's being framed may come to believe he will be imprisoned.

Delusion of Reference- Misinterpretation of events and incidences in the outside world as having direct personal reference to oneself. They may believe that people are talking about them, people are spying on them. They try to interpret situations in self reference. Frequently seen in paranoid patients; ideas of reference may occasionally be observed in normal persons.

Delusions of Persecution- The person believes that people around him are against him and are trying to harm him in one way or the other. The supposed persecutors of the deluded patient may be people in the environment (such as members of the family, neighbours or former friends) or may be political or religious groups, of varying degrees of relevance to the patient.

Here one should also know about delusional misidentification syndromes. The core of these syndromes is delusion of persecution and a framework is developed around that.

Delusional misidentification syndromes are a group of delusional phenomena in which patients misidentify familiar person, objects, or self, and believe that they have been replaced or transformed. These syndromes are delusional because the mis-identifications are false and are not correctable by experience or reason. So far, many different sub-types have been identified, but most authors categorize them into four main syndromes:

- Capgras Syndrome: Belief that a familiar person or object has been replaced by a nearly identical duplicate or impostor.
- Fregoli Syndrome: Belief that a familiar person acquires different physical identities while the psychological identity remains the same, i.e., the familiar person disguises him self as others.

- Syndrome of Inter-metamorphosis: Belief that another person has changed both his physical and psychological identities, i.e., has been transformed into another.
- Syndrome of Subjective Doubles: Belief that another person has been physically transformed into the patient's own self.

Delusion of control- A delusion in which feelings, impulses, thoughts, or actions are experienced as being under the control of some external force rather than being under one's own control.

Delusions of infidelity (Othello syndrome/ Morbid or pathological jealousy)- The person believes that his/her spouse is unfaithful and is having an illicit relationship with someone else. The spouse may be interrogated and may be kept awake for hours at night. A jealous husband, for example, may interpret common phenomena as 'evidence' of infidelity; for example, he may insist that his wife has bags under her eyes as a result of frequent sexual intercourse with someone else, or may search his wife's underclothes for stains and claim that all stains are due to semen. This behaviour may progress to violence against the spouse and even to murder. Delusions of infidelity may develop gradually, as a suspicious or insecure person becomes more and more convinced of their spouse's infidelity and finally the idea reaches delusional intensity. Delusions of infidelity may occur in Schizophrenia, affective psychosis, organic brain disorders and alcohol dependence syndrome.

Delusions of love (de Cleramboult syndrome)- This condition has also been described as 'the fantasy lover syndrome' and 'erotomania'. The patient is convinced that some person is in love with them although the alleged lover may never have spoken to them. They may pester the victim with letters and unwanted attention of all kinds. If there is no response to their letters, they may claim that their letters are being intercepted, that others are maligning them to their lover, and so on. Seen in Schizophrenia, delusional disorder and in abnormal personality states.

Grandiose delusions- The patient has a sense of inflated worth, power, knowledge, identity, or special relationship to a deity or famous person. These occur in Schizophrenia, drug dependence, bipolar affective disorders and organic brain disease. Three variants are usually explained

- Delusion of grandiose ability: Patient believes that he has unusual talents, he is able to read people's thoughts, he is particularly good at helping people, he is much more clever than anyone else, he has invented machines, composed music, solved mathematical problems and so on, beyond most people's comprehension.
- Delusion of grandiose identity: Patient believes that he is famous, rich, chosen for a special mission, related to prominent people. He may believe that he is a changeling and his real parents are royalty.
- Delusion of grandiose mission: As an extension to grandiose identity, patient believes that he had been sent to earth for a special mission with supernatural powers. His role may be to save earth from a possible alien attack.

Delusions of ill health- Individuals with delusions of ill health may believe that they have a serious disease, such as cancer, TB, AIDS, a brain tumor, and so on. Delusions of ill health may involve the patient's spouse and children. A mother may believe that she has infected her children or that she is mad and her children have inherited incurable insanity. This may lead her to harm or even kill her children in a belief that she is putting them out of their misery. Delusions of ill health are a characteristic feature of depressive illnesses, but are also seen in Schizophrenia and abnormal personality developments.

Delusions of guilt- Patient is self-reproachful and self-critical and believes that they are a bad or evil person and have ruined their family. They may claim to have committed an unpardonable sin and insist that they will rot in hell for this. The sin is usually masturbation or extramarital sexual intercourse. Seen in depressive psychosis.

Nihilistic delusions (Cotard Syndrome) - Nihilistic delusions or delusions of negation occur when the patient denies the existence of their body, their mind, their loved ones and the world around them. They may assert that they have no mind, no intelligence, or that their body or parts of their body do not exist; they may deny their existence as a person, or believe that they are dead, the world has stopped, or everyone else is dead. These delusions tend to occur in the context of severe, agitated depression, especially in so called involutional melancholia and also in Schizophrenia and subacute delirium.

Delusions of enormity- Patient believes that they can produce a catastrophe by some action (e.g. they may refuse to urinate because they believe they will flood the world.

Delusions of poverty- The patient with delusions of poverty is convinced that they are impoverished and believe that destitution is facing them and their family. These delusions are typical of depression but appear to have become steadily less common over the past decades.

Hypochondriacal delusions- Recurrent or persistent belief that one might be having one of the serious mental or physical illness in spite of repeated assurance by the doctor and negative test reports. Seen in hypochondriasis, depression. Somewhat similar to these delusions are the delusional preoccupations with facial or bodily appearances, when the subject is convinced that their nose is too big, their face is twisted, or disfigured with acne, and so on. Contemporary classification systems (ICD-10) tend to place some of these patients in the category of delusional disorders, which includes delusional dysmorphophobia.

Bizarre delusions: Bizarre delusions are defined as "*clearly implausible and not understandable and not derived from ordinary life experiences*". A patient can be diagnosed with Schizophrenia if she/he has any one of Schneider's FRS that are specifically named or if she/he has "persistent delusions of other kinds that are culturally inappropriate and completely impossible, such as religious or political identity, superhuman powers and ability". Examples of delusions rated as both Schneiderian and bizarre are, "Even people in Venezuela can hear my thoughts" and "There is a camera in my tooth which is taking photographs of contents of my mind."

Resolution of delusions: Over the period of time, various studies have attempted to look for the possible fates of delusion, after the acute psychosis phase settles. Brett-Jones et al. (1987) have observed three outcomes in their study that might be called 'recovered'.

- Integration- These patients show low conviction in their belief, but remain preoccupied by the ideas. They try to actually 'integrate' the experience into their lives and to redefine them in non-psychotic terms. These patients are usually keen to discuss and try to understand what had happened in non-psychotic terms.
- Sealing over- In this, the patients completely reject their belief and the preoccupation drops to zero. Thus these patients try to seal over their psychotic experiences. They show strong reluctance to discuss their experiences.
- Encapsulation- Certain patients retain high conviction but preoccupation and interference decreases.

Double book keeping- A similar albeit different construct; it is a phenomenon first identified by Bleuler, refers to the patient's ability to, as it were, live in two worlds at the same time. On the one hand is the world of voices, visions, and delusions, and on the other hand, and quite coincident with this psychotic world, is the world as perceived by others. To the patient both worlds seem quite real. For example, a patient may hear a voice as clearly as the voice of the physician and believe it just as real, yet at the same time acknowledge that the physician does not hear it. Or the grandiose patient who fully believed that a coronation was imminent may yet continue to work at a janitor's job and go on doing so, living in two worlds, and feeling little if any conflict between them.

Perceptual disorders: Perception is defined as the conscious awareness of elements in the environment by the mental processing of sensory stimuli. Perception refers to the way the world looks, sounds, feels, tastes or smells. For much of what we perceive the sensory input provides only raw materials for experience but the perceptual system searches for the percept which is most consistent with the sensory data, thus making perception an active process rather than a passive one. (Morgan & King)

Sensory distortion- These are changes in perception that are the results of a change in the intensity and quality of the stimulus or the spatial form of the perception.

Changes in intensity

- Hyperaesthesia- increased intensity of sensations.
- Hyperacusis- increased sensitivity to noise.
- Hypoacusis- increased threshold for all sensations

Changes in quality

- Xanthopsia
- Chloropsia
- Erythropsia

Changes in spatial form

- Micropsia- The visual perception that the objects are smaller than they actually are (DSM-IV-TR); False perception that the objects are smaller than they really are. Sometimes called 'Lilliputian Hallucinations' (CTP); Micropsia is a visual disorder in which the patient sees objects smaller or farther away than they really are (Fish)
- Macropsia/ megalopsia- The visual perception that the objects are larger than they actually are (DSM-IV-TR); In macropsia, objects seem larger than original (SIMS); Micropsia is a visual disorder in which the patient sees objects larger or nearer than they really are (Fish). These conditions occur in acute organic states, epilepsy, visual disorders
- Parropsia- the experience of retreat of objects into the distance without any change in size.
- Meglopsia- describe objects that are larger/ smaller or one side than the other.

Sensory deception

Hallucination- Perception without an adequate external stimulus (Esquirol). A false perception which is not a sensory distortion or a misrepresentation, but which occurs at the same time as real perceptions. (Jasper). A hallucination is an exteroceptive or interoceptive percept which does not correspond to an actual object (Smythies)

According to Slade, three criteria are essential for an operational definition: a) percept like experience in the absence of an external stimulus; b) percept like experience that has the full force and impact of a real perception; c) percept like experience that is unwilling, occurs spontaneously and can't be readily controlled by the perceiver.

According to Cutting, a hallucination is a perception without an object (within a realistic philosophical framework) or the appearance of an individual thing in the world without any corresponding material event (within a Kantian framework)

A hallucination is a percept experienced in the absence of an external stimulus to the corresponding sense organ. It differs from illusions in being experienced as originating in the outside world or from within the person's body. Hallucinations can't be terminated at will (SOTP). A sensory perception that has the compelling sense of reality of a true perception but that occurs without an external stimulation of relevant sense organ. The person may not have an insight into the fact that he/she is having a hallucination (DSM-IV-TR)

Hallucinations of individual senses

- Hearing- Auditory hallucination
- Vision- Visual hallucination
- Smell- Olfactory hallucination
- Taste- Gustatory hallucination
- Touch- Tactile hallucination

Auditory hallucination- A hallucination involving the perception of sound, more commonly of voices. No distinction is made as to whether the source of the voices is perceived as being inside or outside (DSM-IV-TR). These are elementary in form of noises or partly organised as music or completely organised as hallucinatory voices (Fish). False perception of sound, usually voices but also other noises such as music. These are most common hallucinations in Schizophrenia (CTP)

Visual hallucination- A hallucination involving sight, which may consist of formed images such as people or unformed images such as flashes of light. (DSM-IV-TR). Hallucinations primarily involving the sense of sight (CTP). Visual hallucinations characteristically occur in organic states rather than in functional psychoses (SIMS)

Olfactory hallucination- A hallucination involving the perception of odour, such as burning rubber or decaying fish (DSM-IV-TR). Hallucinations primarily involving smell or odour, most commonly in medical disorders, especially temporal lobe (CTP). Smell may/may not be unpleasant, but usually has special and a personal significance.

Gustatory hallucination- A hallucination involving the perception of taste (usually unpleasant) (DSM-IV-TR). Hallucinations primarily involving taste (CTP). Hallucinations of taste occur in Schizophrenia and acute organic states (Fish).

Tactile hallucination- A hallucination involving the perception of being touched or of something being under one's skin. The most common tactile hallucinations are the sensations of electric shock and formication (the sense of something creeping or crawling under the skin) (DSM-IV-TR)

Kinaesthetic hallucinations- Kinaesthetic hallucinations are those of muscle or joint sense. The patient feels that the limbs are being bent or twisted or muscles squeezed. Such hallucinations are often linked with bizarre somatic delusions (SIMS)

Pseudo- hallucination- They are a type of mental image that, although clear and vivid, lack the substantiality of perception, they are seen in full consciousness, known to be not real perceptions and are located not in objective space, but in subjective space. (Fish).

Pseudohallucination is a perceptual experience which is figurative, not concretely real, and occurs in inner subjective space, not in external objective space. It may have clear and vivid details. It may be retained for sometime and it can not be deliberately evolved. It is sometimes described as 'as if' phenomenon or in similar explanatory terms (SIMS)

Pseudo hallucination as a separate form of perception from true hallucination. Pseudo-hallucination is more like sense perception (or true hallucinations) than fantasy. So the image seems to have definite edges, to be vivid, coloured, constant over some time and not created voluntarily. (Kandisky).

Functional hallucinations- A stimulus causes a hallucination but the stimulus is experienced as well as the hallucination. In other words the hallucination requires the presence of another real sensation. For example, a patient with Schizophrenia first

heard the voice of God as her clock ticked; later she heard voices coming from the running tap and voices coming from the chirping of the birds. So both the noises and the voices were audible (Fish)

Reflex hallucinations- A stimulus in one sensory field produces a hallucination in another. For example, a patient felt a pain in her head (somatic hallucination) when she heard other people sneeze (the stimulus) and was convinced that sneezing caused the pain (Fish).

Extracampine hallucinations- The patient has a hallucination that is outside the limits of the sensory field. For example, a patient sees somebody standing behind them when they are looking straight ahead or hear voices talking in London when they are in Liverpool (Fish).

Autoscopy (phantom mirror-image)- It is the experience of seeing oneself and knowing that it is oneself. It is not just a visual hallucination because kinaesthetic and somatic sensation must also be present to give the subject the impression that the hallucination is oneself (Fish). Seen in parietal lobe lesions, normal persons.

Illusion- A misperception or misinterpretation of a real external stimulus such as hearing the rustling of leaves as the sound of voices (DSM-IV-TR). Perceptual misinterpretation of a real external stimulus (CTP). Misinterpretation of stimuli arising from an external stimulus. In illusion, stimulus from a perceived object is combined with a mental image to produce a false perception. It can occur as a result of set, lack of perceptual clarity, intense emotions. Illusions are not morbid in origin as they can, usually be corrected. Illusion occurs in the severe depressive illness, delirium and even in Schizophrenia (Fish).

Fantastic illusions- Patients see extraordinary modifications to their environment. Eg.: Patient seeing head of a pig in mirror. The cause of such illusions is exaggeration and confabulation (Fish)

Pareidolia- This is an interesting type of illusion, in which vivid illusions occur without the patient making any effort. These illusions are the result of excessive fantasy thinking and a vivid visual imagery. They cannot therefore be explained as the result of affect or mind-set, so that they differ from the ordinary illusion. Pareidolias occur when the subject sees vivid pictures in fire or in clouds, without any conscious effort on his part and sometimes even against his will (Fish)

Imagery (SIMS)- It is the internal mental representation of the world and is actively drawn from memory. Imagery underlines our capacity for many crucial cognitive activities such as mental arithmetic, map reading, visualizing, imaging places previously visited, and recollecting spoken speech. Jaspers 1962 describes the formal characteristics of images.

- Are figurative and have a character of subjectivity.
- They appear in inner subjective space.

- Are not clearly delineated and come before us incomplete.
- Although sensory elements are individually the equal of those in perceptions, mostly they are insufficient.
- Images dissipate and always have to be recreated.
- Images are actively created and are dependent on our will.

Other psychotic phenomena

Somatic passivity- It is the belief that outside influences are playing on the body. The patient is a passive and invariably a reluctant recipient of bodily sensations imposed upon him by some external agency. According to Jaspers the perception is simultaneously experienced as being both a bodily change and externally controlled. It is a single experience and not simply the delusional interpretation of an abnormal perceptions bodily sensation. These somatic perceptions may be due to haptic, thermic or kinesthetic hallucinations or at times admixture of different hallucinations.

Made phenomenon- This includes made act, made affect and made impulse, which are defined below:

Made act- The patient experiences his actions as being completely under the control of an external influence. The movements are initiated and directed throughout by the controlling influence, and the patient feels he is an automaton, the passive observer of his own actions.

Made affect- The patient experiences feelings, which do not seem to be his own. The feelings are attributed to some external source and are imposed upon him.

Made impulse- A powerful impulse overcomes the patient to which he almost invariably gives way. The impulse to carry out this action is not felt to be his own, but the actual performance of the act is, i.e. the action is admitted to be the patient's own, but he feels that the impulse that precipitated him into doing it was not his own.

Other phenomena

Depersonalization-derealization

Depersonalization- Depersonalisation is the term used to designate a peculiar change in the awareness of self, in which the individual feels as if he is unreal (Sedman, SIMS). It is a subjective state of unreality in which there is a feeling of estrangement, either from a sense of self or from the external environment. (Fewtrell, SIMS). The symptom is based on a detachment from or loss of the emotional colouring that accompanies the perception of self (SCAN). Sensation of unreality concerning oneself, parts of oneself or one's environment that occurs under extreme stress or fatigue. Seen in Schizophrenia, depersonalization disorders (CTP). A change in the awareness of one's own activity occurs when the patient feels that they are no longer their normal natural self and this is known as 'depersonalisation'. Depersonalization may also be reported in association with Schizophrenia, depressive illness, organic brain diseases or substance misuse. (Fish)

Derealization- Sensation of changed reality or that surroundings have altered. Usually seen in Schizophrenia, panic attacks and dissociative disorders (CTP). In derealization, the emotional component is detached from perception so that respondent experience their surroundings as unreal. The experience has an 'as if' quality (SCAN). A feeling of unreality so that the environment is experienced as flat, dull and unreal (Fish).

Body image disturbance- The body image or body schema is a person's subjective representation against which the integrity of his body is judged and the movement and positing of its parts assessed. Parietal lobes play a major role, but the somatic aesthetic afferent system and the thalamus are also involved.

Hyperschemazia- It is perceived magnification of body parts. It can occur with a variety of organic (partial paralysis, peripheral vascular disease) and psychiatric conditions (hypochondriasis, anorexia nervosa)

Aschemazia or hyposchemazia- The perception of body parts as absent or diminished is known as aschemazia or hyposchemazia respectively and is most likely to occur in parietal lobe lesions. It must be differentiated from nihilistic delusions.

Paraschemazia- It is the distortion of body image is described as a feeling that parts of the body are distorted or twisted or separated from the rest of the body and can occur in association with hallucinogenic use, in epileptic aura and rarely in migraine.

Hemisomatognosia- It is a unilateral lack of body image in which the person behaves as if one side of the body is missing and it occurs in migraine, during an epileptic aura.

Somatoparaphrenia- Some patients show bizarre attitudes to their paralysed limb or delusional beliefs about the body. They may have too many, they may be distorted, inanimate, severed or in other ways abnormal. They may claim the limb belongs to a specified other person. This is called as somatoparaphrenia.

Paramnesia

Déjà vu- Illusion of visual recognition in which a new situation is incorrectly regarded as a repetition of a previous experience (CTP). It is a disturbance in which the associated feeling of familiarity that normally occurs with previously experiences events, occurs when the event is experienced for the first time (SIMS). Here the subject has the feeling that he has seen or experienced the current situation before. The sense of recognition in déjà vu is never absolute so that misidentification does not occur. These experiences occur occasionally in normal persons but they may become excessive in temporal lobe lesions (Fish).

Jamais vu- In jamais vu, an experience which the patient knows he had experienced before is not associated with the appropriate feeling of familiarity. The patient may also have the feeling that some important memory is about to be recalled, although it

does not actually arrive (SIMS). Paramnesic feeling characterised by a false feeling of unfamiliarity with real situation that one has previously experienced (CTP).

Deja entendu- Illusion that what one is hearing, one has heard previously (CTP)

Confabulation- Unconscious filling of gaps in memory by imagining experiences or events that have no basis in fact. Commonly seen in amnesic syndromes (CTP). This is a detailed false description of an event which is alleged to have occurred in past (Fish). This is a falsification of memory occurring in clear consciousness in association with an organically derived amnesia.

Retrospective falsification- Memory becomes unintentionally distorted by being filtered through a person's present emotional, cognitive and experiential state (CTP). The subject modifies his memories in terms of his general attitudes (Fish). Common in depression.

False recognition- Syn. Misidentification- Seen in Schizophrenia; It may be

- a) Positive- Recognition of strangers as one's friends or relatives
- b) Negative- Friends or relatives are viewed as strangers or strangers in disguise

Hyperamnesia- This refers to the exaggerated degree of retention or recall. It may be elicited by hypnosis. Condition- Obsessive compulsive disorder

Judgement: Judgement is a complex mental process whereby a person forms an opinion, makes a decision, or plans an action or response after first analysing the issue and comparing choices with acceptable social behaviour. (Strub & Black). It is an act of comparing and evaluating a proper course of action. It includes the concepts of ethical values and insight in its meaning.

Social judgement- Here the person's interaction with the other social members and the interviewer is assessed. It is usually assessed from the history given by the informants.

Personal judgement- the individual's personal expectations, plans and attitudes are assessed.

Test judgement- It aims to assess the course of action that a person might take in a socially difficult or disastrous situation.

Conditions causing impaired judgement are organic brain damage, anxiety state, Mania, Schizophrenia

Insight: It is a patient's degree of awareness and understanding about being ill. Patient may exhibit complete denial of their illness or may show some awareness that they are ill but place the blame on others, on external factors, or even organic factors. They may acknowledge that have an illness but ascribe it to something unknown or mysterious in themselves. (SOP).

It is one's ability to understand either oneself or an external situation. (Strub & Black).

In psychopathology, the term insight refers to awareness of morbid changes in oneself and a correct attitude to this change including in appropriate cases, a realization that there is a mental disorder.

Intellectual insight- It is the personal ability to comprehend the cause of a particular problem. In connection to psychiatric disorders, it is the understanding of the psychodynamics of a particular phenomenon. There is presence of knowledge but that knowledge is not utilized to the benefit of the patient.

True emotional insight is present when patient's awareness of their own motives and deep feelings leads to a change in their personality or behaviour patterns. (SOP). It is the deeper level of understanding of the problem with due motivation to bring about a positive change in behaviour or personality.

Insight is rated on a 6-point scale from one to six.

Grade-1 Complete denial of illness.

Grade-2 Slight awareness of being sick and needing help, but denying at the same time.

Grade-3 Awareness of being sick, but it is attributed to external or physical factors.

Grade-4 Awareness of being sick, due to something unknown in self.

Grade-5 Intellectual insight- awareness of being ill and that the symptoms/ failures in social adjustment are due to own particular irrational feelings/ thoughts; yet doesn't apply this knowledge to the current/ future experiences.

Grade-6 True emotional insight- it is different from intellectual insight in that the awareness leads to significant basic changes in the future behaviour.

Assessment of insight

Insight is most usefully inquired about and reported as a series of health beliefs. Usually a series of questions are asked to assess insight.

Do you believe that the experiences that you are having, are symptoms?

Do you believe that these symptoms are attributable to illness?

Do you believe that the illness is psychiatric?

Do you believe that psychiatric treatment might benefit you?

Will you be willing to accept advice from a doctor regarding your treatment

MSE of an unco-operative patient (Kirby's Method)

The difficulty of getting information from unco-operative patient should not discourage the psychiatrist from making and recording certain observations. These may be of great importance in the study of various types of cases and give valuable data for the interpretation of different clinical reactions. It is hardly necessary to say that the time to study negativistic reactions is during the period of negativism, the time to study a stupor is during the stuporose phase. To wait for the clinical picture to change or for the patient to become more accessible is often a miss an opportunity and leave a serious gap in the clinical observation. Obviously, it is necessary in the examination of such cases to adopt some other plan than that used in making the usual 'mental status'. The following guide was devised to cover in a systematic way the most important points for the purpose of clinical differentiation.

- I. General reaction and posture
 - a. Attitude voluntary or passive
 - b. Voluntary postures comfortable, natural, constained or awkward
 - c. What does the patient do if placed in awkward or uncomfortable positions
 - d. Behaviour towards physician and nurses: resistive, evasive, irritable, apathetic, compliant
 - e. Spontaneous acts: Any occasional show of playfulness, mischeviousness or assaultiveness. Defence movements when interfered with or when pricked with pin. Does he eat voluntarily or must be fed? Does he dress and undress himself or he needs assistance? Attention to bowels and bladder. Is he neat or untidy? Do the movements show only initial retardation or are they consistent throughout?
 - f. To what extent does the attitude change? Is the behaviour constant or variable from day to day? Do any special occurrences influence the condition?
- II. Facial expressions: Is the expression alert, attentive, smiling, placid, vacant, stolid, sulky, scowling, averse, perplexed, distressed etc. Any play of facial expression or signs of emotion: tears, smiles, flushing, perspiration. On what occasions?
- III. Eyes: Open or closed? If closed, resist having lid raised. Movements of eyes: absent or obtained on request; give attention and follow the examiner or moving objects; or shows only fixed gazing, furtive glances or evasion.
Rolling of eyeballs upwards. Blinking, flickering or tremor of lid. Reaction to sudden approach of threat to stick pin in eye. Sensory reaction of pupils (dialation from painful stimuli or irritation to skin of neck)
- IV. Reaction to what is said or done: Commands: show tongue, move limbs, grasp with hand (clinging, clutching etc.)
Motions slow or sudden. Reaction to pin-pricks. Automatic obedience: tell patient to protude the tongue to have pin stuck into it.
Echopraxia: imitation of actions or others
- V. Muscular reactions: Test for rigidity: Muscles relaxed or tense when limb or body is moved.

Catalepsy, waxy flexibility. Negativism shown by movements in opposite direction or springy or cog-wheel resistance.

Test head and neck by movements forward and backwards and to side.

Test also the jaw, shoulders, elbows, fingers and the lower extremities.

Does distraction or command influence the reaction?

Closing of mouth, protrusion of lips ('*Schnauzkrampf*')

Holding of saliva, drooling.

VI. Emotional Responsiveness: Is feeling shown when talked to of family or children?

Or when sensitive points in history are mentioned or when visitors come?

Note whether or not acceleration of respiration or pulse occurs; also look for flushing, perspiration, tears in eyes, etc. Do jokes elicit any response?

Effect of unexpected stimuli (clap hands, flash of electric light).

VII. Speech: Any apparent effort to talk, lip movements, whispers, movements of head.

Note exact utterances with accompanying emotional reaction (may indicate hallucinations)

VIII. Writing: Offer paper and pencil. Irresponsive or partially stuporose patients will often write when they fail to talk.

Diagnostic formulation

Diagnostic formulation is a process by which the features of an individual case can be discussed between two or more professionals and evaluated to consider a series of possibilities, which will guide the management. Lack of consensus on many aspects of the diagnostic formulation also leads to comments from the listener like – ‘this is a summary, not a formulation’, ‘the formulation does not include management’, ‘give us your formulation in two minutes, please’. Such comments are not only anxiety provoking for the presenter but also reflect an unnecessarily rigid point of view. Instead, it is more appropriate to provide flexible guidelines on formulation for listeners (usually examiners) and presenters (usually candidates), in order to facilitate a discussion about the patient and his problems, which after all, is the purpose of a clinical examination.

Diagnostic Formulation is better defined as an interviewer’s assessment of the case rather than a re-statement of facts. Its length, layout and emphasis will vary considerably from one patient to another. A good formulation is based on the facts of the case and not on speculation. A formulation is concerned with not only the disease concepts, but also with the understanding how the patient’s lifelong experiences have influenced his personality and his ways of reacting to adversity.

Structure:

Introductory comments: Present the salient socio-demographic features of the patient (e.g. Index patient Mrs. R is a 30-year-old married Hindu, graduate, housewife from middle socio-economic family of sub-urban Jharkhand).

Family and personal history: state any relevant family and person history and premorbid personality (.... with family history suggestive of depressive illness in father on treatment and maintaining well; personal history suggestive of anxious avoidant traits premorbidally)

Positive medical history of significance: (... was detected to be hypothyroid a year ago and is on treatment for that...).

Past history of psychiatric disorder, its treatment and outcome: (... had an episode suggestive of depression, after her son’s birth 4 years back; was treated with antidepressant medication and became completely well in about two months...).

Account of informant: State the relation of informant, whether patient is brought or has come on his/her own (.... has come now, accompanied by her husband, history being reliable and adequate)

Presenting problems: Start with duration of presenting problems, precipitating factor if any, its onset, course and progress (.... with illness of 4 month duration which was preceded by a medical termination of pregnancy about which patient was very ambivalent with acute onset, continuous course, deteriorating progress of illness) state the main problems excluding irrelevant details (... characterized by being increasingly depressed, with loss of energy, self-reproaches and crying spells). Briefly mention how the patient’s life has been affected by the problems (.... and she has not been going for work and has also been unable to do the

housework or take care of her child). Avoid long lists of minor or transient symptoms and negative findings except those that will help in the differential diagnosis.

Treatment history: include briefly any treatment received for current illness (.... with no treatment taken for current illness)

Physical examination: state salient features of physical examination (.... Physical examination reveals goitrous growth in the neck, rest within normal limits)

Mental status examination: Mention important findings only. Use labels for psychopathological findings at this stage, (.... MSE findings reveals downcast gaze, decreased psychomotor activity, scant low volume speech with increased reaction time, cognitive functions intact, depressed, appropriate communicable affect, delusions of guilt, depressive cognition, second person auditory hallucinations, impaired judgement and grade I insight). Details of these findings should have already been described during the detailed presentation prior to the formulation and if helpful, could be mentioned again during the discussion of the differential diagnoses.

Diagnosis: Mention whether confident diagnosis, provisional diagnosis, tentative diagnosis.

Differential diagnosis: If there is little doubt about the diagnosis, say so and say why. Do not present an irrelevant differential diagnosis for the sake of giving one. If diagnosis is not clear, embark on a careful discussion of the possibilities in the order of likelihood, and discuss points in favor of and against each of them. This is done using descriptive psychopathology (e.g. first-rank symptoms) elicited during history taking and mental status examination. Details of symptoms collected earlier could be used to support a diagnosis (e.g. content of auditory hallucinations to differentiate between Schizophrenia and depressive illness). Information on the course of illness is also useful (e.g. 'Though the acute psychotic symptoms are remitted with medication, the patient never reached his premorbid level of functioning at work or in social interactions'). Differential diagnosis tests one's ability to make a discriminating clinical judgment. Do not give a long list of differential diagnoses that cover the whole of ICD-10; think twice before giving more than three or four.

If a patient's history and findings justify diagnosing two or more conditions that co-occur, mention those with supporting evidence (e.g. Moderate depressive episode with somatic syndrome with mental and behavioural disorder due to use of alcohol, dependence syndrome, active dependence).

Management: In each case specify which procedure/tests you would organize and its justification

- a) Further investigations:
 - i. Includes information from key relatives/employer/teachers
 - ii. Review of past case records
 - iii. Laboratory investigations.
 - iv. Psychometry.
- b) Immediate management plans
 - i. Is the patient to be treated as an inpatient or outpatient?
 - ii. If as an inpatient, why?

- iii. Management of suicide risk/violence - where indicated
- iv. Medication - specify type/justification/dosage/route/ expected response/side effects and their managements.
- c) Long-term management plans
 - i. Somatic: Medication - type/dosage/duration
 - ii. Psychological: Psychotherapy - indications/type/focus
 - iii. Social: Involvement of the family/rehabilitation measure

Prognosis: This should not be a general pronouncement, based merely on the type of disorder (such as Schizophrenia). Discuss instead the good (e.g. acute onset; affective symptoms) and poor (e.g. poor drug compliance in the past; poor social support) prognostic factors. Prognosis can also be described under the headings of short term (e.g. 'Chances of recovery from the present episode is good with antidepressant treatment') and long term ('risk of relapse and recurrence is high because of the significant marital discord and patient's reluctance to take medicines on a long-term basis'). Come to a reasonably firm final conclusion rather than using vague terms like 'guarded'.

How is a summary different from a diagnostic formulation?

The terms 'summary' and 'diagnostic formulation' are often used together and cause confusion to many candidates who take them to be synonymous. However, there are subtle but important differences and being aware of them is helpful in making a good diagnostic formulation.

Summary is a concise description of all the important aspects of the case, whereas formulation is an assessment of the case rather than a restatement of facts. Summary should be written in such a way that it provides all the necessary information that will assist in the follow-up care of the patient by the same, or other medical team. It should include:

- Demographic data like name, age, gender
- Reasons for referral to psychiatry
- History of present illness
- History of previous illnesses
- Family history
- Personal history - birth and development, childhood, education, occupation, sexual and marital history
- Premorbid personality
- Physical examination
- Investigations - physical and psychological
- Diagnosis
- Treatment and progress
- Prognosis
- Plans for further management

Adapted from: Kuruvilla, K. and Kuruvilla, A. (2010). Diagnostic formulation. Indian Journal of Psychiatry 2010, 52, 78-82.

Diagnosis and Diagnostic nomenclature systems

When the requirements laid down in the diagnostic guidelines are clearly fulfilled, the diagnosis can be regarded as "*confident*". When the requirements are only partially fulfilled, it is nevertheless useful to record a diagnosis for most purposes. It is then for the diagnostician and other users of the diagnostic statements to decide whether to record the lesser degrees of confidence (such as "*provisional*" if more information is yet to come, or "*tentative*" if more information is unlikely to become available) that are implied in these circumstances.

Two diagnostic classificatory systems that are used in psychiatry are

1. The International Classification of Diseases (ICD)-10- {Chapter V(F) of ICD-10}
2. Diagnostic and Statistical Manual of Mental Disorders- Fifth Edition (DSM-5)

ICD-10: With the introduction of operationalized diagnostic systems the multi-axial approach became a more important issue. The proposed multi-axial system of ICD-10 consists of three axes:

- Axis I- Psychiatric diagnoses, made according to the ICD-10 CDDG or DCR
- Axis II (Disability Diagnostic Scale, DDS)- Impairment of psychosocial functioning.
- Axis III- Environmental/circumstantial & personal lifestyle management factors rated.

DSM-IV-TR system of diagnosis uses a five-axes model. Axes 1-3 are compulsory, whereas axes 4 and 5 are optional, although are usually included as well for a more reliable diagnosis.

- Axis I measures the clinical psychiatric disorder
- Axis II measures personality disorders and mental retardation
- Axis III assesses general medical conditions, which are relevant to the condition or treatment
- Axis IV measures psychosocial and environmental problems (life problems that influence the psychological wellbeing of the patient) – examples include homelessness, family issues and unemployment
- Axis V gives a score for the global assessment of functioning, whereby a score from 1 to 100 is used to classify the patient based on an evaluation of how well the individual functions socially, occupationally and psychologically. Generally, a score of 50 or below indicates severe symptoms.

DSM-5 moved to a nonaxial documentation of diagnosis, combining the former Axes I, II, and III, with separate notations for psychosocial and contextual factors (formerly Axis IV) and disability (formerly Axis V).

Special population

The gross outline for psychiatric history taking will remain same across majority of patients, but a few changes need to be made with regard to certain special population (such as children and elderly) and in certain diseases such as substance use disorder, neurological complains, epilepsy). Here one by one, brief description of changes to be made in approach are delineated under the following headings

Child and adolescent

The baseline assessment is done on similar lines as in case of a normal adult psychiatry case work-up. Emphasis needs to be given on a few domains. The major differences are enumerated below

Socio-demographic profile:

Ask exact date of birth, if such details are available. This will help in knowing the exact age of the child at the time of presentation.

Include both parents name and their educational qualification and occupation. In case patient is brought up by some other guardians, their details should be mentioned along with the reason for the same, whether child was abandoned; or the parents died at very early period in his/her life, or the child was adopted out of his biological parents' family along with adequate reason for the same.

History of presenting complains:

In majority of cases (unless the presenting complains have no role whatsoever in childhood period), one should try to start history of present illness dating back to the birth of child and should progress accordingly.

Impact of illness on routine activities here should include interpersonal relation with parents, other adults, peers, interest in work/study, play behaviour apart from other relevant details as probed for in a general case work-up.

While asking for negative history, make sure to rule out any childhood disorder which might not be the reason of prime attention for the informants such as hyperactivity, attention deficits, impulsivity, disobedience, lying, stealing, truancy, eating difficulties, fears, sleep disorders, somatization, temper tantrums, attention seeking behaviour, enuresis, encopresis, tics and unusual habits

Family history:

Try to include household composition including all members like grand parents, parents, siblings and relatives, whosoever lives in some living hood as the child.

Personal history:

In case of children, personal history becomes more important to probe into because many a childhood disorders have their roots in some or the other event occurring in prenatal, natal or postnatal period. Even if such periods are uneventful then early childhood period must have

some or other deviation from normal. So as enumerated in personal history part of general case work-up, these all issues should be dealt with adequate care and caution.

Detailed *immunization history* should be asked for and should be tabulated if adequate details are available. Here, one should be aware that ministry of health and family welfare (MoHFW) Government of India runs a 100% centrally funded programme named as Universal Immunization Programme (UIP) which has its guidelines for immunization of all children at all government/ PSU/ Local/ autonomous health facilities. According to UIP, the schedule is as follows:

	Birth	6wk	10wk	14wk	9-12m	16-24m	5-6yr	10yr	16yr
BCG	1								
OPV	0	1	2	3		Booster			
Hep-B	0	1	2	3					
DPT		1	2	3		B1	B2		
Measles					1				
TT								B3	B4
JE					1	2			
Hib		1	2	3					

BCG- Bacillus Calmette–Guérin vaccine for Tuberculosis

OPV- Oral polio vaccine

DPT- Diphtheria, pertussis, tetanus toxoid vaccine

Hep-B- Hepatitis B vaccine

TT- Tetanus toxoid vaccine

JE- Japanese Encephalitis vaccine

Hib- Haemophilus influenza B

JE vaccine (in selected high disease burden districts) is currently being used in 113 districts and additional 62 new JE endemic districts have been identified.

Hib vaccine is given as pentavalent vaccine (Hib+DPT+Hepatitis B) introduced in 8 states i.e. Tamil Nadu, Kerala, Haryana, J&K, Gujarat, Karnataka, Goa and Puducherry.

Indian academy of pediatrics (IAP) recommends further addition of pneumococcal, rotavirus, varicella, typhoid, hepatitis A, influenza, cholera, MMR (measles, mumps, rubella) HPV and meningococcal vaccines which are even used currently by a number of private practitioners in the country. In 2014 it was announced by the government of India that four vaccines would be added to the immunization programme; namely rotavirus, rubella and Japanese encephalitis, as well as the injectable polio vaccine.

While gathering *information about home environment*, one needs to be sensitive to patterns of parental functioning, which may affect the child directly or indirectly. It should be assessed on following domains: Permissiveness/ Rigidity; Consistency/ Inconsistency; Strictness/ Liberality of discipline; Approval/ Disapproval of interests; Protectiveness/ Non-protectiveness; Tolerance/ Non-tolerance of deviance; Expectations from the child.

Scholastic history assumes more importance in childhood disorders. One should probe in following areas: What was the type of school (normal/ special/ religious school/ studied at home)? What was the age of entry in school (reasons if admitted late than expected)? Provide schooling details (mention changes in schools, durations with reasons). How was scholastic performance (Good/ Average/ Poor)? How was his attendance in school (regular/ irregular/ discontinued)? What were the reasons for irregularity or discontinuity (school refusal/ wanders/ fearful/ financial problems/ poor progress/ behavioural problems/ Request of school authorities)? How was his relation with peer group and authorities? Make a mention of failures and disciplinary actions, if any.

Play behaviour should be enquired about. Whether he enjoys play/ not interested in play/ observes others while playing. What is his play preferences- plays alone/ with older/ younger/ peer group/ animals/ no preferences. Does he have knowledge of games governed by rules? How does he behave while playing in group-situations? Does he have any special likes and dislikes? Mention reasons for poor play behaviour (No company/ Siblings or peer group not interested in playing with the child/ quarrelsome/ overprotected by parents or care-takers/ poor play facilities)

Pre-morbid functioning (Temperament): One should assess pre-morbid temperament of the child by asking behaviour in the first few weeks and months of his life how was he like?

Activity level: (How much did the baby move around?): Completely still/ moderately/Always on move.

Adaptability: (How did the child respond to changed circumstances?): Did not accept/ moderately difficult to adapt/ Adapt quickly.

Rhythmicity: (By the age of 2-3 months, could you tell the time of the day during which it would be hungry, sleepy or wake? About what time and how often during the day did the baby have its bowel movements?): The routine was very and always variable/ moderately so/ predictable.

Approach withdrawal: (How did the baby behave with new events, such as first tub bath, offered new foods or looked after by an unfamiliar person?): Frightened/ do nothing/ seem to like it.

Threshold level: (How would you estimate the baby's sensitivity to noise, heat and cold, things he saw and tasted, and texture of clothing?): Low/ Moderate/ High.

Intensity of reaction: (How would the baby react when he was hungry/ didn't like something?): Hardly any reaction/ moderate reaction/ roaring with laughter, screaming with anger, crying loudly.

Quality of mood: (How would the mood usually be? How would you know that the baby liked or disliked something?) Always crying, angry, irritable/ moderately so/ always laughing, giggling, happy.

Distractibility: (If the child was sucking, would he stop if he heard a sound or would he continue sucking? If the baby was crying could you divert him easily and stop him from crying by holding him or giving him a toy?): Low distraction/ moderate/ highly distractible.

Persistence and attention span: (Would you say that the baby usually continued with an activity for a long time or only for a moment?): No persistence/ moderately so/ continued till he achieved the goal.

Identify attributes that appear extreme in their manifestations, and those, which seem clearly related to its current pattern of deviant behaviour. Enquire into the expression of these temperamental attributes at succeeding periods (age and stage of development).

The final step in assessing the child's temperament is the evaluation of its current temperament characteristics. The inquiry into the present behaviour, while attempting to cover all temperamental categories, should concentrate on those, which appear most pertinent to the present symptoms.

Physical examination: General appearance part of examination becomes a little more important. One should be careful in looking for any obvious or otherwise minor physical anomaly, shape of head (normal/ micro/ macro/ dolicho/ oxycephalic), facies (normal/ mongoloid/ gargolism/ round/ elongated). Also comment on other remarkable facial features.

Mental status examination: A formal mental status is possible in adolescent, whereas it becomes difficult or at times impossible to do a mental status in children (especially who are too young or those suffering from childhood disorders). Thus it becomes very important to attempt a proper *behavioural observation* in such cases. Following headings are used for behavioural observation of the child.

General Appearance and behavior: Physical appearance, appropriate to age, body built and size, dress and physical handicaps, if any.

Relationship capacity and sociability: Response to separation from parents, reaction to interview situation and relation with interviewer.

Spontaneous motility and activity level: Retard, hyperkinetic, quiet.

Motor behaviour: Stereotypes, tics, self-injurious behaviour.

Speech: Flow, form, level of development of speech, inappropriate use of speech.

Affective behaviour: Any evidence of anxiety, fear, depression, shyness, including child's attitude towards examiner.

Attitude towards family members, school and playmates: This should be enquired from informants.

Attention span and distractibility: Assign some simple task as drawing, writing; also from ability to engage in conversation.

Intellectual capacity: General information, calculation, comprehension of commands of varying complexity

Stated interests and content of thought: What is the child's evaluation of problems; any disturbance of thought.

Motivational insight: What is the child's knowledge of reasons for problems, desire for help, sense of own capacity for change.

Play Behaviour: Leave the child with toys in a room with observer; parent can be allowed to be with the child initially.

Substance use disorders

The major differences are enumerated below

History of presenting complains: Here the focus shifts to the substance/s that is/are being used by the patient. There can be other behavioural problems in the patient, which also should not be overlooked. For eliciting drug use, the basic structure of history taking is as follows

One should ask for age of onset, setting, amount, type, frequency, pattern of use, determinants for starting. What was the feeling after first intake, what did the patient think at that time about future intakes? What subsequently happened to intake pattern: progress including change in pattern, frequency and amount? One should also ask for any other substances used by the patient. All relevant details regarding those substances should be noted.

Now it is also equally important to know the current intake pattern; including average amount, maximum amount, last intake.

One by one criterion for dependence (for diagnosis of dependence, 3 out of 6 should be present in a given individual) should be asked for which includes

- Compulsion- there is a strong desire or sense of compulsion to take the substance.
- Difficulty in controlling- there is difficulty in controlling substance-taking behaviour in terms of its onset, termination, or levels of use.
- Withdrawal- a physiological withdrawal state when substance use has ceased or been reduced, as evidenced by the characteristic withdrawal syndrome for the substance or use of the same (or a closely related) substance with the intention of relieving or avoiding withdrawal symptoms.
- Tolerance- A need for markedly increased amounts of substance to achieve intoxication or desired effect or a markedly diminished effect with continued use of the same amount of substance.
- Neglect- there is progressive neglect of alternative pleasures or interests because of the substance use; there is increased amount of time necessary to obtain or take the substance or to recover from its effects.

- Awareness/ignorance about harm- there is ‘persisting with substance use’ despite clear evidence of overtly harmful physical or psychological consequences. Efforts should be made to determine that the user was actually, or could be expected to be, aware of the nature and extent of the harm.

Here, one should be aware about harmful use pattern of substance intake as well. This entity should be entertained when the intake pattern is not fulfilling the criteria for dependence, but still the intake is causing identifiable damage to health. The damage may be physical (as in cases of hepatitis from the self-administration of injected drugs) or mental (e.g. episodes of depressive disorder secondary to heavy consumption of alcohol).

Special mention should be made about intoxication, blackouts, binge drinking, fits and flashbacks associated with drug use.

History of abstinence should be asked for. Any periods of abstinence should be acknowledged and should include all relevant details- period, setting, duration, outcome determinants of abstinence and relapse etc.

Consequences of substance intake should be dealt with in following headings

- Psychological changes in the individual should be looked into. It should also be specified whether symptoms are present under the effect of substance or other wise; amounting to induced psychosis or not; or it is independent psychosis precipitated by substance use. Here, it is also important to know when to call the psychosis induced or independent. The guidelines of ‘an organic cause leading to psychosis’ are followed here (detailed elsewhere). ICD-10 DCR mentions following diagnostic criteria for substance induced psychotic disorder.
 - ✓ Onset of psychotic symptoms during or within two weeks of substance use
 - ✓ Persistence of the psychotic symptoms for more than 48 hours.
 - ✓ Duration of the disorder not exceeding six months.
- Physical changes include neurological, cardiovascular, musculoskeletal, gastrointestinal, respiratory, dermatological, endocrine, sexual consequences due to substance use. For example, use of alcohol will lead to neuropathies, cardiovascular morbidity, acid peptic disease or gastric ulcer, myopathies amongst other systemic manifestation. Chronic cigarette or bidi smoking will lead to chronic bronchitis or other respiratory ailments.
- Accidents under the effect of substance are a common presentation. Chances of accident increase many-folds when a person drives in acute intoxication states.
- Social effects of substance use are mainly
 - ✓ Marital (discord, separation, divorce, sexual relation, assault etc.)
 - ✓ Family (high noise level, strained IPRs, improper parenting, disownment)
- Problems at work, is also commonly seen in substance users. One should comment on absenteeism, suspension, dismissal, unemployment etc.)
- Financial consequences should be seen in terms of per day expenses on substance (as compared to income per day), methods of procuring, debts etc.
- Legal consequences such as any police or court case etc. should be enquired for.

Personality assessment: Personality assessment becomes an important issue in those who are substance users, especially those who are taking multiple substances. Going by the temperament variables, they might have a high novelty seeking and low harm avoidance. Chances of (multiple) substance use intake are more in individuals with dissocial personality or histrionic personality. Borderline personality disorder people can resort to substance intake to combat the periods of mood fluctuations. Even individuals with anxious avoidant personality may resort to substance use (especially alcohol or anxiolytics) to relief their anxiety and thus may end up becoming addicted.

Physical examination: Not to be over-emphasized, physical examination remains a very important thing whenever a patient is seeking a doctor for any problem. A few common physical findings are stained teeth in tobacco chewers, stains over palms in ganja or tobacco users, skin lesions in intravenous drug users; hepatomegaly or hepatic tenderness, engorged neck veins, abdominal tenderness, cerebellar signs in chronic alcoholics, wheeze or bronchial breath sounds in chronic smokers; signs of withdrawal such as tremors, sweating, autonomic sensitivity in alcohol withdrawal state; slurred speech, alcoholic smell, altered consciousness after acute alcohol intake, delirium like picture in case of delirium tremens.

Mental status examination: It should proceed like a general psychiatric case work-up. One should try to look for presence of co-morbid or induced psychotic disorder in patients with substance use disorder. Two entities that warrant a special mention here include

Stage of motivation: 6 stages have been identified (Prochaska & DiClemente).

- Pre-contemplation stage- Individuals who are not considering change in their problem behaviour are described as being in pre-contemplation.
- Contemplation stage- This stage entails the person's beginning to consider both the existence of a problem and the feasibility & costs of changing the problem behaviour.
- Determination stage- As the individual progresses, he or she moves on to the stage where the decision is made to take action and change.
- Action stage- Once the individual begins to modify the problem behaviour, he or she enters the action stage, which normally continues for 3-6 months.
- Maintenance stage- After successfully negotiating the action stage, the individual moves to this stage of sustained change.
- Relapse- If these efforts fail, a relapse occurs, and the individual begins another cycle.

Locus of control (a concept from Rotter's social learning theory): It refers to a person's belief as to whether control over valued reinforcements is internal or external to the person. It is the extent to which individuals believe they can control events affecting them.

- Internal locus of control: According to the individual, the reinforcements are the results of his actions or his enduring characteristics. Thus, he believes that he can control his life.
- External locus of control: According to the individual, the reinforcements are the results of forces outside of him. This person believes that his decisions and life are controlled by environmental factors (which he cannot influence) or by chance, luck or fate.

Epilepsy

Definitions: A few definitions should be known to start with (Ref: ILAE glossary)

Ictus: It is a sudden neurological occurrence such as stroke or an epileptic seizure

Convulsion: (Primarily a lay term): Episodes of excessive and abnormal muscle contractions, usually bilateral, which may be sustained or interrupted.

Epileptic seizure: Manifestations of epileptic (excessive and/or hyper-synchronous) usually self-limited activity of neurons in the brain.

Epileptic disorder: A chronic neurological condition characterised by recurrent epileptic seizures.

Epilepsies: Those conditions involving chronic recurrent epileptic seizures that can be considered epileptic disorders.

Focal (syn. Partial): A seizure whose initial semiology indicated or is consistent with initial activation of only part of one cerebral hemisphere (ILAE Glossary). Focal epileptic seizures are conceptualized as originating within networks limited to one hemisphere. They may be discretely localized or more widely distributed. Focal seizures may originate in subcortical structures. For each seizure type, ictal onset is consistent from one seizure to another, with preferential propagation patterns that can involve the contralateral hemisphere. In some cases, however, there is more than one network, and more than one seizure type, but each individual seizure type has a consistent site of onset (Berg et al., 2010). The clinical characteristics of the seizure reflect the part of brain affected, and a wide variety of symptoms may thus occur. Partial seizures are divided into three main categories (Duncan and Shorvon).

- Simple partial seizures (SPS), in which there is no alteration of consciousness
- Complex partial seizures (CPS), in which consciousness is altered or lost
- Partial seizures evolving to secondarily generalized seizures

Simple partial seizures: These are short lived, lasting a few seconds or so. Clinical features include focal signs (motor, sensory or psychic), sudden onset and cessation, no alteration of consciousness and no amnesia. These are due to focal cortical pathology and focal signs/symptoms reflect the anatomical origin of seizure and thus are useful in localizing the underlying pathology.

Complex partial seizures: These seizures have sudden onset and gradual recovery. Four remarkable components are (4As) aura, altered consciousness, amnesia after the attack and automatism. Altered consciousness is essential feature for CPS and that differentiates these seizures from SPS. Like SPS, CPS also have a focal cortical pathology and arise most commonly in temporal lobe, but can also occur in other lobes especially frontal.

Generalized (syn. Bilateral): A seizure whose initial semiology indicated or is consistent with more than minimal involvement of both cerebral hemispheres (ILAE Glossary, 2001). Consciousness is almost invariably impaired from the onset of the attack (owing to the

extensive cortical and subcortical involvement) (Duncan & Shorvon). Generalized epileptic seizures are conceptualized as originating at some point within, and rapidly engaging, bilaterally distributed networks. Such bilateral networks can include cortical and subcortical structures, but do not necessarily include the entire cortex. Although individual seizure onsets can appear localized, the location and lateralization are not consistent from one seizure to another. Generalized seizures can be asymmetric (Berg et al. 2010)

Absence seizures (typical): The seizure comprises an abrupt sudden loss of consciousness (the absence) and cessation of motor activity (e.g. speaking, eating, walking). There is no warning or aura; tone is usually preserved and falling does not occur. The patient is unaware, inaccessible and motionless. The eyes appear glazed, usually staring ahead. The attack ends as abruptly as it started and previous activity is resumed as if nothing had happened. There is no confusion, but the patient is often unaware that an attack has occurred.

Myoclonic: Sudden, brief (<100ms) involuntary single or multiple contraction(s) of muscle(s) or muscle group of variable topography (axial, proximal limb, distal)

Clonic: Myoclonus that is regularly repetitive, involves the same muscle groups, at a frequency of 2-3 cycles/sec, and is prolonged. Syn: rhythmic myoclonus.

Tonic: A sustained increase in muscle contraction lasting a few seconds to minutes.

Tonic-clonic: A sequence consisting of tonic followed by a clonic phase. Variants such as clonic-tonic-clonic may be seen.

Atonic: Sudden loss or diminution of muscle tone without apparent preceding myoclonic or tonic event lasting >1-2 seconds, involving head, trunk, jaw or limb musculature.

Dystonic: Sustained contractions of both agonist and antagonist muscles producing athetoid or twisting movements, which, when prolonged, may produce abnormal postures

Astatic: Loss of erect posture that results from an atonic, myoclonic or tonic mechanism. Syn: Drop attack.

Versive: A sustained, forced conjugate ocular, cephalic, and/or truncal rotation or lateral deviation from midline.

Jacksonian March: Traditional term indicating spread of clonic movements through contiguous body parts unilaterally.

History of epilepsy: Following points are kept in mind while taking history of a patient suffering from attack of fits.

Age of onset and total duration of epilepsy at the time of presentation should be asked.

Interval between first and second episode gives a rough idea of baseline frequency of seizure attacks

Frequency of seizures: Overall frequency of seizures as well as frequency of full blown and abortive attacks needs to be mentioned separately. Also mention any increase or decrease in frequency with reasons for the same.

Abortive attack (in respect to a full blown attack): A patient suffering from generalized tonic-clonic seizures (either primary or secondarily), may experience attacks, which might not be associated with loss of consciousness (unlike the full blown attacks) and may include attacks such as movements of certain body parts such as limbs (viz. partial seizures) or absence seizures; such attacks are termed as abortive attacks and history of such attacks should be adequately asked for.

Change in pattern of seizures since onset of illness, if any, all seizure types should be adequately explained.

Cluster attacks: Incidence of seizures within a given period (usually one or a few days) that exceeds the average incidence over a longer period for the patient. So any clustering of attacks currently or in past should be mentioned here.

Status attacks: A seizure that shows no clinical sign of arresting after a duration encompassing the great majority of seizures of that type in most patients or recurrent seizures without inter-ictal resumption of baseline central nervous system functions. So history of status attacks and their treatment should be mentioned.

Last attack: Here mention both abortive and full-blown attacks separately.

Time of attack: One should try to look for any diurnal pattern of seizure attacks, whether the attacks are nocturnal, day time, early in the morning, sleep related (if so, whether occur during falling asleep or during sleep or while getting up from sleep) or these attacks occur anytime in the day. Some particular seizure types are known to occur at a specific time in the day. For e.g. myoclonic attacks usually occur during sleep.

Precipitants (if any) for the seizure attack: Try to find whether any antecedent precipitates the attacks. If present, then try to find exact reason. Possible reasons can be: reactive seizures (in association with transient systemic perturbation such as fever, lack of sleep or emotional stress) catamenial seizures (occurring primarily or exclusively in any one phase of menstrual cycle), state dependent seizures (occurring primarily or exclusively in various stages of drowsiness, sleep or arousal), associated with emotional factors (such as anger, worry, fear, laughter, weeping, frustration, stress), toxic and metabolic causes (such as alcohol, drugs, hypoglycaemia, fatigue), and skipping of drug.

Reflex seizures: Objectively and consistently demonstrated to be evoked by a specific afferent stimulus or by the activity of the patient. Stimuli can be elementary (such as light flash, startle, monotone) or elaborate (such as symphony). Even activity can be elementary (a motor movement) or cognitive function (reading, playing chess) or both (reading aloud).

Try to find any factor that prevents or aborts an attack. If present, elaborate

Prodrome: Prodrome is a preictal phenomenon. It is a subjective or objective clinical alteration that heralds the onset of an epileptic seizure but does not form a part of it. It may range from minutes to hours. This may include headache, irritability, lack of interest, cheerfulness, mood swings, impaired attention and concentration, forgetfulness, change in appetite, abdominal discomfort, change in behaviour. Describe the symptoms in detail.

Aura: Aura is defined as a subjective ictal phenomenon that in a given patient may precede an observable seizure (ILAE). According to Duncan and Shorvon, aura is simple partial seizure and can take any of the forms of SPS (motor, sensory or psychic). It is usually short lived, lasting a few seconds or so, although in rare cases a prolonged aura persists for minutes, hours or even days. Patients often describe the same features occurring in isolation as self-limited simple partial seizures. The occurrence of an aura prior to complex partial seizures may be noticed by an alert witness, but not subsequently recalled by the patient.

Make a mention of type of aura, if present and describe in detail. Possible types of aura are enumerated below

Focal motor: Movement of different muscle groups can be seen. Specify the part of body that shows motor activity. It can be eyelids (unilateral or bilateral), angle of mouth (specify which side), thumb (specify side), finger (specify which finger and which side), toe (specify which toe and which side), versive movements (specify side) or jacksonian march of events

Somato-sensory: It can present as numbness, tingling or paraesthesia of different body parts. Specify exact area where these sensations are felt during aura.

Visual: It can present as sensory distortions such as micropsia, macropsia, megalopsia or as sensory deceptions such as illusions, visual hallucinations (elementary or formed). It can also be seen as complex memory processes such as déjà vu, jamais-vu; as depersonalization-derealization or even as diplopia or blurred vision.

Auditory: Again it can present as sensory distortion such as change in intensity of sound (louder or fainter), deceptions such as auditory hallucinations (elementary or formed)

Olfactory: Hallucination in form of pleasant or unpleasant smell

Gustatory: Hallucination in form of pleasant/sweet or unpleasant/bitter taste

Vertiginous: It can be giddiness (objective space) or dizziness (subjective space)

Speech: There can be arrest of speech, vocalization or dysphasia

Disturbance of awareness: It can present with blank staring look, impaired (partial/ delayed/ irrelevant/ no) responsiveness to stimuli or a dreamy state

Automatism: A more or less coordinated, repetitive, motor activity usually occurring when cognition is impaired and for which the subject is usually amnesic afterwards. This often resembles a voluntary movement and may consist of inappropriate continuation of ongoing preictal motor activity. There can occur various bodily movements such as masticatory

movements, lip smacking, swallowing, tapping (finger/foot), fumbling, rotatory (clockwise/anticlockwise), volvular (clockwise/anticlockwise), bending body, stiffening of limbs, posturing or cursive movements.

Emotions: Various emotions can be seen as a part of aura. To mention a few, it can be fear without any reason, running to catch hold of some person/object, euphoria or laughter (seen in gelastic seizures).

Pain: Pain can be associated with seizures and can be cephalic pain (holocranial, bifrontal, bioccipital, right or left hemicranial), Unilateral ictal pain (either sided arm, leg, face or trunk), abdominal pain

Postural: Slow loss of posture or adoption of a posture that may be bilaterally symmetrical or asymmetrical (such as fencing posture) can be associated with a seizure attack

Epigastric aura: It can take the form of abdominal discomfort such as emptiness, tightness, churning, butterflies, sense of ball of gas rising up, borborygmi, belching, flatulence, nausea, vicarious or insatiable appetite

Autonomic aura: It can be seen in form of flushing or blanching, pallor of face, sweating (can be profuse or focal), palpitation, tachypnea, sub-sternal distress, chocking, feeling of temperature sensation (warmth/cold), pilo-erection or pupillary dilatation.

Cognitive aura: It can be seen as sudden difficulty in thinking coherently, evocation of thoughts, thought block, forced thinking, distortion of time sense, epileptic fugue, twilight state.

Autoscopy: It can be ictal or post ictal and is termed as Doppelganger phenomenon. If present, then examiner should enquire about time of occurrence, whether entire body or parts seen, associated movements, location in space, whether coloured or transparent and about emotional reaction to the process of autoscopy.

Description of ictal phenomenon: After elaboration of aura (if present), examiner should give a full description of the ictal attack. Emphasis should be on how a typical attack starts and progresses. One should also explain average duration of the attack.

Semiology is defined as branch of linguistics concerned with signs and symptoms

If properly delineated, this helps to identify whether the attack is focal or generalized, if focal, then whether simple or complex.

Post-ictal phenomenon: A transient clinical abnormality of central nervous system function that appears or becomes accentuated when clinical signs of the ictus have ended. Make a comment on whether post-ictal phenomenon is present or not. If present, then what usually occurs during the post-ictal period and how long it lasts and how does it recover. Possible phenomenon include headache, bodyache, dullness, lethargy, excessive sleep, sleeplessness, confusion, excitement, irritability, mood changes (such as euphoria or depression), automatic

behaviour, dysphagia, paraesthesia, amnesia, anorexia, nausea, vomiting, fever, abnormal behaviour, Todd's palsy or post ictal psychosis.

After explaining in detail about a typical attack (or all types of attacks, if more than one type of attacks occur in a given patient), one should make a mention about effects of illness on the patient such as effect on daily activity, role functioning (such as school/vocational) etc. Also see if any behavioural problems are present amounting to post-ictal or inter-ictal psychosis.

Is there any personality change due to illness: Associated personality change can be varied such as stubbornness, stickiness, circumstantiality, perseveration, retardation, decreases general interest, humourlessness, hypergraphia etc.

Are there any changes in mental function due to illness: Look for any changes in attention and concentration, comprehension, memory, reasoning and judgement, intelligence, learning, scholastic or occupational performance.

Injuries associated with seizure attacks: Make a comment whether injuries occur in the patient, if so, then how often? What is type of injury that usually occurs? It may be minor soft tissue injury, tongue/cheek bite, head injury (contusion/laceration), fracture (facial bone/ mandible/ vertebrae/ ribs/ scapula/ clavicle/ skull bones/ teeth loss/ any other site), joint dislocation (shoulder/ hip/ other joints), sprain (ankle/ wrist/ elbow/ knee/ other joints). There may even be history of burns associated with seizure attack which may be first degree or even second or third degree. The injury may or may not have received hospitalization for treatment. Also enquire, where the injury was acquired: at work place, at home or while walking or driving on road.

Treatment history: treatment history proceeds as in a normal case work-up. One should be acquaint with the possible side effects of anti-epileptic drugs.

Past history: Special emphasis needs to be given on presence or absence of birth trauma or asphyxia, febrile convulsions in early childhood, head injury or CNS infections anytime prior to onset of seizures, any other insult to the brain such as chronic headache or stroke.

Family history: History of epilepsy, psychiatry disorders in family need to be asked for.

Personal history: Dietary habits such as such use of pork or food contaminated with animal excreta need to be looked into. Also ask for use of substance of abuse such as alcohol or opioid and any relation of these substances to seizure attacks.

Physical examination: A careful examination in such patients should include fundus examination using ophthalmoscope, plantar reflexes. Body weight should be recorded invariably.

Investigation reports: Any report available should thoroughly seen, both for the accuracy and the suggested findings. Investigations that may be of help include EEG, CT scan, MRI, ELISA (for neuro-cysticercosis and TB), psychological; testing (such as neuropsychological assessment)

Diagnosis: Here one should be verse with multiaxial diagnosis system recommended by ILAE and should diagnose a case those multiple axes.

Multi-axial diagnosis: There is a proposed diagnostic scheme for people with epileptic seizures and with epilepsy (Engel, 2001). That diagnostic scheme is divided into five parts, or Axes which are as follows

Axis 1: Ictal phenomenology—from the Glossary of Descriptive Ictal Terminology, can be used to describe ictal events with any degree of detail needed.

Asix 2: Seizure type: from the List of Epileptic Seizures. Localization within the brain and precipitating stimuli for reflex seizures should be specified when appropriate.

Axis 3: Syndrome: from the List of Epilepsy Syndromes, with the understanding that a syndromic diagnosis may not always be possible.

Axis 4: Etiology: from a Classification of Diseases Frequently Associated with Epileptic Seizures or Epilepsy Syndromes when possible, genetic defects, or specific pathological substrates for symptomatic focal epilepsies.

Axis 5: Impairment: this optional, but often useful, additional diagnostic parameter can be derived from an impairment classification adapted from the WHO ICIDH-2.

ILAE has proposed several classifications from time to time. The ones that are accepted are:

1. Classification of seizures, 1981
2. Classification of epilepsies and epilepsy syndromes, 1989

Several other classifications and amendments have been proposed subsequently, but none has been accepted officially.

Those classifications, which are worth knowing, are mentioned in appendices.

Appendices

Appendix 1

What all questions one should ask if a particular complain is put forward?

A method can't be taught to decipher each and every symptom a patient or his informants can present with. Here are a few question one can ask in given complains. Further expertise is acquired over period of time and with seeing more and more patients

Aggressive behaviour: From when/ on what matters/ in which situations/ with whom (known or unknown or anyone)/ provoked or unprovoked/ if provoked, whether on trivial issues or major issues/ planned or not/ associated with grievous injuries to others or self/ any police or court case thereof/ reaction after the behaviour/ any feelings of guilt or remorse/ any other related information/ progression of behaviour

Suspiciousness: From when/ how did it started/ how it was notices/ usually on whom (family members or spouse or relatives or neighbors or work mates or strangers or anyone or everyone)/ on which matters/ how exactly is the behaviour, give examples/ does it lead to verbal or physical altercation/ progression of behaviour

Increased talk: From when/ how did it started/ understandable or not/ with known or strangers or anyone or everyone/ talks how much in whole day/ gets tired of talking or not/ content of talks/ whether says one thing again and again or talks on different issues/ associated with tall claims/ whether stoppable or not/ what was behaviour used to when he/she was interrupted/ sings songs (even if no-one is there to listen)/ progression of behaviour

Demanding behaviour: From when/ from whom (any specific individual or family members or strangers or anyone or everyone)/ if demands are not fulfilled then/ associated with excessive spending or spree buying/ distributes to friends or stangers.

Muttering: From when/ how did it start/ what was noticed in initial days/ what does he/she says when asked about it/ whether audible or not/ whether it looks like as if he/she is talking to someone/ associated with smiling to self- softly or loudly/ whether makes hand gestures along/ whether associated with sudden anger outbursts and speaking out abuses/ whether he/she tells what he/she is muttering/ what time in the day is it more/ does it occurs in night also/ whether he/she is able to sleep/ progression of behaviour

Wandering: From when/ how did it start/ where he/she used to wander initially and subsequently/ how frequently he wanders around: daily or on some days/ what time in the day does he/she sets out/ whether comes back on own or has to be traced and brought back/ what does he/she do when outside/ does he/she interacts with people around when he/she roams/ collects unwanted or dirty things or garbage/ what does he/she says when enquired about this behaviour/ progression of behaviour.

Long standing postures: From when/ how did it start/ whether stands also in sunshine or rain till he/she is not removed by others from there/ how frequently does it happen/ for how long

in one instance/ whether stands like an idol or does some action/ what he/she says when asked about it

Poor interaction: From when/ how did it start/ what was noticed in initial days/ whether stays away from people/ how he/she behaves when someone else tries to interact/ whether gets irritated/ maintains eye contact with talking or not/ whether he/she expresses will to interact/ what is the behaviour in times of festival or get-togethers or times of sorrow/ how does he/she behaves with guests at home/ progression of behaviour

Repetitive acts: From when/ how did it start/ what was noticed in initial days/ how many times do you do an act/ for how much time/ whether able to stop or not/ associated with any thoughts/images/impulses (probe for obsessions, which usually precede the acts)/ how much time is spent in cleaning/ any checking behaviour/ any difficulty in deciding/ progression of behaviour

Low mood: From when/ how did it start/ how it was in initial days/ what is the thought in the mind/ what is future like/ any ideas of hopelessness/helplessness/worthlessness/ any suicidal thoughts/acts/ how is self confidence/ progression from time of onset

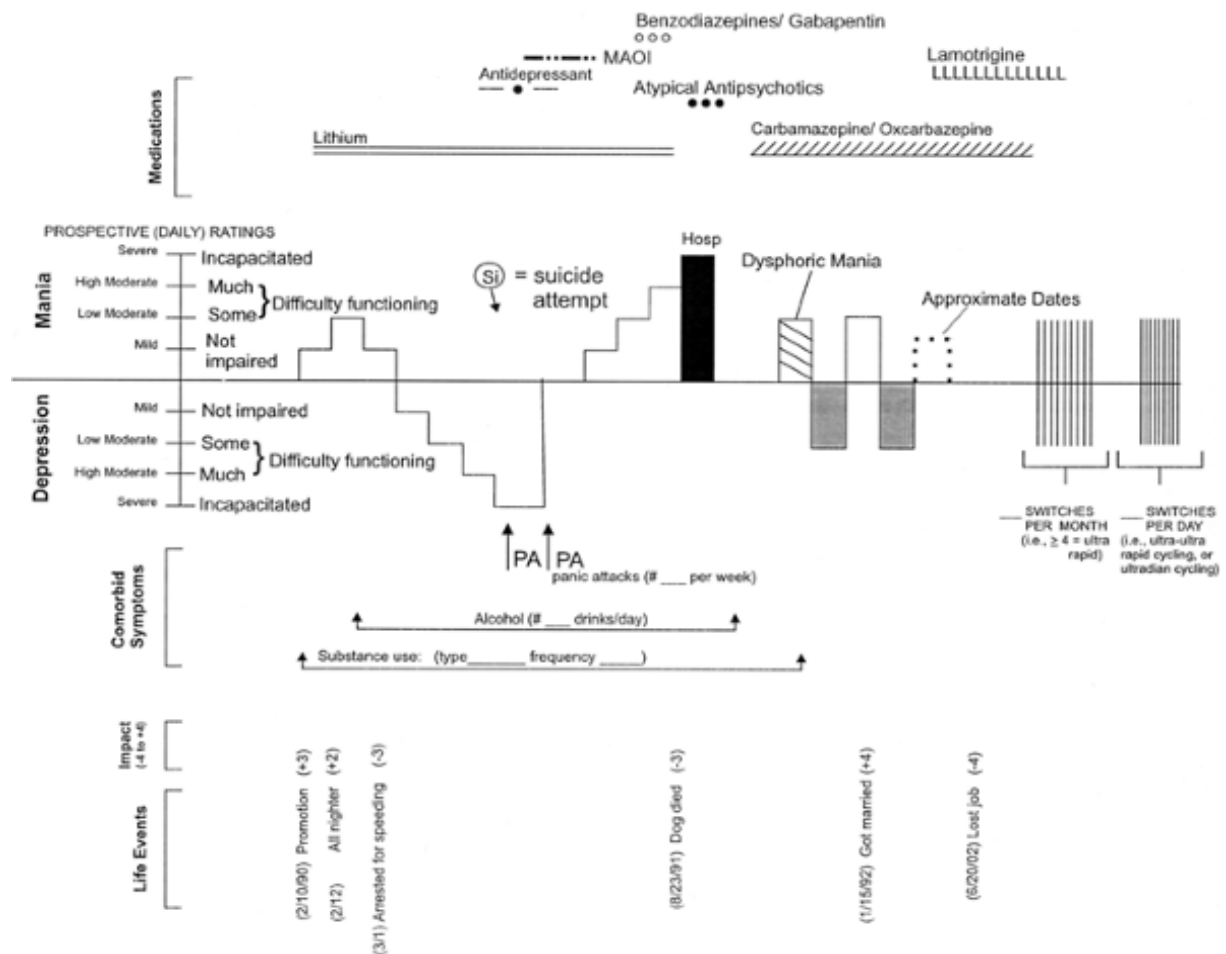
Sleep disturbance: From when/ duration of sleep before and now (more or less than usual)/ if less then whether problem in falling sleep or maintaining or gets up earlier than before/ at what time you go to bed/ at what time you actually fall asleep/ when do not fall asleep what do you do/ whether there are breaks in-between/ if breaks, what do you do/ at what time u get up/ whether sleep is refreshing/ how to you feel after getting up/ whether you have to take medicine/ dreams (if any), comment.

Appendix 2

Mood graph & Illness graph

Mood graph (Ref. CTP 9th Edition)

It is suggested that a rough graphic depiction of illness patterns be done as part of the initial work up and be the primary mode of recording a patient's history, even preferable to an extensive written account. In this way, the patient and the mental health professional are immediately and systematically focused on the longitudinal course of the illness and its variation over time, rather than being sidetracked by focus only on acute symptoms and their improvement. The graphic approach and its associated temporal landmarks can also facilitate recall of important events, dates, and even entire prior episodes that would otherwise be obscured or forgotten, as well as psychosocial precipitants. In this fashion, the mood chart may facilitate the formulation and institution of appropriate psychotherapeutic interventions as well. This chart can be further prospectively and longitudinally, updated at each patient visit.



Illness graph: On the similar lines, in all other psychiatric illness, be it psychosis or neurosis and stress related illness, substance use disorders, personality disorder or childhood disorders, the evolution of symptoms can be drawn on a graph. The lines on graph would represent the fluctuations in various symptoms over the period of time. X- axis would represent time frame and Y-axis would represent severity of a given symptom. Any other important information that has some significance in the course of illness would be presented in time frame. Examples include:

- Any stressor or life event (such as loss of job, death of a relative etc.) in course of illness
- Use of medications and their effect on the course of symptoms
- Event such as suicidal attempt, self harm

This way, it would be easy to find out the evolution of various symptoms, role (if any) of various factors such as precipitating factors, perpetuating factors, limiting or modifying factors. Treatment received would be incorporated in the graph itself and would help in determining future treatment in the given case.

Appendix 3

Cognitive function assessment (Ref. Strub & Black)

Before starting to assess cognitive functions, one should keep in mind the profile of the patient. When attempted as a part of overall mental status examination, these variables are already entered in socio-demographic profile of the patient. So one should be verse with patient's name, address, age, sex, education (highest level, failures or honors, age at completion) and occupation.

Also make a remark on any neurological or neurosurgical diagnosis such as hemiplegia, hemianopia etc., whether recovered or not, any deficits if present). Any investigation reports available such be entered such as EEG, MRI, CT Scan, Angiogram

Detailed examination will be covered under following headings

I. Behavioural Observation:

- a. History of behaviour change, memory difficulties, bizarre behaviour, change in work habits, and the like
- b. Physical Appearance:
- c. Emotional Status:
- d. Frontal Lobe Test Results:
- e. Denial or Neglect

II. Level of Consciousness: It is advisable to check consciousness by applying Glasgow Coma scale. Patient is given a score from 3-15 on a 15 point scale and is assessed for

- a. Rate: Alert/Lethargic/Stupor/Coma
- b. Describe the stimulus necessary to arouse patient, and record the response

Eye Opening Response

Spontaneous- open with blinking at baseline-	4
To verbal stimuli, command, speech-	3
To pain only (not applied to face)-	2
No response-	1

Verbal Response

Oriented-	5
Confused, but able to answer questions-	4
Inappropriate words-	3
Incomprehensible speech-	2
No response-	1

Motor Response

Obeys commands for movement-	6
Purposeful movement to painful stimulus-	5
Withdraws in response to pain-	4
Flexion in response to pain (decorticate posturing)-	3
Extension in response (decerebrate posturing)-	2
No response-	1

III. Orientation: Orientation to time, place, person and passage of time is looked for. Following questions provide a structure for assessment of orientation.

a. Time:

What is the day today?

What is the date?

Which month is this?

Which year is this?

What is the weather right now?

b. Place:

At which floor of the building we are sitting now?

Which building is this?

Which city are we in?

Which state are we in?

Which country are we in?

c. Person:

What is your name?

What is your age?

When is your birthday? (day, month, year)

Who has come with you?

Who can I be?

d. Sense of passage of time:

For how are we talking now?

For how many days are you in this hospital?

How much time it takes from your home to this place?

IV. Handedness

Which hand do you often use for combing your hair?

Which hand do you often use to open a tap?

Which hand do you often use to lift a bucket?

Which foot do you often use to kick something?

Take an X-Ray film, fold it and see through it (See which eye he uses)

Does anybody in your family is left-handed?

V. Attention

a. Observation of the patient during examination

b. Digit Repetition: First ask the patient that which language is he comfortable in? Then start with the procedure for testing Digit forward and digit backward. Digit Forward (DF): Instruct that I will speak out a few numbers, listen to then carefully and repeat after I finish. Make sure that the instructions are clearly understood by the patient. It is better to give an example before starting the test. Speak digits at a rate of one per second, and avoid associations between numbers. It is thus advisable to keep a list handy at the time of assessment. Stop once the patient fails twice at a given number of numerical. Last successful repetition is taken as digit forward span. Normal DF- 7 ± 2

Digit backward (DB): Instruct the patient that now I will speak similar strings of numbers. You will have repeat in opposite sequence. For example, if I say 3-7, you will say 7-3. Normal DB span is 5 ± 2 .

Test items for DF and DB are as follows (respectively)

3-7	9-2
2-4-9	1-7-4
8-5-2-7	5-2-9-7
2-9-6-8-3	6-3-8-5-1
5-7-1-9-4-6	2-9-4-7-3-8
8-1-5-9-3-6-2	4-1-9-2-7-5-1
3-9-8-2-5-1-4-7	8-5-3-9-1-6-2-7
7-2-8-5-4-6-7-3-9	2-1-9-7-3-5-8-4-6

- c. Sustained attention/Concentration: Ask the following questions

Speak the spellings of word "W-O-R-L-D" in opposite sequence. We can use "ह-म-स-फ-र" in Hindi.

How many days are there in a week? Can you tell the names in reverse sequence?

Do you know addition/subtraction? Can u subtract 7 from 100 and then again subtract 7 from the remainder. Keep on doing that till I stop you. (Stop after 5 subtractions).

- d. Vigilance: Repeat letters at a rate of one per second. Instruct as follows:

I will speak some alphabets in English/Hindi. Whenever you hear the alphabet "A"/ "अ", kindly indicate by tapping on the table.

L-T-P-E-A-O-A-I-C-T-D-A-L-A-A-A-N-I-A-B-F-S-A-M-R-Z-E-O-A-D-P-A-K-L-A-U-C-J-T-O-E-A-B-A-A-Z-Y-F-M-U-S-A-H-E-V-A-A-R-A-T

ल-त-प-ए-अ-औ-अ-ई-च-त-द-अ-ल-अ-अ-अ-न-ई-अ-ब-फ-स-अ-म-र-ज-ए-औ-अ-ड-प-अ-क-ल-अ-उ-च-ज-त-ओ-इ-अ-ब-अ-अ-ट-य-फ-म-ह-स-अ-ह-ख-व-अ-अ-र-अ-त

Make a comment on errors of omission and errors of commission

VI. Language:

- a. Spontaneous Speech: Describe, including fluency, articulation, and presence of paraphasias.

Can you tell something about your village?

- b. Verbal fluency:

Animal naming test: Normal- $18-22 \pm 5-7$

I know that you must be aware of names of animals. Kindly tell me names of some animals, as many as you can in a period of 1 minute.

FAS test (फ, प, म test): Normal- 36 to 60. Minimum 12 words per alphabet

I would give you three alphabets. With each alphabet, try to tell as many words as you can in a span of 1 minute. Take care not to tell names of persons or places. (One by one give three alphabets- F, A, S or फ, प, म in Hindi)

c. Comprehension:

Patient's response to pointing commands: Ask the patient to point to one, two, three, then four room objects or body parts in sequence. Continue to test until consistent failure occurs. Normal person succeeds in pointing to 4 or more objects.

I would name certain things in this room. Kindly point towards them and let me know:

Pen (Look for response)

Chair and then fan (look for response)

Window then table and then your nose (look for response)

Door then book then your hand and then watch (look for response)

Patient's response to yes-no questions: Ask at least 7 questions. Correct answers should alternate between "yes" and "no" randomly.

Whatever, I will ask, just reply in yes or no:

Is it raining today?

Is Mr. Manmohan Singh still the Prime Minister of India?

Is this a hotel?

Does sun rise in the east?

Do we use fan in winter?

Are there 7 days in a week?

Do we eat medicine during illness?

d. Repetition: Tell the patient to repeat words or phrases. Look for paraphasias, grammatical errors, omissions and additions. Normal people can repeat sentences of 19 or more syllables

I would say some words or phrases. Kindly repeat after me:

Ball

Help

Airplane

Hospital

Mississippi River

The little boy went home.

We all went over there together.

The old car wouldn't start on Tuesday morning.

The fat short boy dropped the china vase.

Each fight readied the boxer for the championship bout.

- e. Naming & Word Finding: Tell the patient to name the simple colors and objects. Total 20 items; a normal individual will know all objects except parts of object, for which average score is 4.5 (± 0.8)

I would show you certain colors or objects. Kindly tell me the name.

I would start with the colors first. So you need to tell me which color is this. (Point towards five different colors around viz. Red, Blue, Yellow, Pink, Purple).

Now I would show you body parts and you tell what this body part is? (point towards five different body parts viz. Eye, Leg, Teeth, Thumb, Knuckles)

Now tell about these clothes or objects. (Point towards five clothing and room objects viz. Door, Watch, Shoe, Shirt, Ceiling)

Now you will need to tell me the exact part of the object that I point to. (Point towards five parts of objects viz. Watch stem (winder), Coat lapel, Watch crystal, Sole of shoe, Buckle of belt)

- f. Reading: It is important to know patient's educational background before testing reading. Both reading comprehension and reading aloud ability is tested. Describe level of adequacy (words, phrases, sentences, paragraphs) and note types of errors

I will give you a newspaper. Can you just read aloud this news for me? (Subsequently ask questions from what is written in paper so that the comprehension is checked)

- g. Writing: Describe level of adequacy and note types of errors

I would dictate certain words to you. Can you write those words on this piece of paper?

Can you just write down names of a few objects or body parts?

Can you write a short sentence telling about how the weather is?

- h. Spelling: Describe performance to dictation and note errors

I would say certain words. You will have to tell me the spelling of those words. (Spell words in increasing difficulty)

VII. Memory

- a. Immediate Recall (Short term memory): Digit Repetition (DF, DB)

- b. Recent memory: Orientation as already tested also checks for recent memory. To add to that, can ask

What you ate in breakfast today?

With whom have you come here?

- c. Remote Memory: checked by following two kind of information

Personal Information:

Where were you born?
Where did you go to school?
When did you attend school?
Where is your school located?
What do you do for work?
When did you work at those places?
What is your wife's name?
How old is your wife?
What was your mother's maiden name?

Historic Facts:

When did India get its Independence?
Who was the first prime minister of India?
Ranchi is in which state?
How many brothers Pandavas were?

d. New Learning Ability:

Four Unrelated Words: Normal patients under 60 years would remember all 4 words at 10-minute time. There are significant variations with age. Score must be interpreted in light of overall history and performance on other subtests.

I am going to tell you four words that I would like you to remember. (Have the patient repeat the 4 words after they are initially presented, correct any error made on immediate repetition). I will ask you these words after some time. (Continue with examination, and at intervals of 5, 10, and 30 minutes, ask the patient to recall the words. Use semantic ("one word is a color") or phonemic ("one word begins with B") cues if he or she is unable to recall the word spontaneously on any trial. Describe types of cues if necessary). Use one of these three lists:

<i>Brown</i>	<i>Fun</i>	<i>Grape</i>
<i>Honesty</i>	<i>Carrot</i>	<i>Stockings</i>
<i>Tulip</i>	<i>Ankle</i>	<i>Happiness</i>
<i>Eyedropper</i>	<i>Loyalty</i>	<i>Toothbrush</i>

Verbal story for Immediate Recall: Look for number of correct memories and describe confabulation if present. Story contains 26 relatively separate ideas. Normal individuals under 70 years should be expected to produce at least 10 of these items on immediate recall. In patients with good immediate recall, it may be useful to ask for another recall after 30 minutes, which is a sensitive method to test short-term verbal recall.

I am going to read you a short paragraph. Listen carefully, because when I finish reading, I want you to tell me everything that I told you. (Read the story slowly and carefully, but without pausing at the slash marks).

“It was July / and the Gupta family / had packed up / their four children / in the maruti car / and were off / on vacation. They were taking / their yearly trip / to the beach / at Goa. This year / they were making / a special / one-day stop / at the Durga temple / in Pune. After a long day’s drive / they arrived / at the hotel / only to discover / that in their excitement / they had left / the twins / and their suitcases / in the front yard.”

Now tell me everything that you can remember of the story. Start at the beginning and tell me all that happened. (Record the number of correct memories i.e. information within the slashes).

Visual Memory (Hidden Objects): Look for number of hidden objects found, number of hidden objects named but not found and number of locations indicated but objects not named. Normal people under 60 years should find 4-5 of hidden objects after a 5-minute delay without difficulty.

I am going to hide some objects around the office; desk or bed and I want you to remember where they are. (Hide five common, small, easily recognizable objects (keys, pen, comb, coin and reflex hammer) in various areas in the patient’s sight. Name the items when they are being hidden. After a delay of five minutes, ask the patient to find the objects. Ask patient to name those items he or she is unable to find).

Paired Associated Learning: Normal-3 out of 4 in 1st trial and 4/4 in 2nd trial

I am going to read a list of words two at a time (in pairs). You listen to them carefully. You will be expected to remember the words that go together. Then once I say the first word of the pair, you will be expected to tell the second word. (After reading the first list, test for recall by presenting the first recall list. Give the first word of the pair and ask for the word that went with it. Correct the incorrect responses and proceed to the next pair. After the first recall has been completed, allow a 10 second delay and continue with the second presentation and recall lists).

Presentation lists:

1st list

Weather-Box

High-Low

House-Income

Book-Page

2nd list

House-Income

Weather-Box

Book-Page

High-Low

Recall lists:

1st list

House-

High-

Weather-

Book

2nd list

High-

House-

Book-

Weather-

VIII. Constructional ability

- a. Reproduction Drawings: Ask the patient to copy the drawing in the space provided. (Examiner can use pre-drawn figures or draw and show to the patient to copy). Scores of 0-4 for poor-excellent drawings. Mention total score.

Vertical Diamond

Two-dimensional cross

Three-dimensional cube

Three-dimensional pipe

Triangle within a triangle

- b. Drawing to Command: Scoring same as reproduction drawing. Mention total score.

I would now like you to draw some simple pictures on this paper. Draw each picture as well as you can:

Clock with all numbers and hands on it

Daisy in flowerpot

House in perspective

- c. Block Designs: The use of this test requires four multi-colour cubes (used in WAIS-III or Kohs block test) and four stimulus designs. Because of need of extra equipment, this test is considered ancillary. Details can be learnt from block test instructions used in intelligence testing.

IX. Higher Cognitive Functions:

- a. Fund of Information: The questions are asked in order of increasing difficulty. Examiner should continue to ask questions until the test is complete or until three consecutive failures. Average patient should answer 6 out of these 10 questions. Results vary according to educational status.

How many weeks are there in a year?

Why do people have lungs?

What is the capital of Bihar?

Where is Bangladesh?

How far is it from Ranchi to Kolkata?

Why are light colored clothes cooler in the summer than dark colored clothes?

Name four Prime Ministers since Independence.

What causes rust?

Who wrote National Anthem?

How many members are there in Lok Sabha?

- b. Calculations: Each response should be scored as correct or incorrect. Thus patient's overall level of calculation ability and area of adequacy and deficit can be determined.

We are going to do some arithmetic examples. Try your best on each

Verbal rote examples:

Addition: (4 + 6)

Subtraction: (8 – 5)

Multiplication: (2 × 8)

Division: (56 ÷ 8)

Verbal complex examples: 20 seconds for each response

Addition: (14 + 17)

Subtraction: (43 – 38)

Multiplication: (21 × 5)

Division: (128 ÷ 8)

Written complex examples: 30 seconds for each response

Addition: 108+79

Subtraction: 605 - 86

Multiplication: 108 ×36

Division: 559 ÷ 43

c. Proverb Interpretation:

Do you know of any proverbs that are used in your place? Can you tell a few of them and their meaning?

If patient does not spontaneously comes up with proverbs of his choice then ask commonly used proverbs in his culture. Continue asking till two consecutive failures. Score of 0-2 is given after assessing whether the response is concrete/semi-abstract/abstract. Total score of less than 5 is suspicious.

Don't cry over spilled milk.

Rome wasn't built in a day

A drowning man will clutch at a straw.

A golden hammer breaks an iron door.

The hot coal burns, the cold one blackens.

d. Similarities: Score of 0-2 for incorrect-correct responses.

I will tell you a pair of things. Can you just tell me the similarity between the two?

Turnip & Cauliflower:

Car & Airplane

Desk & Bookcase:

Poem & Novel:

Horse & Apple:

X. Related Cortical Functions:

- a. Ideomotor Apraxia: Ask the patient to carry out motor acts to commands. Indicate if imitation or use of the real object was necessary to facilitate performance.

Blow out a match.

Drink through a straw.

Lick crumbs off your lips.

Salute.

Comb your hair.

Flip a coin.

Wear a chappal

Kick a football

Mop the floor

Bow for prayer

Hit a ball with bat

- b. Ideational Apraxia: Ask the patient to perform on the following motor tasks

Letter-envelope-stamp

Candle-holder-match

Toothpaste-toothbrush

- c. Right-Left Disorientation:

Identification on self:

Show me your right foot

Show me your left hand

Crossed commands on self:

With your right hand touch your left shoulder

With your left hand touch your right ear

Identification on examiner:

Point to my left knee

Point to my right elbow

Crossed commands on examiner:

With your right hand point to my left eye

With your left hand point to my left foot

- d. Finger Agnosia: Describe the adequacy of the patient's nonverbal or verbal performance.

- e. Visual Agnosia: Describe any deficits in visual identification of objects, naming of objects whose use can be demonstrated, color naming, and facial recognition.

- f. Astereognosis: Describe deficits in either of left hand or right hand

- g. Geographic Disorientation:

Describe evidence of disorientation obtained from history:

Map localization: Describe patient's ability to localize well-known cities on a map.

Orientation of self in hospital: Describe patient's ability to orient self within the hospital environment.

- h. Denial & Neglect: If present, describe patient's response.
Does patient frankly deny his illness
Is there evidence of unilateral neglect
Is there evidence of unilateral neglect on drawings?

i. Frontal Lobe Tests:

Alternate sequences-Visual pattern completion test: The patient is given visual patterns to reproduce. Patients with intact motor and sensory systems should be able to complete these sequences without error. A loss of sequence or perseveration in the reproduction of sequences suggests a loss of the ability to move from one motor movement to another and an ability to shift sets efficiently.

See the following drawing and copy it



Alternating motor pattern tests: This test consists of a series of changes in hand position (adapted from Luria), which is first demonstrated to the patient, and then he is asked to perform.

Fist-Palm- Side test: *Like I just demonstrated, hit the top of the table repeatedly, first with a fist, then with an open palm, and then with the side of the hand. Perform it until I ask you to stop. (Performance for 15-20 seconds should suffice to assess the adequacy of these alternating movements)*

Fist- Ring test: *Now see, how I do this one. Extend your arm several times, first with the hand in a fist, and then with the thumb and forefinger opposed to form a ring. With successive extension of the arm, alternate between these two positions.*

Reciprocal Coordination test: *Place both hands on the table, one in a fist and one with fingers extended palm down. Then alternate the position of the two hands rapidly (simultaneously extending the fingers of one hand while making a fist with the other.*

Primitive reflexes: The frontal release signs should be looked for. These primitive reflexes include (briefly described hereunder):

- Glabellar reflex. Failure to extinguish eye blink response to gentle tapping to the center of the forehead right above the nose.
- Grasp reflex. Perhaps the most helpful frontal release sign, as it is fairly specific of frontal lobe injury, and has localizing value to the contralateral supplementary motor area located in the medial frontal

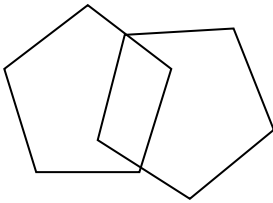
lobe. The grasp reflex occurs when the hand grasps onto an object (or examiner's finger). It is elicited by stroking the inside palm in a distal motion towards the base of the fingers. One may also stroke the proximal surface of the fingers (towards the palm). The grasp can be quite strong, allowing the person's torso to be lifted up from a lying position. Release may be voluntary or in some cases, takes considerable effort to release.

- Palmomenthal reflex. Ipsilateral contraction of the muscle of the chin (mentalis muscle) occurring to an unpleasant stimulus of the thenar eminence (body of the palm just proximal to the thumb). The ipsilateral corner of the mouth may also contract. The stimulus eliciting the reflex is started at the lower wrist and up the base of the thumb. The stimulus can be a tongue depressor or the handle of a reflex hammer.
- Root reflex. The turning of the patient's head ipsilateral to the side of the cheek that is lightly stroked. It is associated with the suck reflex in its adaptability for infants to breast feed.
- Snout reflex. The puckering of the lips to make a "snout" when the top lip is gently tapped (percussed). Typically, the Snout reflex can be elicited by gently tapping on the center of the upper lip when the lips are closed with your finger.
- Suck reflex. Sucking movements of the lips when the lips are generally stroked or touched. The sucking movement can be elicited by stroking the upper or lower corners of the mouth.

Appendix 4

Mini- Mental Status Examination (MMSE)

Instructions: Score one point for each correct response within each question or activity.

Parameter	Item	Score
Orientation (10)	What is the year	1
	What is the season	1
	What is the date	1
	What is the day	1
	What is the month	1
	What state are we in	1
	What country are we in	1
	What town/city are we in	1
	What building are we in	1
	Which floor are we on	1
Registration (3)	Name 3 unrelated objects clearly and slowly, then ask the patient to name all three of them (1 point for each correct item named). Repeat them until patient learns all of them, if possible	3
Attention and concentration (5)	Serial 100-7 (1 point for each correct 5 answer). Alternative: "Spell WORLD backwards." (D-L-R-O-W)	5
Recall (3)	Ask names of three objects told above. (1 point for each item)	3
Language (9)	Ask to identify a pen and a watch	2
	Repeat the sentence – no ifs and buts	1
	Ask to follow a three step command Take 3 the paper in your right hand, fold it in half, and put it on the floor."	3
	Read and obey "Close your Eyes"	1
	Write a meaningful sentence	1
	Please copy this picture. (All 10 angles must be present and two pentagons must intersect.) 	1
Total (30)		30

Ref: Folstein, M. F., Folstein, S. E. and McHugh, P. R. (1975). "Mini-mental state: A practical method for grading the cognitive state of patients for the clinician." Journal of Psychiatric Research, 12, 189-198.

Interpretation of the MMSE:

Method	Score	Interpretation
Single Cutoff	<24	Abnormal
Range	<21 >25	Increased odds of dementia Decreased odds of dementia
Education	21 <23 <24	Abnormal for 8th grade education Abnormal for high school education Abnormal for college education
Severity	24-30 18-23 0-17	No cognitive impairment Mild cognitive impairment Severe cognitive impairment

Interpretation of MMSE Scores:

Score	Degree of Impairment	Formal Psychometric Assessment	Day-to-Day Functioning
25-30	Questionably significant	If clinical signs of cognitive impairment are present, formal assessment of cognition may be valuable	May have clinically significant but mild deficits. Likely to affect only most demanding activities of daily living.
20-25	Mild	Formal assessment may be helpful to better determine pattern and extent of deficits	Significant effect. May require some supervision, support and assistance.
10-20	Moderate	Formal assessment may be helpful if there are specific clinical indications.	Clear impairment. May require 24-hour supervision
0-10	Severe	Patient not likely to be testable.	Marked impairment. Likely to require 24-hour supervision and assistance with ADL.

Appendix 5

Busch Francis Catatonia Rating scale

Bush-Francis Catatonia Rating Scale is a 23-item scale and measures the severity on these items on a scale of 0-3. The method described here is used to complete the 23-item Bush-Francis Catatonia Rating Scale (CRS) and the 14-item Catatonia Screening Instrument (CSI). Item definitions on the two scales are the same. In contrast to CRS, which measures the severity, the CSI measures only the presence or absence of the first 14 signs.

Ratings are to be made solely on the basis of observed behaviour during the examination with the exception of completion of the items for 'withdrawal' and autonomic abnormality', which may be based on directly observed behaviour and for chart documentation.

1. Excitement: Extreme hyperactivity, constant motor unrest which is apparently non-purposeful 0 = Absent 1 = Excessive motion 2 = Constant motion, hyperkinetic without rest periods 3 = Full blown catatonic excitement	2. Immobility/stupor: Extreme hypoactivity, immobile, minimally responsive to stimuli 0 = Absent 1 = Sits abnormally still, may interact briefly 2 = Virtually no interaction with external world 3 = Stuporous, non-reactive to painful stimuli
3. Mutism: Verbally unresponsive or minimally responsive 0 = Absent 1 = Unresponsive to majority of questions, incomprehensible whisper 2 = Speaks less than 20 words/5mins 3 = No speech	4. Staring: Fixed gaze, little or no visual scanning of environment, decreased blinking 0 = Absent 1 = Poor eye contact, repeatedly gazes less than 20 seconds between shifting of attention; decreased blinking 2 = Gaze held longer than 20 s, occasionally shifts attention 3 = Fixed gaze, non-reactive
5. Posturing/Catalepsy: Spontaneous maintenance of posture(s) including mundane (.g. setting or standing for long periods without reacting). 0 = Absent 1 = Less than 1 minute 2 = Greater than one minute, less than 15 minutes 3 = Bizarre posture, or mundane maintained more than 15 mins	6. Grimacing: Maintenance of odd facial expressions. 0 = Absent 1 = Less than 10 seconds 2 = Less than 1 minute 3 = Bizarre expression(s) or maintained more than 1 minute
7. Echopraxia/echolalia: Mimicking of examiner's movements/speech 0 = No mimicking of examiner's movements/speech 1 = Occasional 2 = Frequent 3 = Constant	8. Stereotypy: Repetitive, non-goal-directed motor activity (e.g. finger-play; repeatedly touching, patting or rubbing self); abnormality not inherent in act but in frequency. 0 = Absent 1 = Occasional 2 = Frequent 3 = Constant
9. Mannerisms: Odd, purposeful movements (hopping or walking tiptoe, saluting passers- by or exaggerated caricatures of mundane movements); abnormality inherent in act itself. 0 = Absent 1 = Occasional 2 = Frequent 3 = Constant	10. Verbigeration: Repetition of phrases or sentences (like a scratched record). 0 = Absent 1 = Occasional 2 = Frequent 3 = Constant
11. Rigidity: Maintenance of a rigid position despite efforts to be moved, exclude if cog- wheeling or tremor present. 0 = Absent 1 = Mild resistance 2 = Moderate 3 = Severe, cannot be repostured	12. Negativism: Apparently motiveless resistance to instructions or attempts to move/examine patient. Contrary behaviour, does exact opposite of instruction 0 = Absent 1 = Mild resistance and/or occasionally contrary 2 = Moderate resistance and/or frequently contrary 3 = Severe resistance and/or continually contrary
13. Waxy Flexibility: During reposturing of patient, patient offers initial resistance before allowing himself to be repositioned, similar to that of a bending candle. 0 = Absent 3 = Present	14. Withdrawal: Refusal to eat, drink and/or make eye contact. 0 = Absent 1 = Minimal PO intake/interaction for less than 1 day 2 = Minimal PO intake/interaction for more than 1 day 3 = No PO intake/interaction for 1 day or more

15. Impulsivity: Patient suddenly engages in inappropriate behaviour (e.g. runs down hallway, starts screaming or takes off clothes) without provocation. Afterwards can give no, or only a facile explanation. 0 = Absent 1 = Occasional 2 = Frequent 3 = Constant or not redirectable	16. Automatic obedience: Exaggerated cooperation with examiner's request or spontaneous continuation of movement requested. 0 = Absent 1 = Occasional 2 = Frequent 3 = Constant
17. Mitgehen: "Anglepoise lamp" arm raising in response to light pressure of finger, despite instruction to the contrary. 0 = Absent 3 = Present	18. Gegenhalten: Resistance to passive movement which is proportional to strength of the stimulus, appears automatic rather than willful. 0 = Absent 3 = Present
19. Ambitendency: Patient appears motorically "stuck" in indecisive, hesitant movement. 0 = Absent 3 = Present	20. Grasp reflex: Per neurological exam 0 = Absent 3 = Present
21. Perseveration: Repeatedly returns to same topic or persists with movement. 0 = Absent 3 = Present	22. Combativeness: Usually in an undirected manner, with no, or only a facile explanation afterwards. 0 = Absent 1 = Occasionally strikes out, low potential for injury 2 = Frequently strikes out, moderate potential for injury 3 = Serious danger to others
23. Autonomic abnormality: Circle: temperature, BP, pulse, respiratory rate, diaphoresis. 0 = Absent 1 = Abnormality of 1 parameter [excluding pre-existing HTN] 2 = Abnormality of two parameters 3 = Abnormality of three or more parameters	TOTAL: _____

As a general rule, only rate items, which are clearly present. If uncertain as to the presence of an item, rate the item as '0'.

	Procedure	Examines
1	Observe patient while trying to engage in a conversation	Activity level Abnormal movements Abnormal speech.
2	Examiner scratches head in exaggerated manner	Echopraxia
3	Examine arm for cogwheeling. Attempt to reposition, instructing patient to "keep your arm loose" - move arm with alternating lighter and heavier force.	Negativism Waxy flexibility Gegenhalten
4	Ask patient to extend arm. Place one finger beneath hand and try to raise slowly after stating, "Do NOT let me raise your arm".	Mitgehen
5	Extend hand stating "Do NOT shake my hand".	Ambitendence
6	Reach into pocket and state, "Stick out your tongue, I want to stick a pin in it".	Automatic obedience
7	Check for grasp reflex.	Grasp reflex
8	Check chart for reports of previous 24-hour period. In particular check for oral intake, vital signs, and any incidents.	
9	Attempt to observe patient indirectly, at least for a brief period, each day.	

Ref: Bush, G., Fink, M., Petrides, G., Dowling, F., and Francis, A. (1996). Catatonia. I. Rating scale and standardized examination. *Acta Psychiatrica Scandinavica*, 93, 129–136.

Appendix 6

Abnormal Involuntary Movement Scale (AIMS)

Scoring Procedure

Complete the examination procedure before making ratings.

For the movement ratings (the first three categories below), rate the highest severity observed.

0 = none/ No awareness

1 = minimal (may be extreme normal)/ Aware, No distress

2 = mild/ Aware, mild distress

3 = moderate/ Aware, moderate distress

4 = severe/ Aware, severe distress

According to the original AIMS instructions, one point is subtracted if movements are seen **only on activation**.

Facial and Oral Movements
1. Muscles of Facial Expression (eg: Movement of forehead, eyebrows, periorbital area, cheeks: including frowning, blinking, smiling, grimacing)
2. Lips and perioral area (e.g., puckering, pouting, smacking.)
3. Jaw, e.g., biting, clenching, chewing, mouth opening, lateral movement
4. Tongue. Rate only increase in movement both in & out of mouth, not inability to sustain movement
Extremity Movements
5. Upper (arms, wrists, hands, fingers). Include movements that are choreic (rapid, objectively purposeless, irregular, spontaneous) or athetoid (slow, irregular, complex, serpentine). Do not include tremor (repetitive, regular, rhythmic movements)
6. Lower (legs, knees, ankles, toes), (e.g., lateral knee movement, foot tapping, heel dropping, foot squirming, inversion and eversion of foot)
Trunk Movements
7. Neck, shoulders, hips, (e.g., rocking, twisting, squirming, pelvic gyrations. Include diaphragmatic movement)
Global Judgement
8. Severity of abnormal movements. Based on the highest single score on the above items
9. Incapacitation due to abnormal movements
10. Patient's awareness of abnormal movements. Rate only patient's report
Dental Status
11. Current problems with teeth and/or dentures (0= No, 1= Yes)
12. Does patient usually wear dentures (0= No, 1= Yes)
Total Score Obtained: Minimum Score: 0, Maximum Score: 42(40 for abnormal movements)

Reference: Guy W. ECDEU Assessment Manual for Psychopharmacology, revised ed. Washington, DC, US Department of Health, Education, and Welfare, 1976.

Examination Procedure

Either before or after completing the examination procedure, observe the patient unobtrusively at rest (e.g., in the waiting room).

The chair to be used in this examination should be a hard, firm one without arms.

1. Ask the patient whether there is anything in his or her mouth (such as gum or candy) and, if so, to remove it.
2. Ask about the **current** condition of the patient's teeth. Ask if he or she wears dentures. Ask whether teeth or dentures bother the patient **now**.
3. Ask whether the patient notices any movements in his or her mouth, face, hands, or feet. If yes, ask the patient to describe them and to indicate to what extent they **currently** bother the patient or interfere with activities.
4. Have the patient sit in chair with hands on knees, legs slightly apart, and feet flat on floor. (Look at the entire body for movements while the patient is in this position.)
5. Ask the patient to sit with hands hanging unsupported -- if male, between his legs, if female and wearing a dress, hanging over her knees. (Observe hands and other body areas).
6. Ask the patient to open his or her mouth. (Observe the tongue at rest within the mouth.) Do this twice.
7. Ask the patient to protrude his or her tongue. (Observe abnormalities of tongue movement.) Do this twice.
8. Ask the patient to tap his or her thumb with each finger as rapidly as possible for 10 to 15 seconds, first with right hand, then with left hand. (Observe facial and leg movements.)
9. Flex and extend the patient's left and right arms, one at a time.
10. Ask the patient to stand up. (Observe the patient in profile. Observe all body areas again, hips included.)
11. Ask the patient to extend both arms out in front, palms down. (Observe trunk, legs, and mouth.)
12. Have the patient walk a few paces, turn, and walk back to the chair. (Observe hands and gait.) Do this twice.

Appendix 7

Intoxication states for substances of abuse (Ref: ICD-10 DCR)

Alcohol

Dysfunctional behaviour, as evidenced by at least one of the following:

1. Disinhibition
2. Argumentativeness
3. Aggression
4. Lability of mood
5. Impaired attention
6. Impaired judgment
7. Interference with personal functioning.

At least one of the following signs:

1. Unsteady gait
2. Difficulty standing
3. Slurred speech
4. Nystagmus
5. Decreased level of consciousness (e.g. stupor, coma)
6. Flushed face
7. Conjunctival injection.

When severe, it may be accompanied by hypotension, hypothermia; depression of gag reflex.

Opioid

Dysfunctional behaviour as evidenced by at least one of the following

1. Apathy and sedation
2. Disinhibition
3. Psychomotor retardation
4. Impaired attention
5. Impaired judgement
6. Interference with personal functioning.

At least one of the following signs

1. Drowsiness
2. Slurred speech
3. Pupillary constriction (except in severe overdose when pupillary dilatation occurs)
4. Decreased level of consciousness (e.g. stupor, coma)

When severe, it may be accompanied by respiratory depression (and hypoxia), hypotension and hypothermia

Cannabis

Dysfunctional behaviour or perceptual disturbances that include at least one of these:

1. Euphoria and Disinhibition
2. Anxiety or agitation
3. Suspiciousness or paranoid ideation
4. Temporal slowing (a sense that time is passing very slowly, and/or the person is experiencing a rapid flow of ideas)
5. Impaired judgement
6. Impaired attention
7. Impaired reaction time
8. Auditory, visual or tactile illusions
9. Hallucinations with preserved orientation
10. Depersonalization
11. Derealization
12. Interference with personal functioning.

At least one of the following signs

1. Increased appetite
2. Dry mouth
3. Conjunctival injection
4. Tachycardia

Sedative hypnotic

Dysfunctional behaviour, as evidenced by at least one of the following:

1. Euphoria and Disinhibition
2. Apathy and sedation
3. Abusiveness or aggression
4. Lability of mood
5. Impaired attention
6. Anterograde amnesia
7. Impaired psychomotor performance
8. Interference with personal functioning.

At least one of the following signs

1. Unsteady gait
2. Difficulty standing
3. Slurred speech
4. Nystagmus
5. Decreased level of consciousness (e.g. stupor, coma)
6. Erythematous skin lesion or blisters.

When severe, it may be accompanied by hypotension, hypothermia; depression of gag reflex.

Cocaine or other stimulants

Dysfunctional behaviour or perceptual abnormalities, as evidenced by at least one of these:

1. Euphoria and sensation of increased energy
2. Hyper-vigilance
3. Grandiose beliefs or actions
4. Abusiveness or aggression
5. Argumentativeness
6. Lability of mood
7. Repetitive stereotyped behaviours
8. Auditory, visual or tactile illusions
9. Hallucinations usually with intact orientation
10. Paranoid ideation
11. Interference with personal functioning (most readily apparent from the interactions of the users, which range from extreme gregariousness to social withdrawal).

At least two of the following signs

1. Tachycardia (sometimes bradycardia)
2. Cardiac arrhythmias
3. Hypertension (sometimes hypotension)
4. Sweating and chills
5. Nausea or vomiting
6. Evidence of weight loss
7. Pupillary dilatation
8. Psychomotor agitation (sometimes retardation)
9. Muscular weakness
10. Chest pain
11. Convulsions

Hallucinogens

Dysfunctional behaviour or perceptual abnormalities, as evidenced by at least one of these

1. Anxiety and fearfulness
2. Auditory, visual or tactile illusions or hallucinations occurring in a state of full wakefulness and alertness
3. Depersonalization
4. Derealisation
5. Paranoid ideation
6. Ideas of reference
7. Lability of mood
8. Hyperactivity
9. Impulsive acts
10. Impaired attention
11. Interference with personal functioning.

At least two of the following signs

1. Tachycardia
2. Palpitations
3. Sweating and chills
4. Tremor
5. Blurring of vision
6. Pupillary dilatation
7. Incoordination

Nicotine

Dysfunctional behaviour or perceptual abnormalities, as evidenced by at least one of these:

1. Insomnia
2. Bizarre dreams
3. Lability of mood
4. Derealisation
5. Interference with personal functioning.

At least one of the following signs

1. Nausea or vomiting
2. Sweating
3. Tachycardia
4. Cardiac arrhythmias

Volatile solvents

Behavioural changes which include at least one of the following

1. Apathy and lethargy
2. Argumentativeness
3. Abusiveness or aggression
4. Lability of mood
5. Impaired judgement
6. Impaired attention and memory
7. Psychomotor retardation
8. Interference with personal functioning.

At least one of the following signs

1. Unsteady gait
2. Difficulty standing
3. Slurred speech
4. Nystagmus
5. Decreased level of consciousness (e.g. stupor, coma)
6. Muscle weakness
7. Blurred vision or diplopia.

When severe, it may be accompanied by hypotension, hypothermia; depression of gag reflex.

Appendix 8

Withdrawal states for substances of abuse (Ref ICD-10 DCR)

Alcohol

Presence of any three of the following:

1. Tremor of the outstretched hands, tongue or eyelids
2. Sweating
3. Nausea, retching or vomiting
4. Tachycardia or hypertension
5. Psychomotor agitation
6. Headache
7. Insomnia
8. Malaise or weakness
9. Transient visual, tactile or auditory hallucinations or illusions
10. Grand-mal convulsions.

If delirium is present, the diagnosis of alcohol withdrawal state with delirium ("delirium tremens") should be made.

Cannabis

This is an ill-defined syndrome for which definitive diagnostic criteria are not yet established. It has been reported variously as lasting from several hours to up to seven days following cessation of prolonged high-dose use of cannabis. Symptoms and signs include anxiety, irritability, tremor of the outstretched hands, sweating, and muscle aches.

Opioid

Presence of any three of the following

1. Craving for an opioid drug
2. Rhinorrhea or sneezing
3. Lacrimation
4. Muscle aches or cramps
5. Abdominal cramps
6. Nausea or vomiting
7. Diarrhea
8. Pupillary dilatation
9. Piloerection, or recurrent chills
10. Tachycardia or hypertension
11. Yawning
12. Restless sleep

Sedative hypnotic

Presence of any three of the following

1. Tremor of the outstretched hands, tongue or eyelids
2. Nausea or vomiting
3. Tachycardia
4. Postural hypotension
5. Psychomotor agitation
6. Headache
7. Insomnia
8. Malaise or weakness
9. Transient visual, tactile or auditory hallucinations or illusions
10. Paranoid ideation
11. Grand-mal convulsions

If delirium is present, the diagnosis of sedative-hypnotic withdrawal state with delirium should be made.

Cocaine or other stimulants

Presence of dysphoric mood along with any two of the following symptoms and signs

1. Lethargy and fatigue
2. Psychomotor retardation or agitation
3. Craving for cocaine
4. Increased appetite
5. Insomnia or hypersomnia
6. Bizarre or unpleasant dreams.

Nicotine

Presence of any two of the following symptoms and signs

1. Craving for tobacco (or other nicotine-containing products)
2. Malaise or weakness
3. Anxiety
4. Dysphoric mood
5. Irritability or restlessness
6. Insomnia
7. Increased appetite
8. Increased cough
9. Mouth ulceration
10. Difficulty concentrating

Appendix 9

Classification of Epilepsy, ILAE

Table 1. ILAE Classification of seizure type, 1981

- I. Partial (focal or local) seizures
 - A. Simple partial seizures (SPS)
 - 1. With motor signs
 - 2. With somatosensory or special sensory symptoms
 - 3. With autonomic symptoms or signs (including epigastric aura)
 - 4. With psychic symptoms
 - B. Complex partial seizures (CPS)
 - 1. Simple partial onset followed by impairment of consciousness
 - 2. With impairment of consciousness at onset
 - C. Partial seizures evolving to generalized tonic-clonic convulsions (GTC)
 - 1. Simple partial seizures evolving to generalized
 - 2. Complex partial seizures evolving to generalized
 - 3. SPS evolving to CPS evolving to generalized
- II. Generalized Seizures (convulsive or non-convulsive)
 - A. Absences
 - 1. Typical absences
 - 2. Atypical absences
 - B. Myoclonic
 - C. Clonic seizures
 - D. Tonic seizures
 - E. Tonic-clonic seizures
 - F. Atonic seizures
- III. Unclassified epileptic seizures

Table 2. ILAE's 1989 International Classification of Epilepsies and Epileptic Syndromes

- I. Localization-related epilepsies and syndromes
 - A. Idiopathic
 - 1. Benign childhood epilepsy with centrotemporal spikes
 - 2. Childhood epilepsy with occipital paroxysms
 - 3. Primary reading epilepsy
 - B. Symptomatic
 - 1. Chronic progressive epilepsia partialis continua of childhood (Kojewnikow's Syndrome)
 - 2. Syndromes characterized by seizures with specific modes of precipitation
 - 3. Temporal lobe epilepsies
 - 4. Frontal lobe epilepsies
 - 5. Parietal lobe epilepsies
 - 6. Occipital lobe epilepsies
 - C. Cryptogenic

II. Generalized epilepsies and syndromes

A. Idiopathic

1. Benign neonatal familial convulsions
2. Benign neonatal convulsions
3. Benign myoclonic epilepsy in infancy
4. Childhood absence epilepsy
5. Juvenile absence epilepsy
6. Juvenile myoclonic epilepsy
7. Epilepsy with GTCS on awakening
8. Other generalized idiopathic epilepsies not defined above
9. Epilepsies with seizures precipitated by specific modes of activation

B. Cryptogenic or symptomatic

1. West syndrome
2. Lennox-Gastaut syndrome
3. Epilepsy with myoclonic-astatic seizures
4. Epilepsy with myoclonic absences

C. Symptomatic

1. Non-specific etiology
 - a. Early myoclonic encephalopathy
 - b. Early infantile epileptic encephalopathy with suppression burst
 - c. Other symptomatic generalized epilepsies not defined above
2. Specific syndromes
 - a. Diseases in which seizures are a presenting feature

III. Epilepsies and syndromes undetermined whether focal or generalized

A. With both generalized and focal seizures

1. Neonatal seizures
2. Severe myoclonic epilepsy in infancy
3. Epilepsy with continuous spike-waves during slow wave sleep
4. Acquired epileptic aphasia (Landau-Kleffner syndrome)
5. Other undetermined epilepsies not defined above

B. Without unequivocal generalized or focal features (i.e. – Sleep related GTCS; when the EEG shows both focal and generalized ictal or interictal discharges, and when focal or generalized onset cannot be determined clinically)

IV. Special syndromes

A. Situation-related seizures

1. Febrile convulsions
2. Isolated seizures or isolated status epilepticus
3. Seizures occurring only when there is an acute metabolic or toxic event (alcohol, drugs, eclampsia, nonketotic hyperglycemia)

ILAE Commission on Classification and Terminology, 2005–2009 in its report has suggested some revision in terminology and concepts for organization of seizures and epilepsies. A summary of those suggestions is given in tables that follow

TABLE 3. *Classification of Seizures*

- I. Generalized onset
 - A. Tonic-clonic seizures (in any combination)
 - B. Absences
 1. Typical absences
 2. Atypical absences
 3. Absence with special features
 - a. Myoclonic absences
 - b. Eyelid myoclonia
 - C. Myoclonic
 1. Myoclonic seizures
 2. Myoclonic atonic
 3. Myoclonic tonic
 - D. Clonic seizures
 - E. Tonic seizures
 - F. Atonic seizures
- II. Focal onset (partial)
 - A. Without impairment of consciousness/responsiveness
 1. With observable motor or autonomic components (roughly corresponds to concept of SPS)
 2. Involving subjective sensory or psychic phenomenon only (corresponds to concept of aura)
 - B. With impairment of consciousness/responsiveness (roughly corresponds to concept of CPS)
 - C. Evolving to a bilateral convulsive seizure (involving tonic, clonic or tonic and clonic components, replaces the term secondarily generalized seizure)
- III. May be focal, generalized or unclear
 - A. Epileptic spasms

TABLE 4. *Electro-clinical syndromes and other epilepsies*

Electro-clinical syndromes arranged by age of onset

Neonatal period

- Benign familial neonatal seizures (BFNS)
- Early myoclonic encephalopathy (EME)
- Ohtahara syndrome

Infancy

- Migrating partial seizures of infancy
- West syndrome
- Myoclonic epilepsy in infancy (MEI)
- Benign infantile seizures
- Dravet syndrome
- Myoclonic encephalopathy in non-progressive disorders

Childhood

- Febrile seizures plus (FS+)
- Early onset benign childhood occipital epilepsy (Panayiotopoulos type)
- Epilepsy with myoclonic atonic (previously astatic) seizures
- Benign epilepsy with centro-temporal spikes (BECTS)
- Autosomal-dominant nocturnal frontal lobe epilepsy (ADNFLE)
- Late onset childhood occipital epilepsy (Gastaut type)
- Epilepsy with myoclonic absences
- Lennox-Gastaut syndrome (LGS)
- Epileptic encephalopathy with continuous spike-and-wave during sleep (CSWS) including Landau-Kleffner syndrome (LKS)
- Childhood absence epilepsy (CAE)

Adolescence

- Juvenile absence epilepsy (JAE)
- Juvenile myoclonic epilepsy (JME)
- Epilepsy with generalized tonic-clonic seizures alone
- Progressive myoclonus epilepsies (PME)
- Autosomal dominant partial epilepsy with auditory features (ADPEAF)
- Other familial temporal lobe epilepsies

Less Specific Age Relationship

- Familial focal epilepsy with variable foci
- Reflex epilepsies

Distinctive constellations

- Mesial temporal lobe epilepsy with hippocampal sclerosis (MTLE with HS)
- Rasmussen syndrome
- Gelastic seizures with hypothalamic hamartoma

Epilepsies that *do not* fit into any of these diagnostic categories can be distinguished first on the basis of the presence or absence of a known structural or metabolic condition (presumed cause) and then on the basis of the primary mode of seizure onset (generalized versus focal).

Epilepsies attributed to and organized by structural-metabolic causes

- Malformations of Cortical development (hemimeganencephaly, heterotopias etc)
- Neurocutaneous syndromes (Tuberous sclerosis complex, Sturge-Weber, etc)
- Tumor
- Infection
- Trauma; Peri-natal insults
- Angioma
- Stroke Etc.

Epilepsies of unknown cause

Conditions with epileptic seizures that are traditionally not diagnosed as a form of epilepsy *per se*.

- Benign neonatal seizures (BNS)
- Febrile seizures (FS)